# Virtuoso® Layout Pro: T8 Virtuoso Concurrent Layout Editing

**Course Version IC 25.1**

**Lab Manual** **Revision 1.0**

# Table of Contents

## Virtuoso Layout Pro: T8 Virtuoso Concurrent Layout Editing

# Module 1: About This Course

**There are no labs for this module.**

# Module 2: Setting Up the Concurrent Layout Editing Environment

## Lab 2-1 Setting Up the Working Directory

### Objective: To set up your design environment to run the CLE tasks.

The Virtuoso® Concurrent Layout (CLE *with E for editing*) is a layout editing environment that enables designers to work concurrently on the same cellview within Virtuoso. This helps them parallelize their efforts and, in turn, increases the productivity of the layout design team. You can perform concurrent editing in Layout XL and Layout EXL.

This lab aims to provide an overview of the Concurrent Layout Editing (CLE) flow with a focus on the capability to edit, review, and merge design changes at the top level and in the hierarchical subcells. It also describes different design cases where you can take advantage of the area-based partition during concurrent editing by pushing down the area partitions into the hierarchical subcells to define the Edit Scope for hierarchical concurrent editing.

### Overview of the CLE Flow

CLE allows concurrent editing on the same cellview at the same time. You can use this tool to concurrently edit the top design as well as the hierarchical subcells.

To concurrently edit your designs, you first need to create design partitions. You should create the design partitions based on the task for which you want to use the CLE flow.

### Roles and Responsibilities

A user can play the following two roles in the CLE flow, namely, **Manager** and **Designer**. The responsibilities they follow are based on their current role.

#### Manager

Owner of the cellview. The Manager has the following responsibilities:

- Decide whether to enable concurrent layout editing on the cellview.
- Distribute the cellview by defining design partitions based on the task.
- Review the progress of the developments in the respective design partitions without blocking them. This is called the health-check merge flow.
- Review and merge the edited design partitions submitted by the designers with the top design.

#### Designer

A user or users who work concurrently in the assigned design partition views.

- The Edit Scope for the designers is defined by the design partition region assigned to each design partition view. Alerts are displayed if a designer edits outside the current design partition or when a peer designer tries to edit a design partition assigned to the current designer. At any point during editing, designers can preview or import the changes made by peer designers to view updates made by them.
- While editing in a design partition view, a designer can either save the changes locally or check in the updated design partition view. The status of the design partition view stays as *Not Submitted*. The manager can merge these changes to preview them but cannot permanently save them in the top design.
- After completing the edits, the designer needs to submit the edited design partition view for merge. This changes the status of the design partition view to *Submitted*. The manager can now review and merge the changes and then permanently save them in the top design.

### Concurrent Layout Modes

The commands that are enabled in the *Concurrent* menu depend on the mode in which the design is open. These modes are the following:

#### Manager Mode

It lets you perform various managerial tasks, such as defining the design partition for each designer and merging the respective design partition views back to the top design.

#### Designer Mode

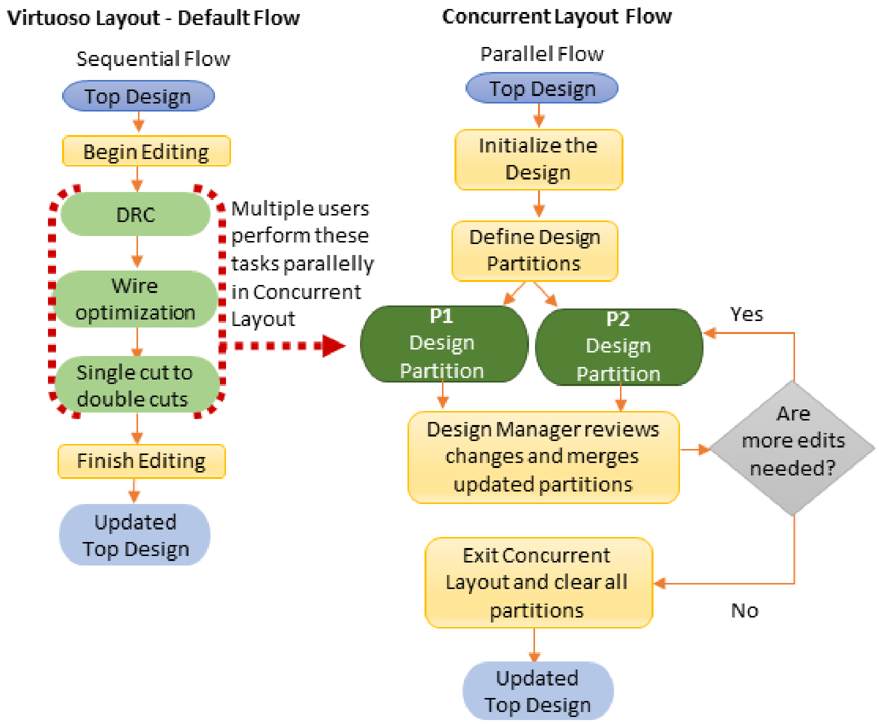
It lets you edit the design in the assigned design partition view and then submit these changes for merging with the top design.

#### Single User Mode

It lets a single user edit a design in a safe environment. You can choose to work on the complete design or in a focused area or focused range of layers. Your changes are implemented only when you save your changes and merge them with the top design.

### The Default Virtuoso Layout Flow and Concurrent Layout Flow

The default Virtuoso Layout flow is sequential, in which each task is carried out after the previous task is complete. However, in the CLE environment, you can create multiple partitions in the design and have designers work in parallel, which helps save time. The following flow chart shows the difference between the default Virtuoso Layout flow and the Concurrent Layout flow.




The basic Concurrent Layout flow involves the following tasks:

1. Open the top design. (Design Manager)
2. Initialize the design for concurrent layout editing. (Design Manager)
3. Define design partitions based on how you want to divide the work among various designers. (Design Manager)
4. Perform various tasks such as DRC checks and wire optimization in parallel on assigned designed partitions. (Designers)
5. Save design partition updates in respective design partition views and submit the updates for merge with the top design. (Designers)
6. Review and merge design partitions. (Design Manager)

7. Repeat steps four to six as needed. A merged design partition is auto reset in memory when it is reopened. (Designers)

8. Exit the CLE environment and clear all design partitions. (Design Manager)

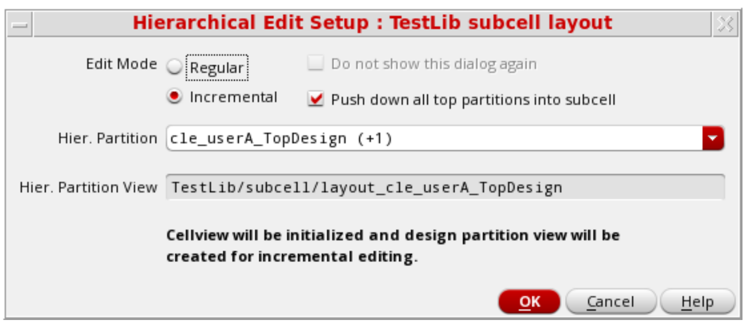
9. Open the updated top cellview in Virtuoso Layout. (Design Manager)

### Overview of Hierarchical Editing in the CLE Flow

Hierarchical editing in the CLE flow lets multiple designers concurrently Edit In Place (EIP) into hierarchical subcells. The Edit Scope is retained because area partitions defined at the top are pushed down the hierarchy. So, a manager can define design partitions without worrying about the area boundaries of the hierarchical subcells.

### Hierarchical Edit Modes in CLE

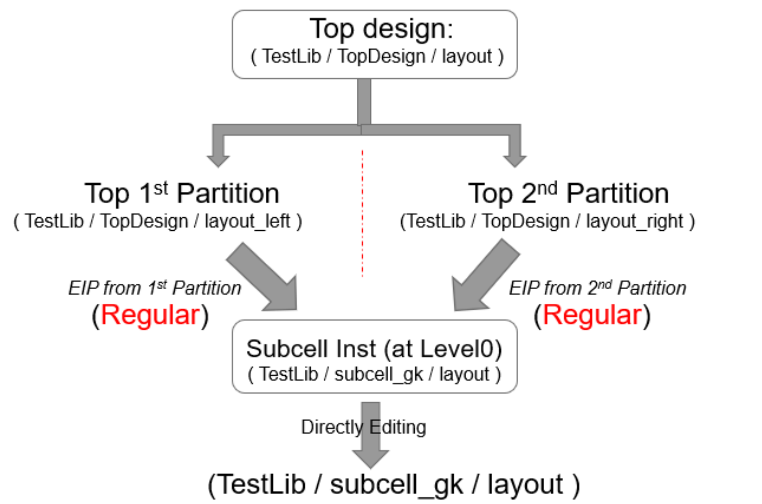
When you EIP into a hierarchical subcell from a top design partition view, CLE provides two hierarchical editing modes – **Regular** and **Incremental**.

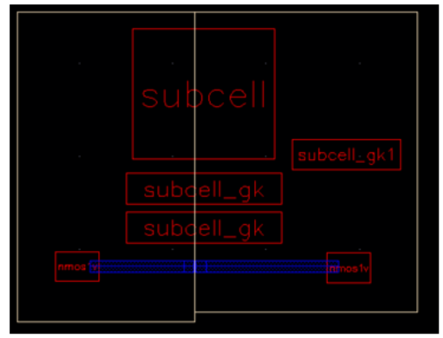


#### Regular Edit Mode

EIP in this mode lets you edit the main cellview, and only one user can make the edits at a time. This means that concurrent editing of the hierarchical subcell is not possible in this mode. After the subcell is edited and saved, the changes cannot be reversed.

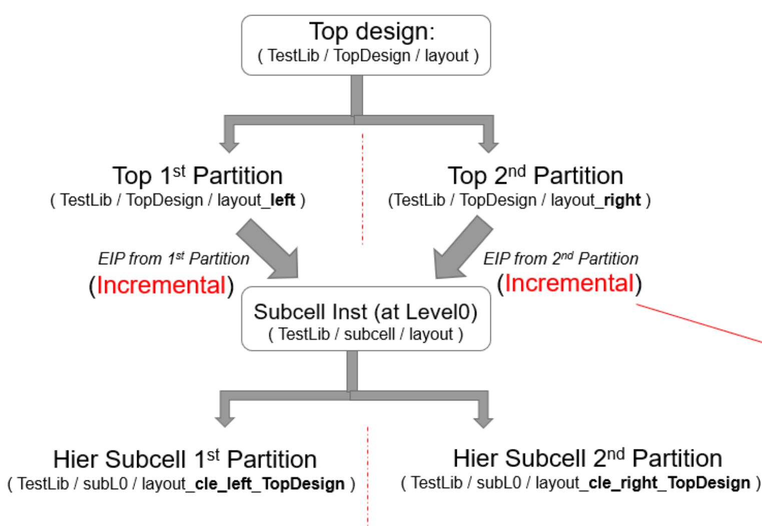


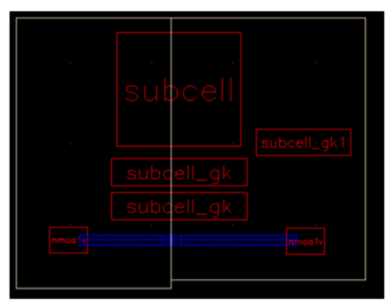



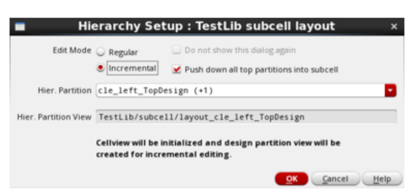

**Note:** *Concurrent Layout* assistant is disabled when it is in Regular mode.

#### Incremental Edit Mode

EIP in this mode keeps the original hierarchical subcell unmodified. A hierarchical design partition view is created when you EIP from the parent top design partition view. The area partitions are pushed down in the hierarchical subcell from the parent top design partition view to keep the same Edit Scope in the hierarchical subcell.



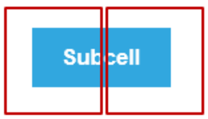




Hier Subcell Partition name:
***cle_<topPartitionName>_<topCellName>***

CLE enables incremental mode by creating hierarchical design partition views and pushing down the area partitions to limit the Edit Scope to be the same as what is set on the top design partition view.

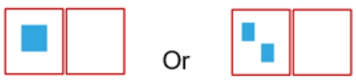
The following cases show which hierarchical edit mode is selected in different scenarios.

Case1. Hierarchical subcell straddles over two or more design partitions and has a single occurrence.



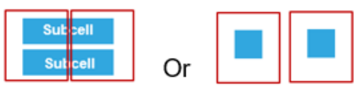

In this case, the hierarchical editing mode can be either Incremental or Regular.

Case2. Hierarchical subcell has one or more occurrences, but all are only in one design partition.




In this case, the hierarchical editing mode is Regular.

Case3. Hierarchical subcell has one or more occurrences in two or more design partitions.




In this case, the hierarchical editing mode is Regular.

### Invoking the *virtuoso* Executable

In this lab, you set up your design environment to run the Concurrent Layout Editing tasks in the subsequent labs.

The following lab requires users in the following roles:

- Manager
- Two designers, userA and userB

If you are a single designer testing the CLE flow, you can play the role of the three users using two VNC sessions.

In the first VNC session, you can be the manager and the first designer, userA, where:

- The Manager will define the design partitions.
- userA will edit the first design partition.

In the second VNC session, you can be the second designer, userB, and edit the second design partition.

In the following steps, you will switch between the two VNC sessions to perform tasks as the manager, userA, and userB.

1. Open the first VNC session, and perform the following steps.

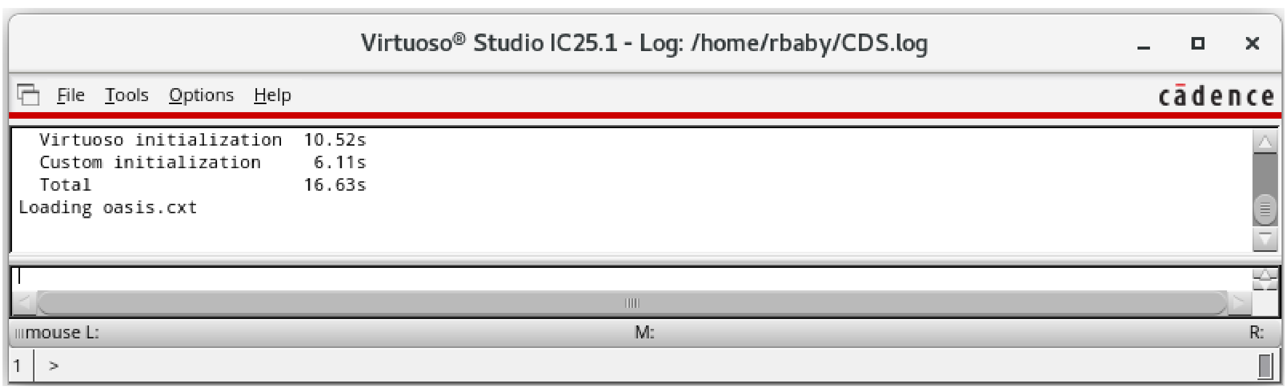
2. In the first VNC session, in a terminal window, enter the following command from your home (login) directory to get to the lab database's first working directory when you initially logged in:

```
cd ./VLPT8_IC_25_1
```

3. To start the Cadence® Virtuoso software, enter this command in the same terminal window:

```
virtuoso &
```

The Command Interpreter Window (CIW) appears.




**Note:** Keep the display to your first VNC session, for example, in your UNIX terminal, enter: `setenv DISPLAY noivl1-rbaby:1`

4. In the CIW, choose **Tools – Library Manager** to display the Library Manager if it is not already open.

5. In the Library Manager, select the following to open the top design cellview later in the lab:

| | |
|---|---|
| **Library** | `TestLib` |
| **Cell** | `TopDesign` |
| **View** | `layout` |

6. Open the second VNC session, and perform the following steps.

7. Go to the second working directory.

```
cd ./VLPT8_IC_25_1/UserB
```

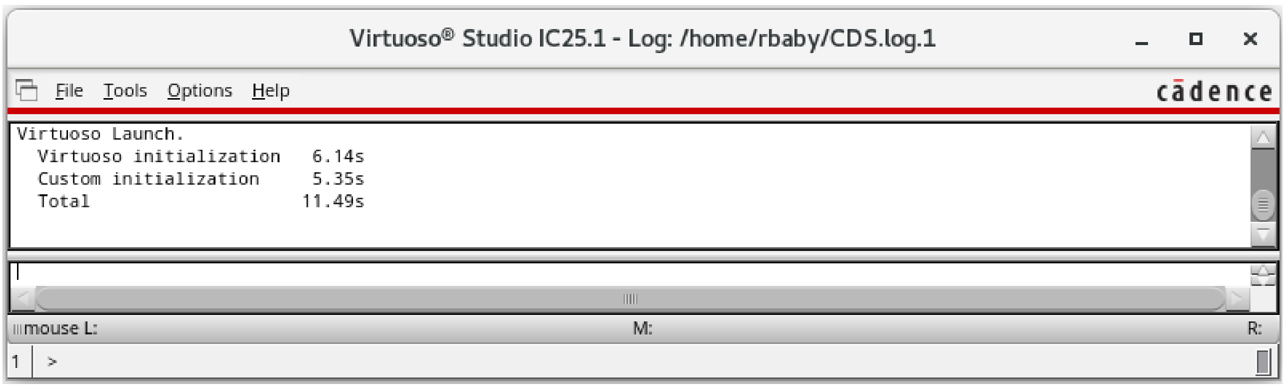
8. Redefine the main library to include it in the *cds.lib* file.

   **Note:** Change the path of the main library in *cds.lib* to refer to the main library. It is already defined, open the *cds.lib* file, observe it, and close the file.

9. To start the Cadence® Virtuoso software, enter this command in the same terminal window:

```
virtuoso &
```

   The Command Interpreter Window (CIW) appears.




   **Note:** Keep the display to your second VNC session, for example, in your UNIX terminal, enter: `setenv DISPLAY noivl1-rbaby:2`

10. Keep both the CIW windows open in both the VNC sessions with the Virtuoso session running and continue with the next lab.

## Summary

In this lab, you learned how to:

- Set up your design environment to run the Concurrent Layout Editing tasks.

# Module 3: Initializing and Partitioning the Top Design in Manager Mode

## Lab 3-1 Initializing the Top Design in Manager Mode

### Objective: To initialize the top design to use in the CLE flow.

In this lab, you will see how to initialize the top design in manager mode to use in the Concurrent Layout Editing (CLE) flow.

From Concurrent Layout Flow, you can see that tasks in Concurrent Layout are dependent on whether you are in manager mode or designer mode. There are certain tasks, such as initializing the design and defining the design partitions, that can be done only in manager mode, while tasks such as editing the design partition or resolving the conflicts can be done only in designer mode.

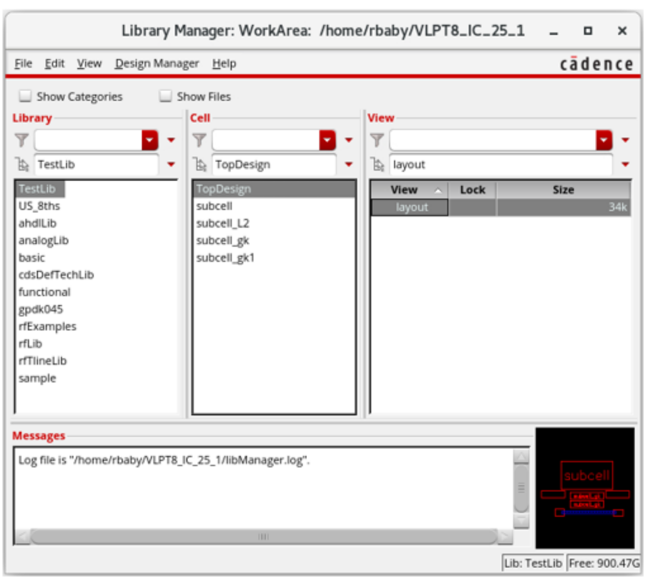
A single user can perform tasks of both manager and designer at a time. You cannot have more than one manager for the top cell on which you are performing concurrent editing. However, you can have multiple designers working in parallel on different design partition views. The manager should have write permission on the top design for creating design partition views.

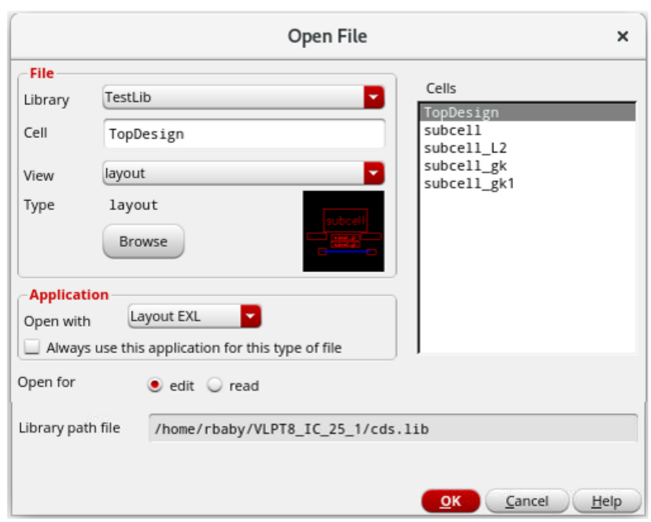
1. Ensure that both the CIW windows are open in both the VNC sessions.

2. Open the first VNC session, and perform the following steps as a manager:

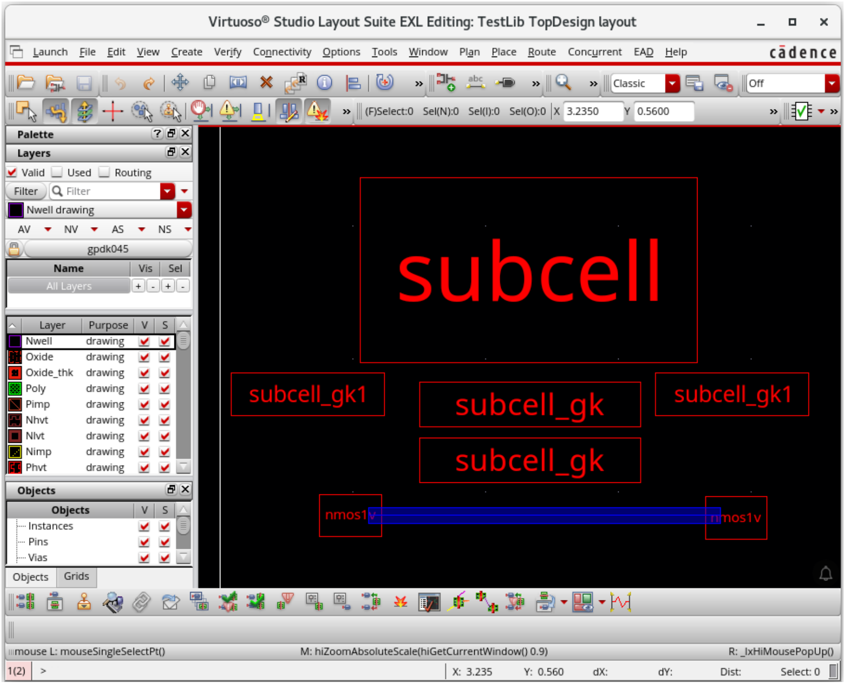
3. Open the following layout in **EXL** from the **Library Manager** or choose **File – Open** from the CIW, as shown here.

| | |
|---|---|
| **Library** | `TestLib` |
| **Cell** | `TopDesign` |
| **View** | `layout` |





The *TestLib/TopDesign/layout* window opens in the **EXL** mode, as shown here.




**Note:** In the layout canvas, from the **Window – Assistants** menu, you can enable/disable the assistants as per your requirement.

**Note:** In the layout canvas, from the **Window – Toolbars** menu, you can make the toolbars appear/disappear as per your requirement.

**Note:** In the layout canvas, from the **Workspace Configuration** cyclic field, you can choose the required workspace.

## Accessing the Concurrent Layout Assistant

The *Concurrent Layout* assistant is a dockable assistant pane that provides various options that let you perform tasks related to concurrent layout editing. The assistant also displays alerts and other information about the design partition you are updating.

1. To access the *Concurrent Layout* assistant in Layout EXL, do one of the following:

   a. In the Layout EXL window, choose **Concurrent – Initialize**.

   b. In the Layout EXL window, choose **Window – Assistants – Concurrent Layout**.

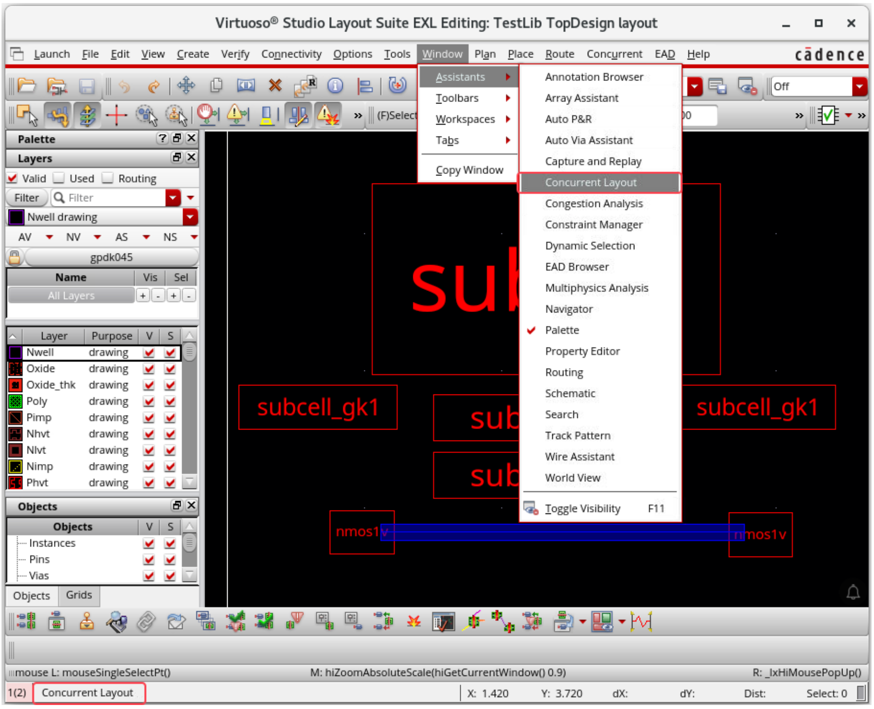
   c. **Right**-click the main Layout EXL menu or toolbar area and select **Concurrent Layout**.

   d. Apply the Concurrent Layout workspace by selecting **Window – Workspaces – Concurrent_Layout**.

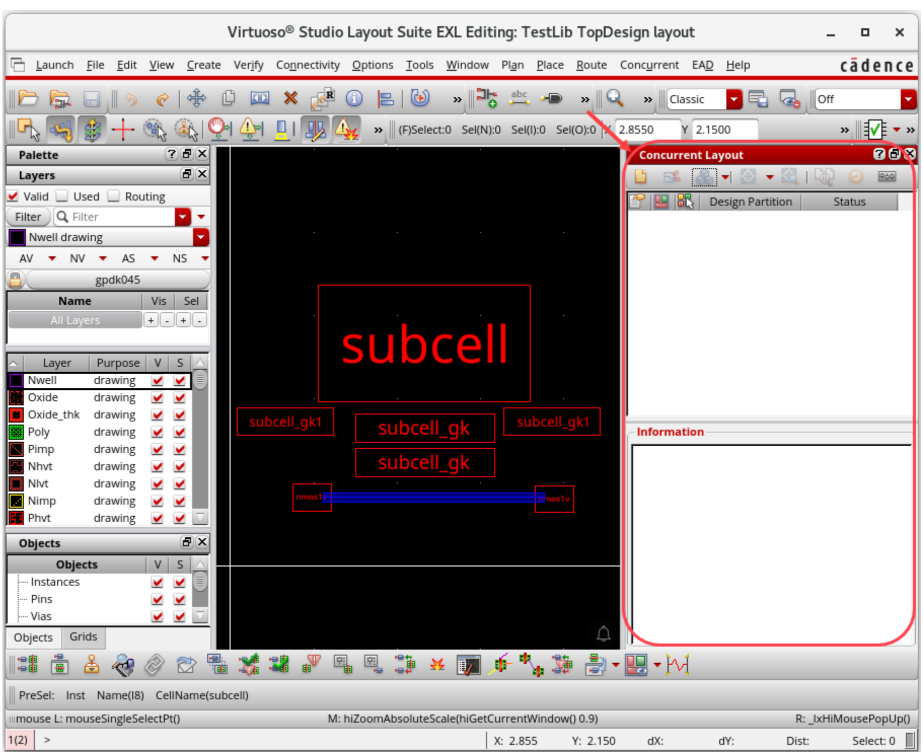
 

e. Select **Concurrent_Layout** from the drop-down combo box on the Workspaces toolbar.

In this example, the **Window – Assistants – Concurrent Layout** menu command is used, and *Concurrent Layout* assistant is displayed, as shown here.



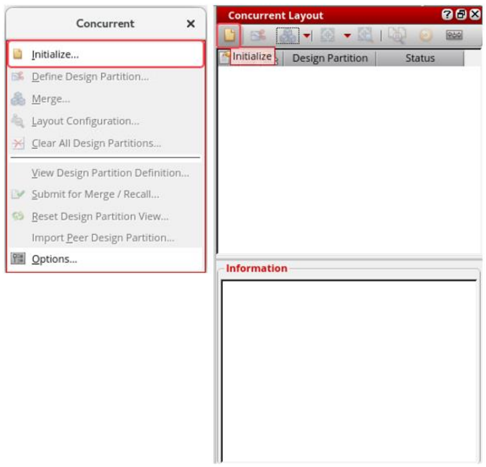

Layout EXL embeds the *Concurrent Layout* assistant as a docked assistant pane within the current session window. By default, the *Concurrent Layout* assistant is positioned on the right side of the session window.



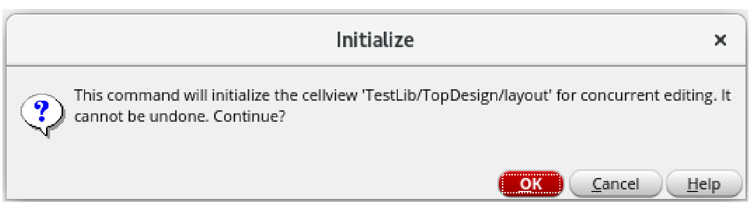
## Initializing the Top Design for Concurrent Layout Editing

The first task for the manager is to initialize the design to get it ready for concurrent editing.

1. To initialize the design for concurrent layout editing, in the layout canvas, choose the **Concurrent – Initialize** menu command from the menu bar or click the **Initialize** icon on the *Concurrent Layout* assistant toolbar.

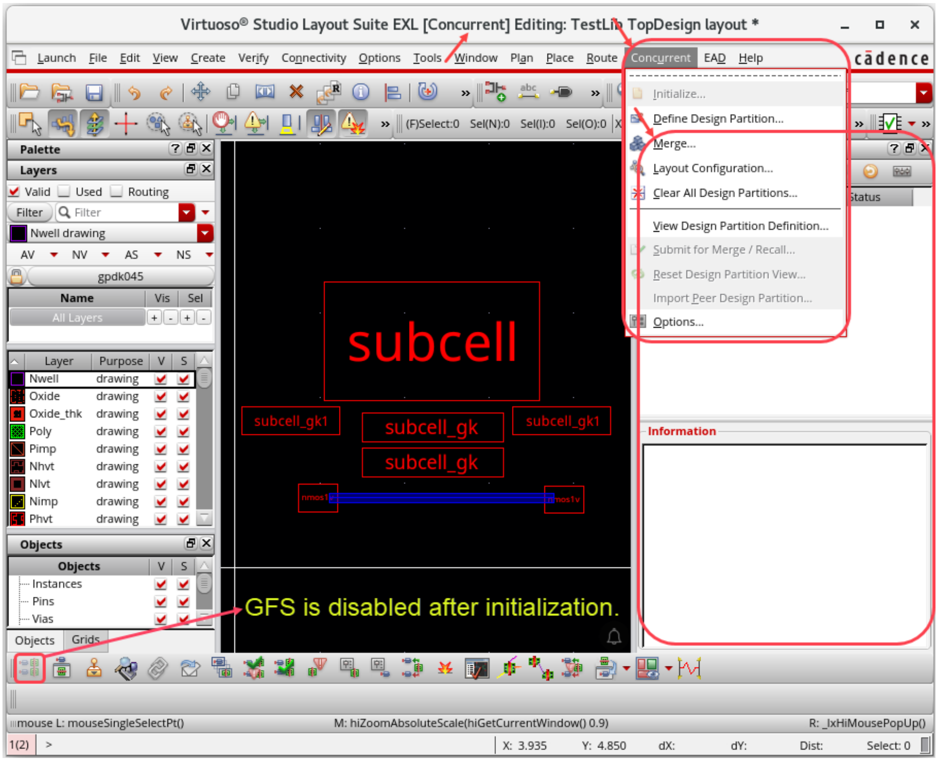



You get the *Initialize* popup window, as shown here.




2. Click **OK**.

The cellview is initialized for concurrent editing, and the *Concurrent Layout* assistant is displayed if it was not displayed earlier, as shown here.




Initialization saves some information about all the objects in the design. It is used as a reference later to identify and merge back the objects edited in a design partition.

After the design is initialized for concurrent editing, additional options will be available on the *Concurrent* menu and the *Concurrent Layout* assistant, as shown here.

Layout configuration cellview *TestLib/TopDesign/layout_cle_layConfig* has been created.

The cellview contains a text file that is automatically updated to list all subcells from top partitions that were incrementally edited during concurrent editing.

The value of the concurrent layout option **DM Mode** is set to **Off** for cellview *TestLib/TopDesign/layout*.

**Note:** *Generate All From Source* is disabled after initialization because it can erase concurrent layout data saved in the initial design. Run this command before initialization.

3. Keep the *TestLib/TopDesign/layout* cellview open and the Virtuoso session running and continue with the next lab.

### Summary

In this lab, you learned how to:

- Initialize the top design to use in the CLE flow.

End of Lab

# Lab 3-2 Partitioning the Top Design in Manager Mode

## Objective: To partition the top design to use in the CLE flow.

In this lab, you will see how to partition the top design in manager mode to use in the Concurrent Layout Editing (CLE) flow.

From the previous lab, the *TestLib/TopDesign/layout* cellview should be opened. If not, open it now.

### Defining an Area-Based Design Partition

After the design is initialized, you can start defining the design partitions based on your editing requirements.

You should partition the design carefully to avoid partition overlapping, which will increase the likelihood of edit conflicts among design partitions. Additionally, try not to use a global container, like a FigSet, across design partitions.

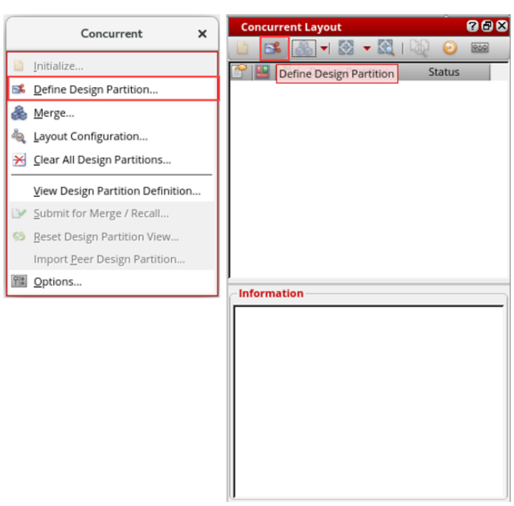
You can define the following types of design partitions:

- Area-Based Design Partition
- Layer-Based Design Partition
- Mixed Design Partition

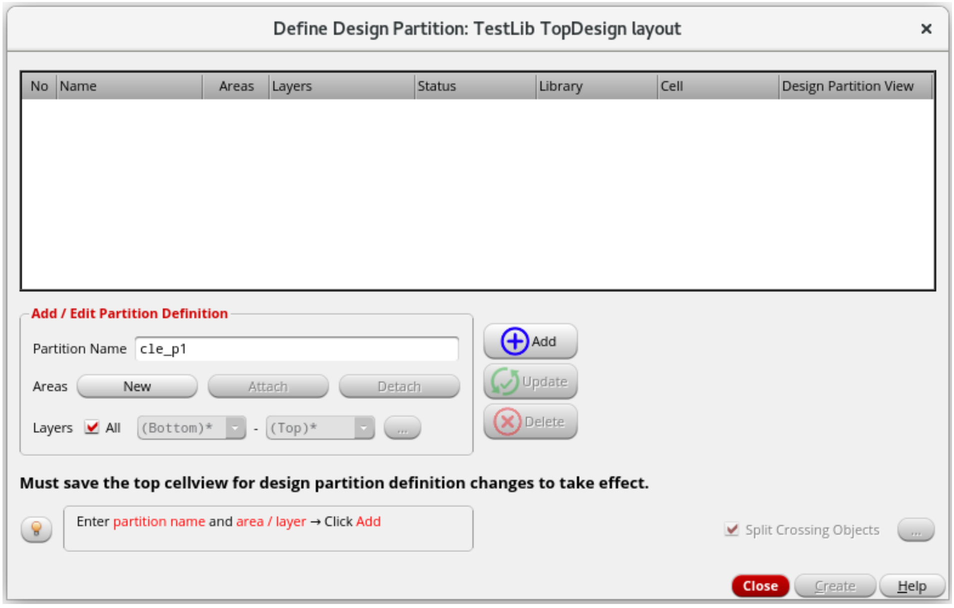
In this lab, you will add the area-based design partition definitions for **userA** and **userB**.

#### To Add the Design Partition Definition for userA:

1. In the layout canvas, choose the **Concurrent – Define Design Partition** menu command from the menu bar or click the **Define Design Partition** button on the *Concurrent Layout* assistant.

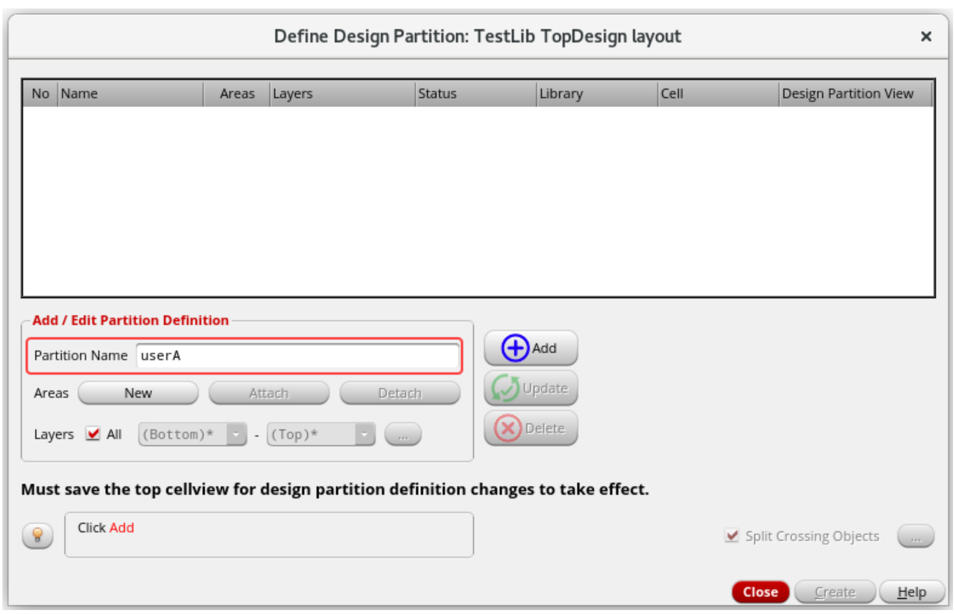


The Define Design Partition form appears.




2. Add **Area-based design partition** in the Define Design Partition form.

3. Specify a name, say, **userA**, in this example, for the new design partition in the *Partition Name* field in the Define Design Partition form.



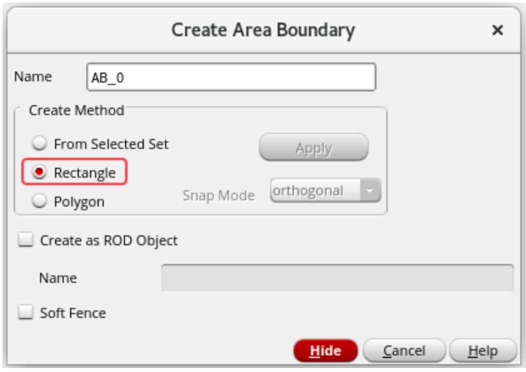

**Note:** If you do not specify a name, the default name in this field is used. By default, the names of the new design partitions and the corresponding design partition views are **cle_px** and **layout_cle_px**, where **x** is a number.

4. Click the **New** button in the *Areas* field in the Define Design Partition form.

   a. Press the **F3** key to display the *Create Area Boundary* form.

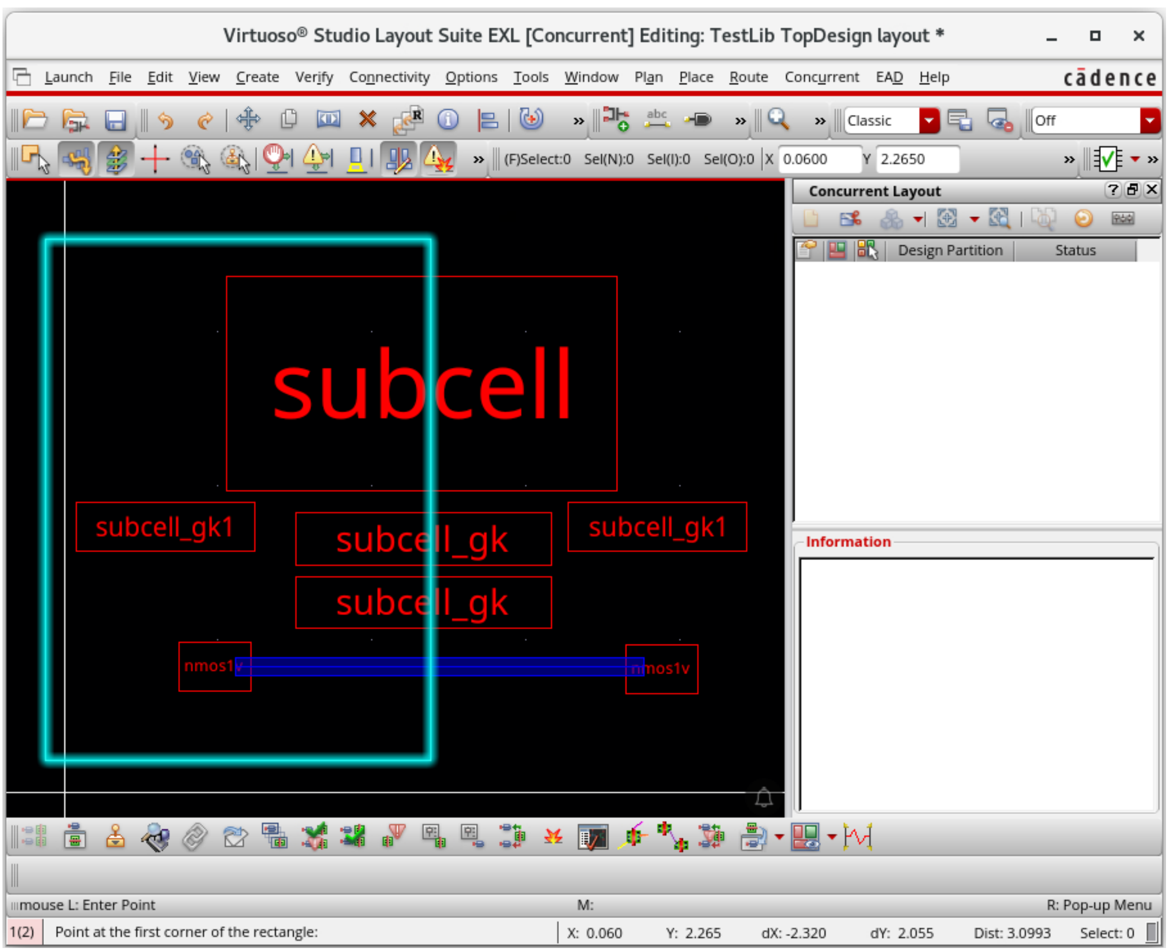
   b. Select **Rectangle** as the *Create Method*.




   c. Now, *Point at the first corner of the rectangle*.

   d. Then *Point at the opposite corner of the rectangle*.

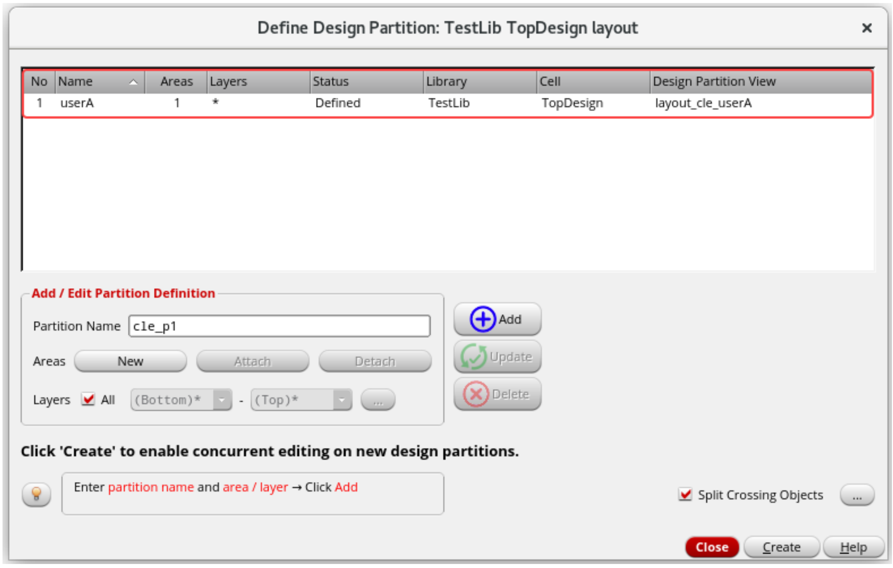
   *Area Boundary* is created for **userA**, as shown here.



5. Click the **Add** button in the Define Design Partition form.

   A design partition, named **userA**, with corresponding design partition view, **layout_cle_userA**, with **1 Area** *Defined*, is added for the **TopDesign** cell in the **TestLib** library in the Define Design Partition form, as shown here.

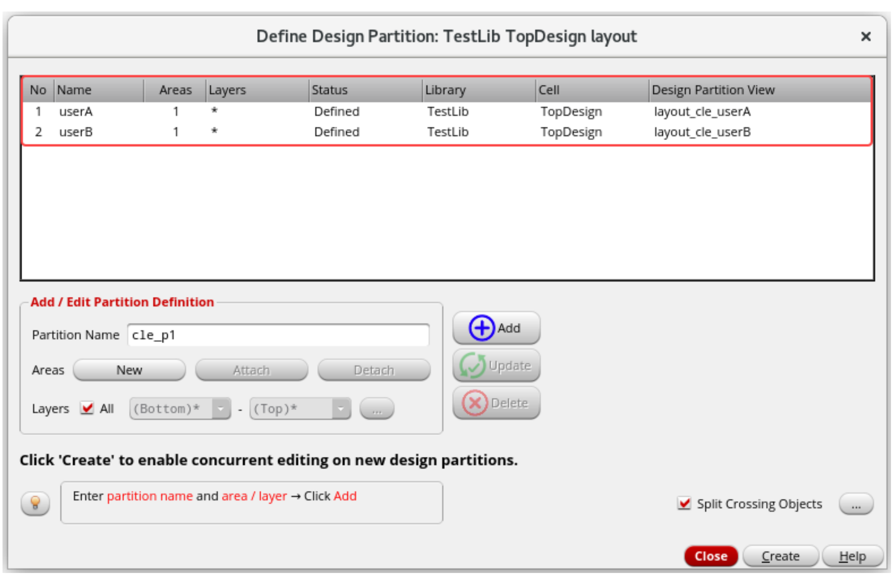
   The status of the newly added design partition, **userA**, is *Defined*. If a design partition view already exists, the status is *Reuse*.




### To Add the Design Partition Definition for userB:

1. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.

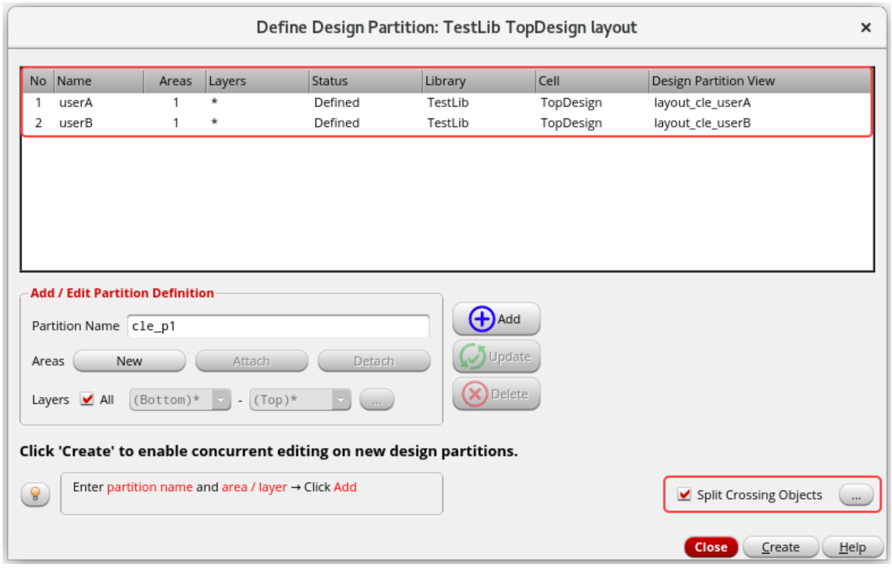
   You see the two design partitions, named **userA** and **userB**, with their corresponding design partition views, **layout_cle_userA**, and **layout_cle_userB**, with **1 Area** *Defined* each, are added for the **TopDesign** cell in the **TestLib** library in the Define Design Partition form, as shown here.



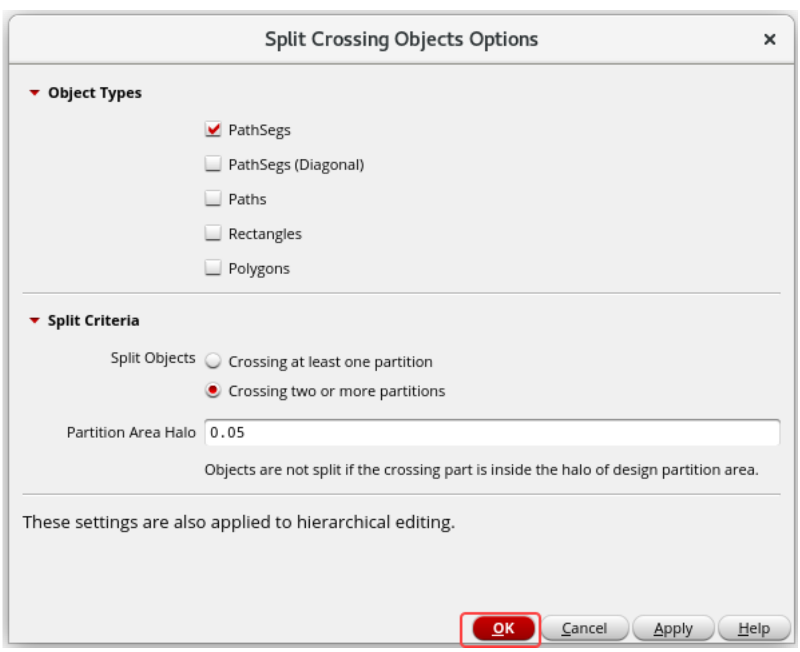
**Note:** Use the *Attach* and *Detach* buttons on the Define Design Partition form to add or remove the area boundary associated with a design partition.

**Note:** Use the *Update* and *Delete* buttons on the Define Design Partition form to update and delete the design partitions.

2. Click the button next to the **Split Crossing Objects** option to display.




The Split Crossing Objects Options form is displayed.

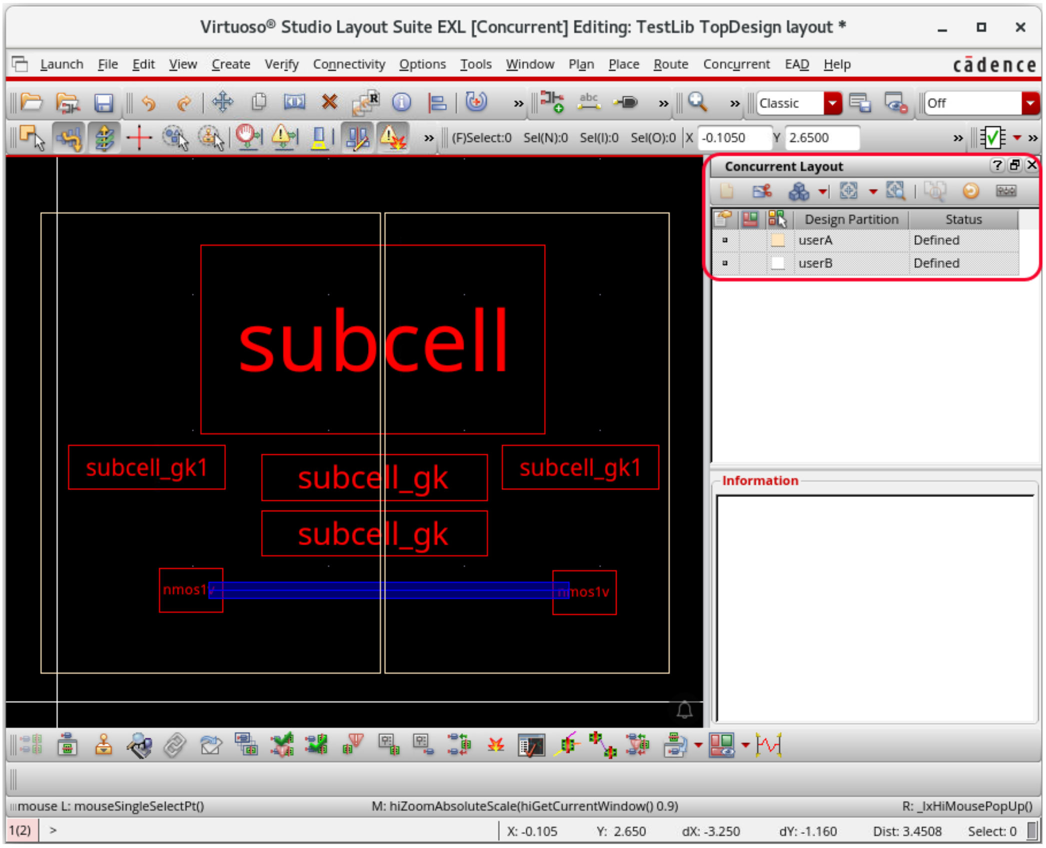



You can use this form to change the settings of how crossing objects are handled when the design partitions are created.

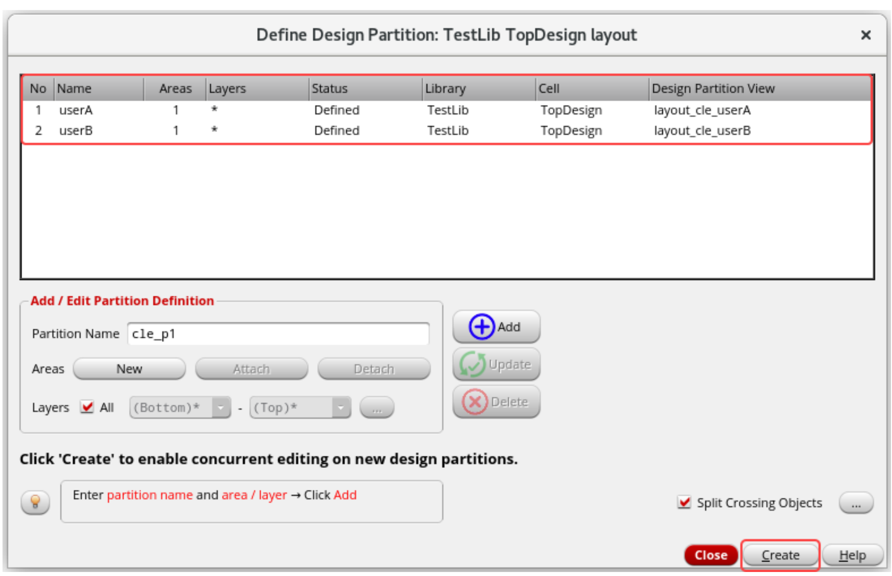
a. Keep the default values and **OK** the form.

You see the *TestLib/TopDesign/layout* cellview with their design partitions, named, **userA**, and **userB** *Defined* and added in the *Concurrent Layout* assistant, as shown here.



3. Click the **Create** button after you have completed defining all the design partitions.

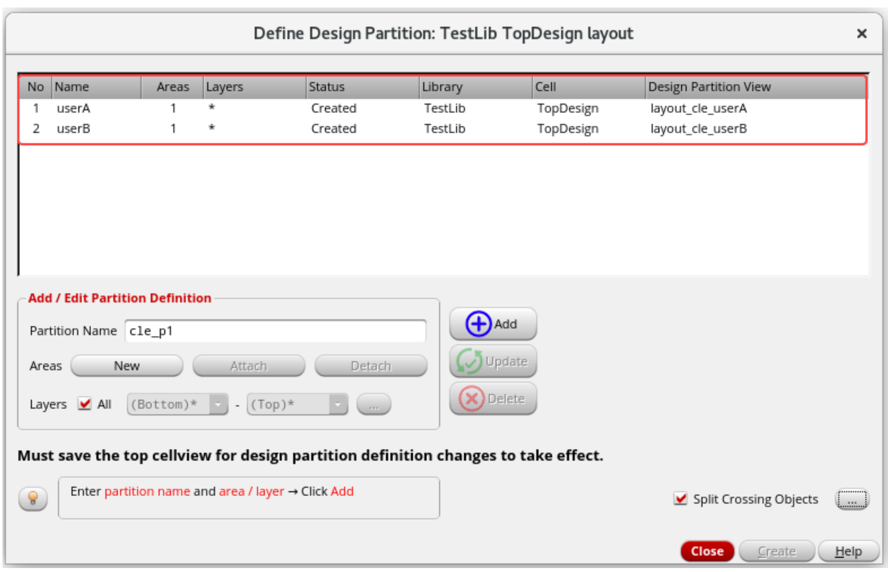


A message box is displayed informing you about the design partitions that will be created.

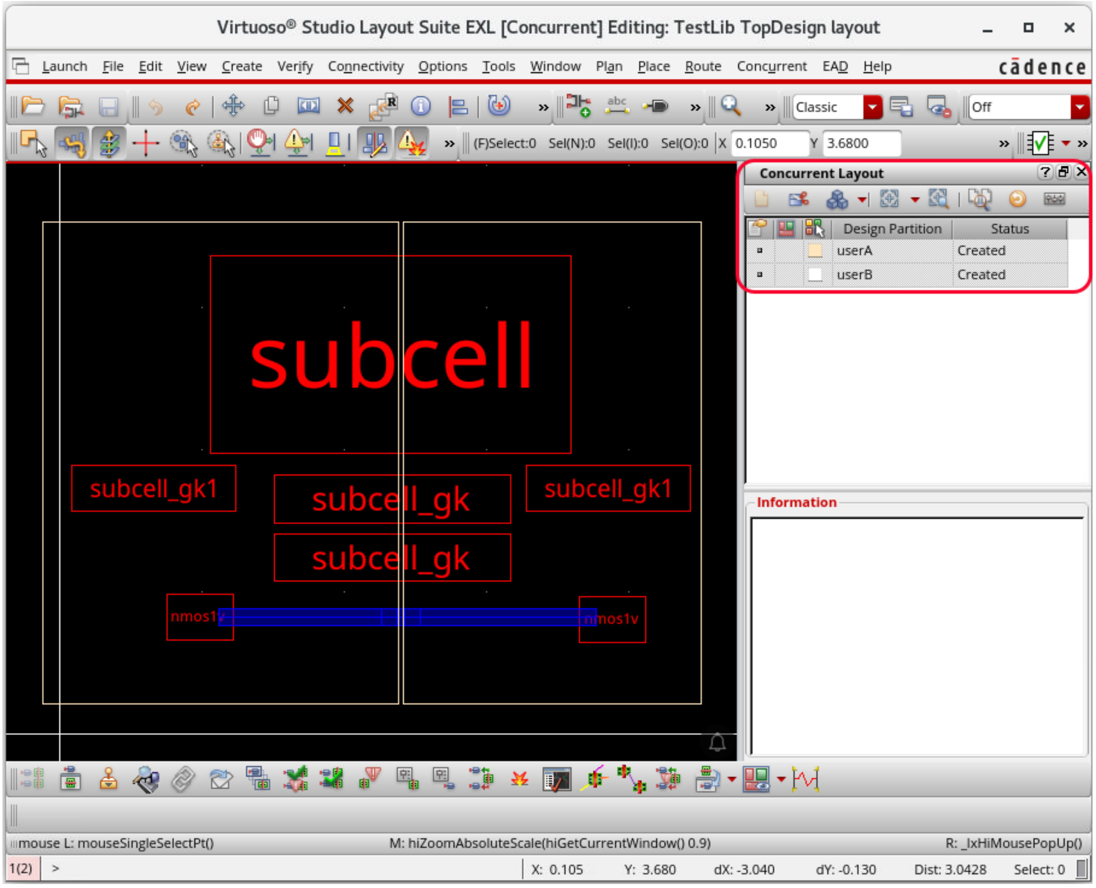

a. Click **OK**.

- Design partition views are created, and the status *Created* is displayed in the Define Design Partition form and in the *Concurrent Layout* assistant.

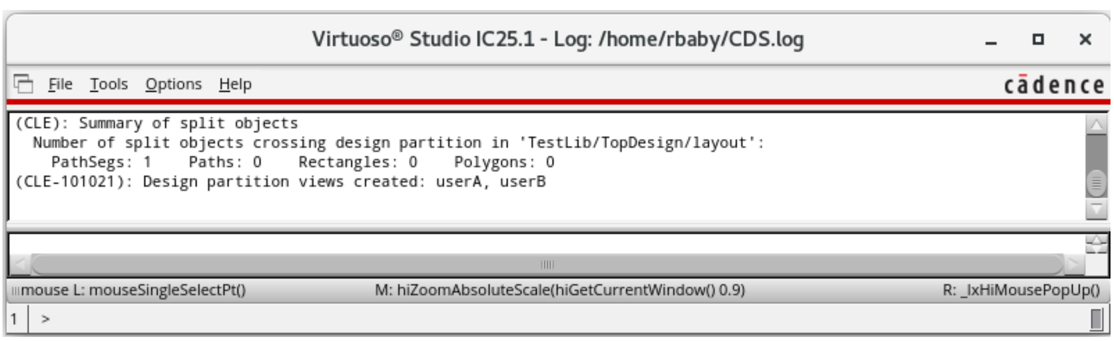
  You see the status *Created* in the Define Design Partition form, as shown here.



You see the status *Created* in the *Concurrent Layout* assistant, as shown here.




- The top design is saved automatically.
- Objects crossing the design partitions are split at the boundary. This happens because the **Split Crossing Objects** option is selected by default. Observe the CIW window for the *Summary of split objects*.

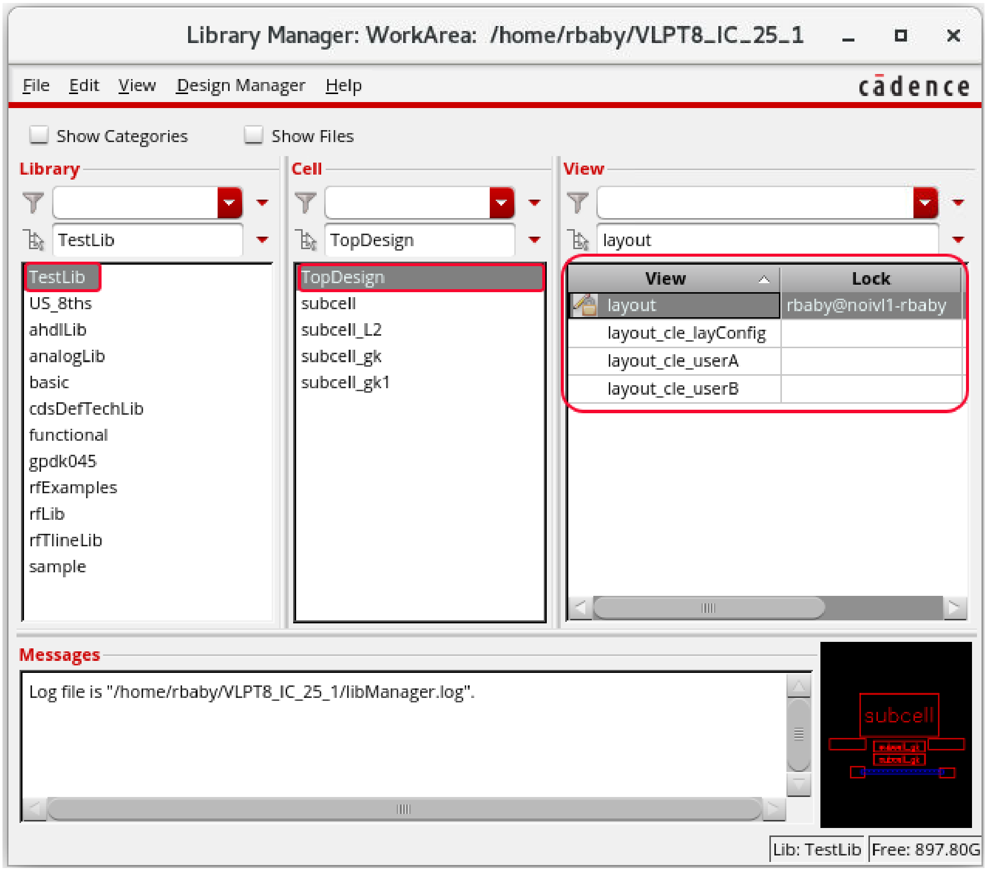



4. Click the **Close** button to close the Define Design Partition form.

   In the Design Management environment, an additional dialog box is displayed to confirm that new design partition views have been checked into the design management system.

   You can exit the Define Design Partition form only by clicking the **Close** button.

5. Open the Library Manager to review the created design partition views and observe the changes.




You see that *TestLib/TopDesign/layout_cle_userA* and *TestLib/TopDesign/layout_cle_userB* design partition views are created, as shown here.

The created design partition views are of minimal size. This is because only the changes made to the associated design partition are saved in this view.

Design partition definitions are saved in the top design, so any modifications in the design partition definition or area boundaries are saved only in the top design to be saved and checked in before the changes are reflected in the design partition view.

6. Close the top design *TestLib/TopDesign/layout* and keep both the Virtuoso sessions running to start concurrent editing in the design partition views you have created.

## Summary

In this lab, you learned how to:

- Partition the top design to use in the CLE flow.

End of Lab

# Module 4: Editing the Design Partition userA in Designer Mode

# Lab 4-1 Setting the Edit Scope for the Design Partition userA in Designer Mode

**Objective: To set the edit scope for the design partition userA in designer mode.**

In this lab, you will see how to set the *Edit Scope* for the design partition **userA** in designer mode to use in the Concurrent Layout Editing flow.

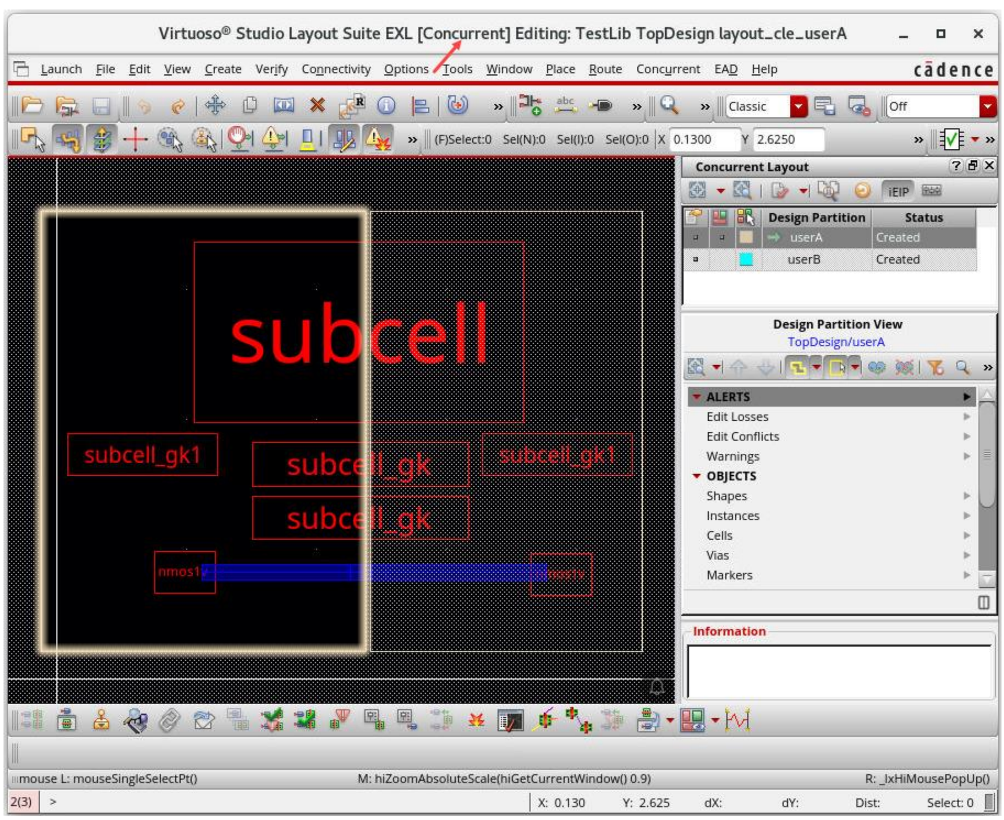
1. Ensure that both the CIW windows are open in both the VNC sessions.

2. In the following steps, you switch to designer mode and continue using the same first VNC session.

3. To start editing as first designer **userA**, open the following first design partition view from the **Library Manager** or choose **File – Open** from the CIW, as shown here.

| | |
|---|---|
| **Library** | `TestLib` |
| **Cell** | `TopDesign` |
| **View** | `layout_cle_userA` |

The *TestLib/TopDesign/layout_cle_userA* cellview opens in the **EXL** mode, which is enabled for **Concurrent Editing**, as shown here.



### Setting the Edit Scope for the Design Partition userA

Concurrent Layout uses the definition you made in the Define Design Partition form to identify which object belongs to which design partition. Information about the objects undergoing modification in a design partition is stored in the associated design partition view.

Edit scope for each design partition is set to select only those objects that are inside the current partition by default.

You can change these settings using one of the following two methods in designer mode:

- Using the *Concurrent Layout Options* form.
- Using the *Concurrent Layout* assistant.

If the *Edit Scope* is set to *Off*, editing objects outside the partition generates alerts in the *Concurrent Layout* assistant. You will have to individually sign off each alert and warning.

The following options can be used to configure the Edit Scope for a designer based on requirement:

- *Off*: Disables the edit mode settings.
- *Only Select Inside Partition*: Lets you select only those objects that are inside the current partition. Objects that are fully or partially inside or have been temporarily moved outside the current partition for editing are also selectable.
- *Only Edit Inside Partition*: Lets you edit only those objects that are inside the current partition. When you specify this option, all edit commands are restricted to the specified design partition. You cannot edit objects outside this design partition.

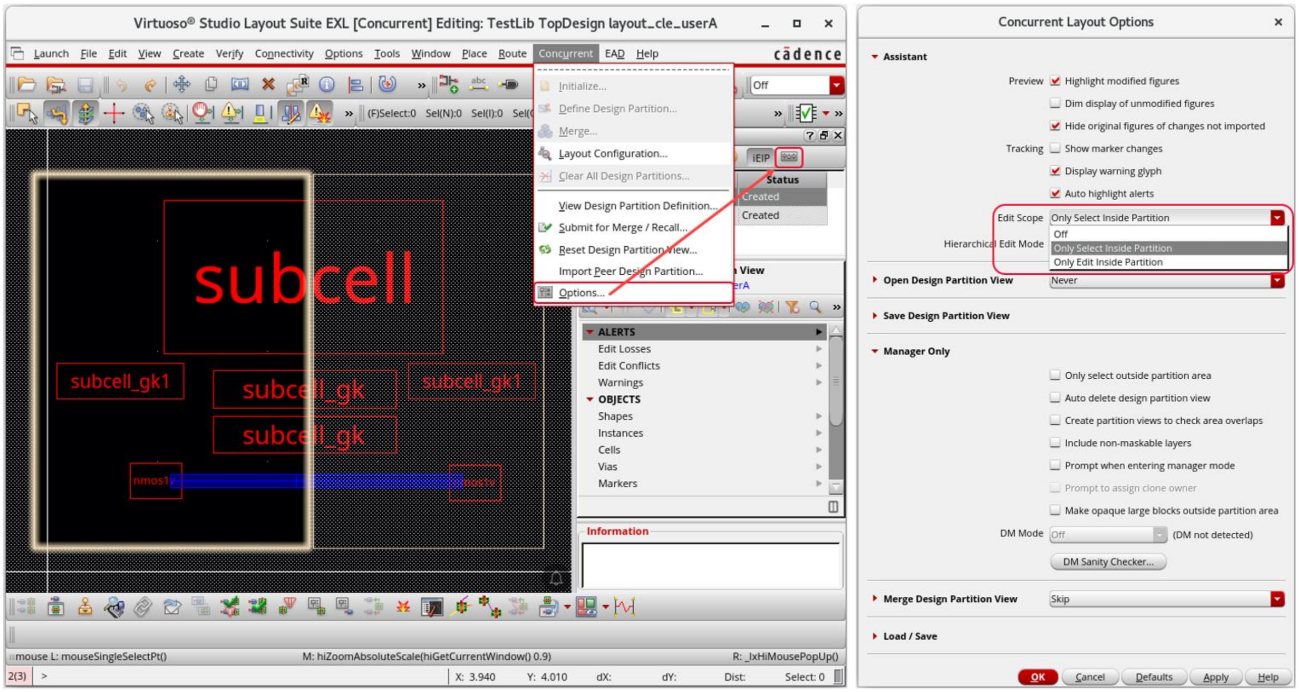
  **Note:** Supported commands are: *Create – Shapes*, *Create – Instance*, *Create – Pin*, *Create – Label*, *Create – Fluid Guard Ring*, *Create – Wiring – Wire*, *Create – Via*, *Edit – Move*, *Edit – Copy*, *Edit – Stretch*, *Edit – Delete*, *Edit – Quick Align*, *Edit – Advanced – Reshape*.

  **Note:** Environment variable: *cleEditMode*

## Method 1: Setting the Edit Scope for userA Using the Concurrent Layout Options Form

You will set the *Edit Scope* for **userA** using the *Concurrent Layout Options* form.

1. To set the Edit Scope for **userA**, perform the following steps:




   a. From the layout window, choose the **Concurrent – Options** menu command or click the **Options** icon in the *Concurrent Layout* assistant to display the *Concurrent Layout Options* form.

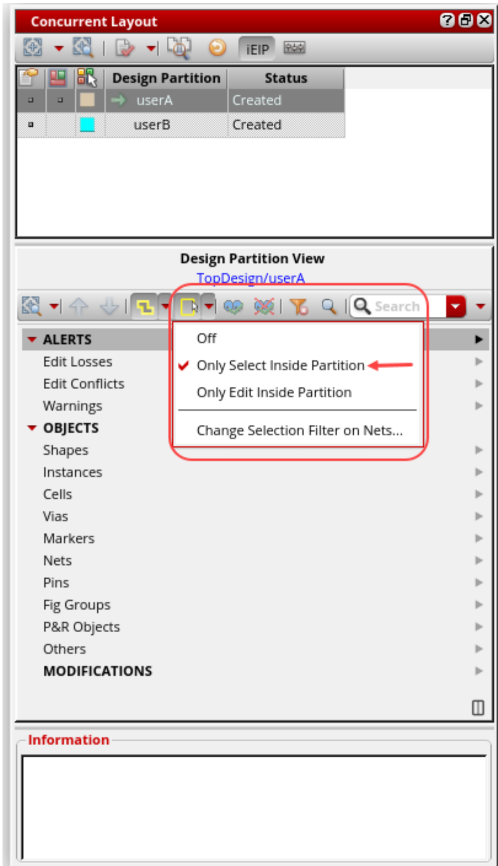
   b. In the *Concurrent Layout Options* form, in the *Edit Scope* cyclic field, choose the required option; in this case, say, select the default option, *Only Select Inside Partition*.

   c. Click **OK** in the form.

### Method 2: Setting the Edit Scope for userA Using the Concurrent Layout Assistant

You will set the *Edit Scope* for **userA** using the *Concurrent Layout* assistant.

1. To set the Edit Scope for **userA**, perform the following steps:




   a. In the *Concurrent Layout* assistant, from the *Edit Scope* cyclic field, choose the required option; in this case, say, select the default option, *Only Select Inside Partition*, as shown here.

2. Keep the *TestLib/TopDesign/layout_cle_userA* cellview open and the Virtuoso session running and continue with the next lab.

### Summary

In this lab, you learned how to:

- Set the *Edit Scope* for the design partition **userA** in designer mode to use in the Concurrent Layout Editing flow.

End of Lab

## Lab 4-2 Editing the Design Partition userA Using the CLE Flow

**Objective: To edit the design partition userA in designer mode using the CLE flow.**

In this lab, you will see how to edit the design partition **userA** in designer mode using the Concurrent Layout Editing flow.

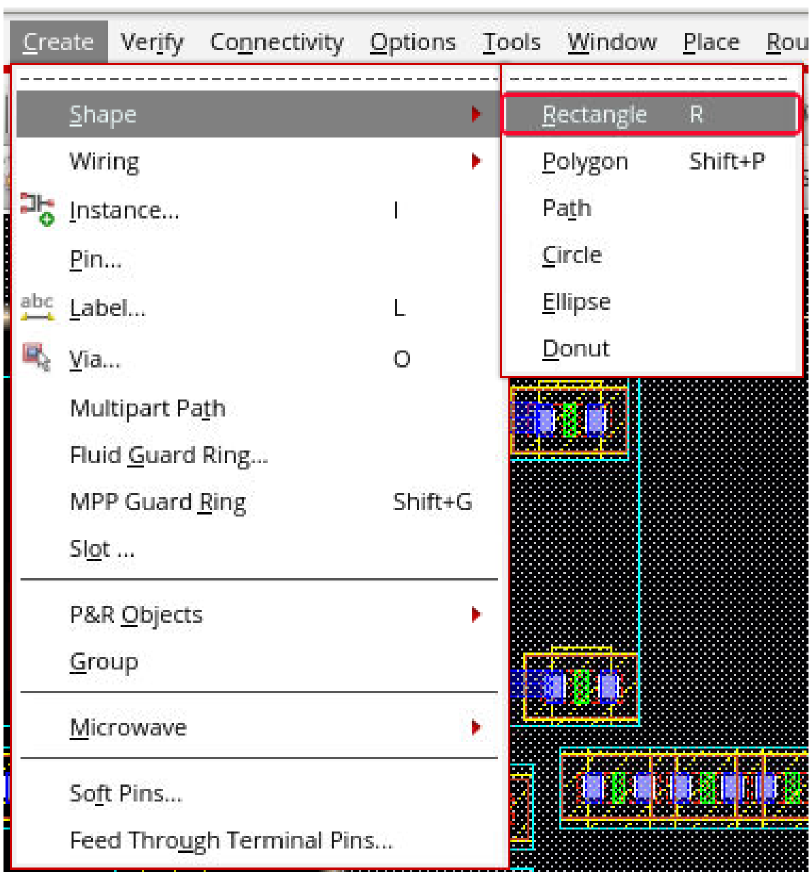
From the previous lab, the *TestLib/TopDesign/layout_cle_userA* cellview should be opened. If not, open it now.

1. In the following steps, you remain in designer mode and continue using the same first VNC session.

### Editing Inside the Design Partition userA Using the CLE Flow

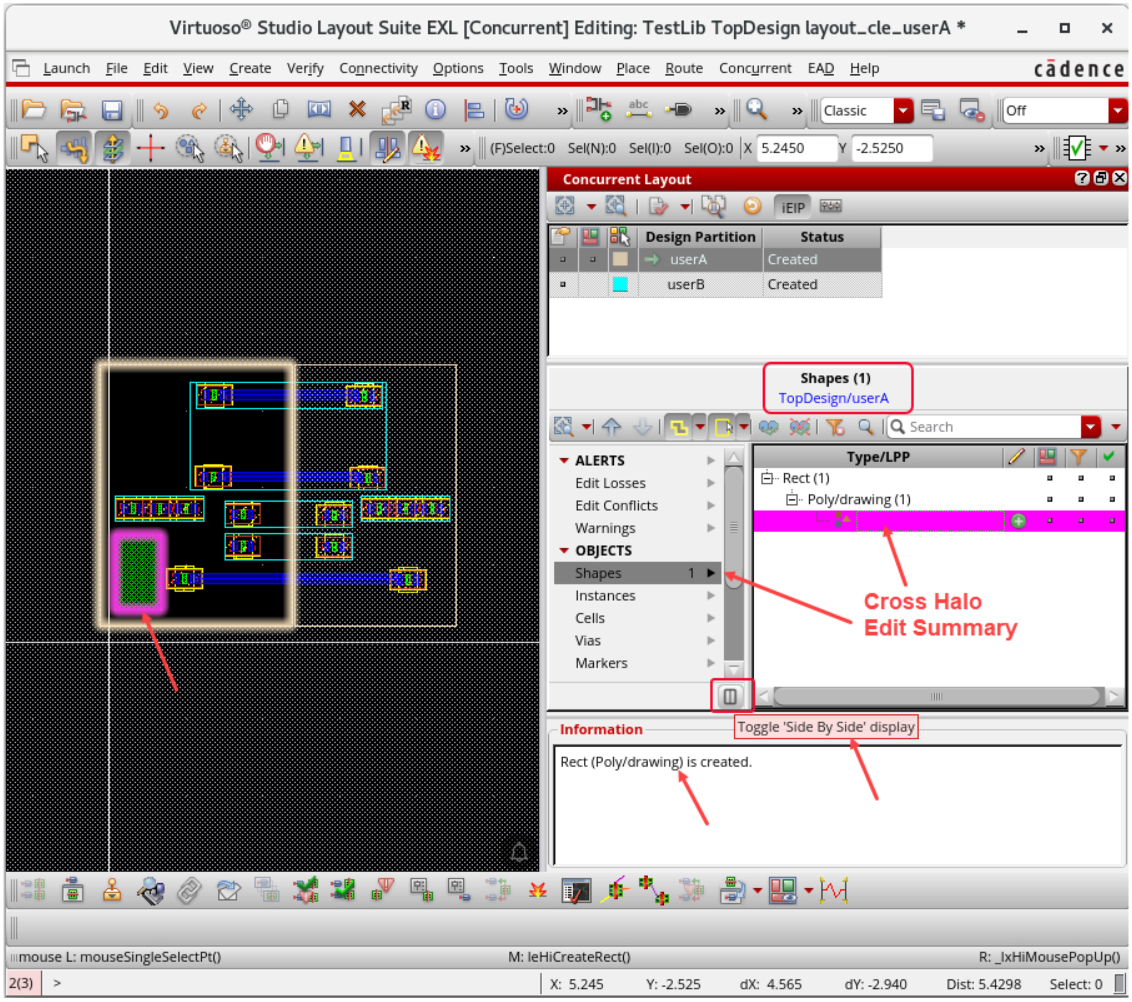
You will make some edits inside the design partition **userA**.

1. In the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.



2. Create a rectangle inside the design partition, as shown here.




Observe the *Edit Summary* and the *Cross Halo* results for the rectangle shape created inside the design partition **userA** in the *Concurrent Layout* assistant.

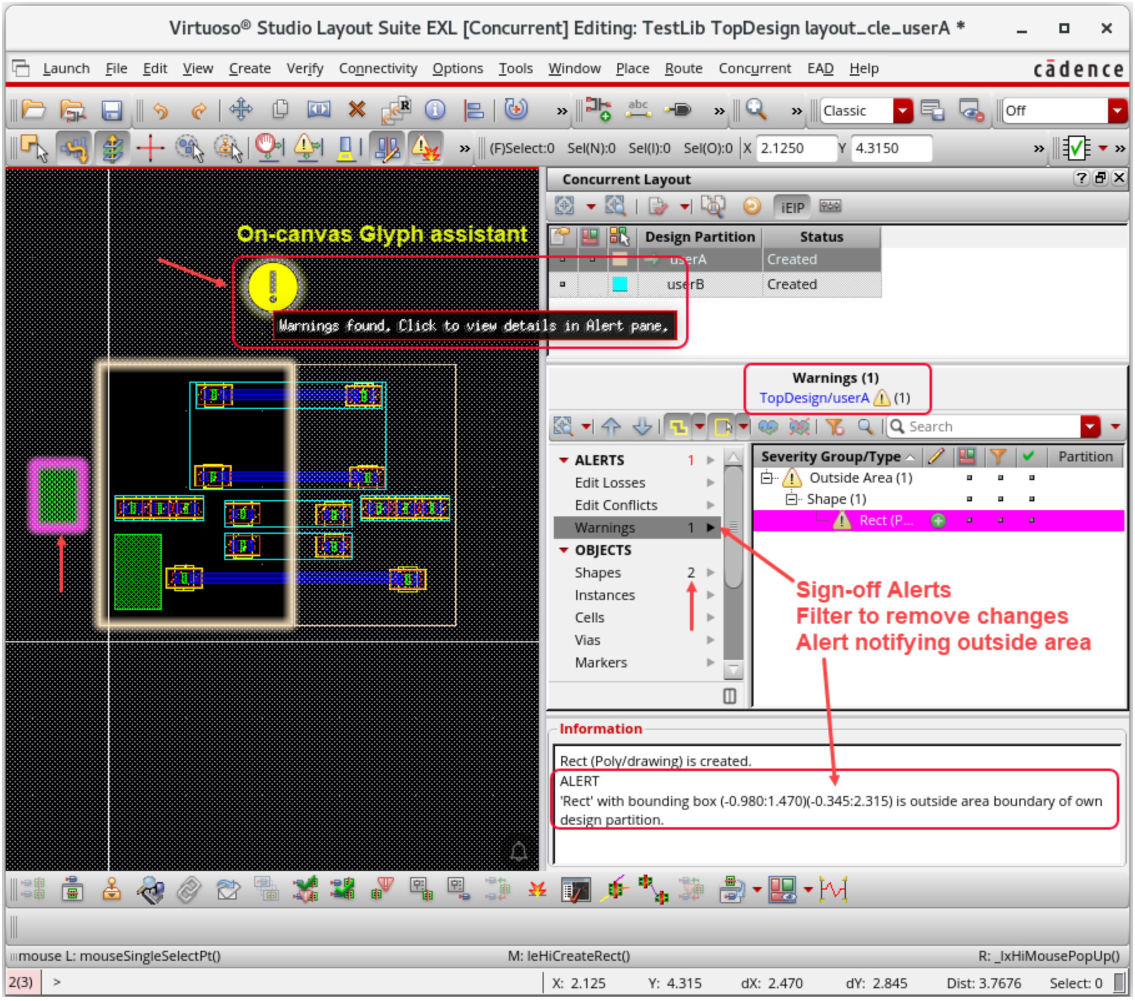
## Editing Outside the Design Partition userA Using the CLE Flow

You will make some edits outside the design partition **userA**.

1. In the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.

2. Create a rectangle outside the design partition, as shown here.

3. Click the *On-canvas Glyph assistant* to view the details in the *Alert* pane.

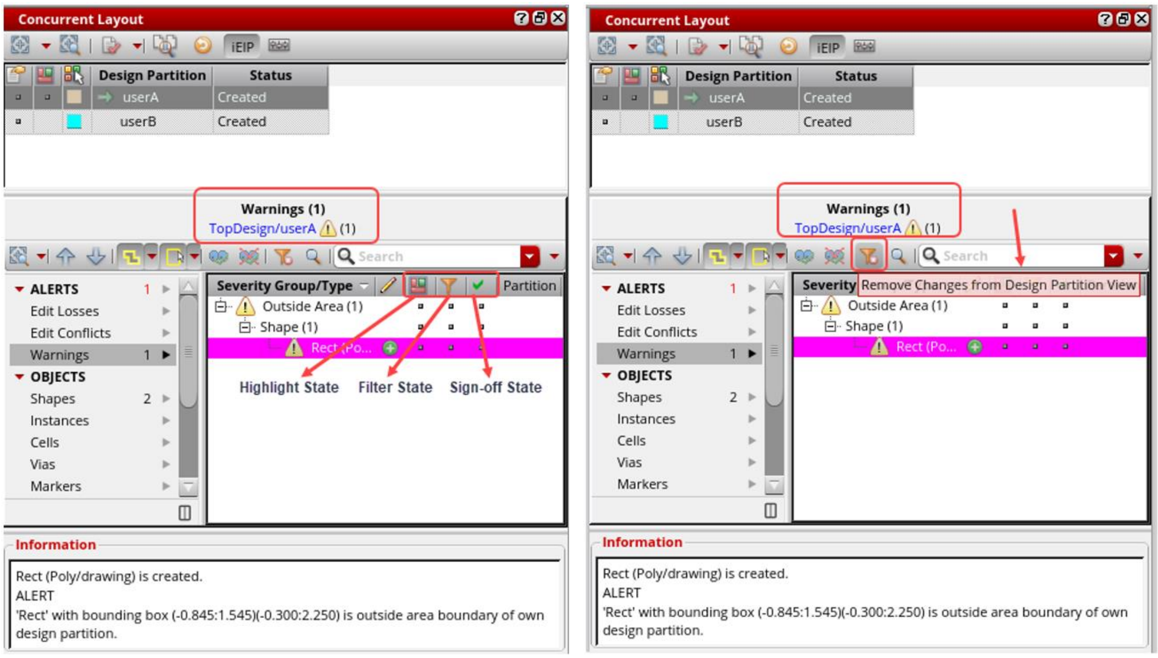



Observe the *Sign-off Alerts* and the *Alert notifying outside area* result for the rectangle shape created outside the design partition **userA** in the *Concurrent Layout* assistant.

**Tip:** Outside area edits need to be signed off to indicate if these edits are to be retained and were done intentionally. However, if any edits are not to be retained, click in the *Filter State* column to first mark these changes in the assistant and then use the *Remove Changes from Design Partition View* button to remove them.

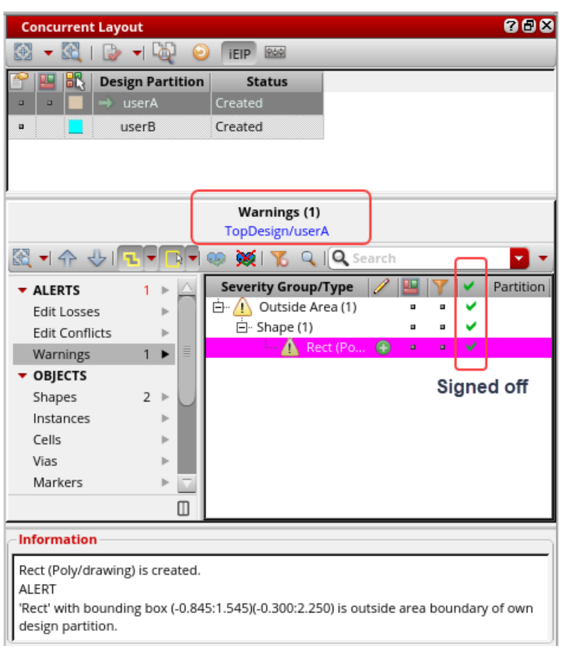
## Options in the Concurrent Layout Assistant

1. Observe the different options in the *Concurrent Layout* assistant, as shown here.




## Signing Off the Outside Area Edits

1. Click the *Sign-off State* column to remove the warning and sign off the outside area edits alert, as shown here.




The outside area edit is signed off, and the alert is removed.

2. Keep the *TestLib/TopDesign/layout_cle_userA* cellview open and the Virtuoso session running, and continue with the next lab.

## Summary

In this lab, you learned how to:

- Edit the design partition **userA** in designer mode using the Concurrent Layout Editing flow.

End of Lab

## Lab 4-3 Editing the Design Partition userA Hierarchically Using the CLE Flow

**Objective: To edit the design partition userA hierarchically in designer mode using the CLE flow.**

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In this lab, you will see how to edit the design partition **userA** hierarchically in designer mode using the CLE flow.

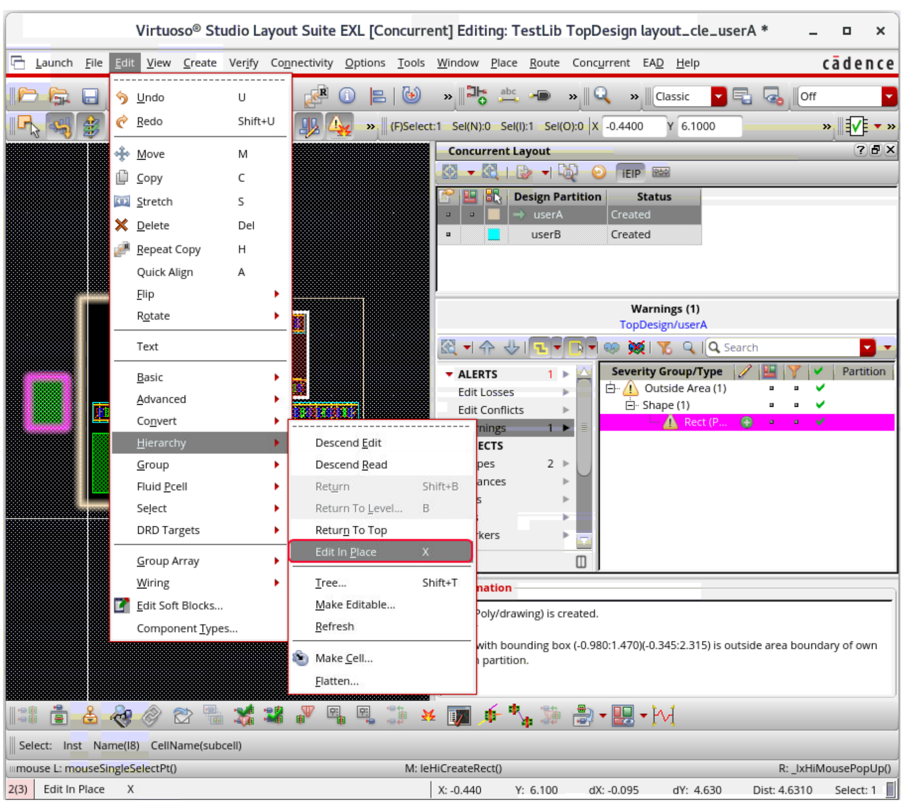
From the previous lab, the *TestLib/TopDesign/layout_cle_userA* cellview should be opened. If not, open it now.

1. In the following steps, you remain in designer mode and continue using the same first VNC session.

### Editing the Hierarchical Instance *subcell* Using the CLE Flow (EIP)

You will make hierarchical editing of the design partition **userA** using the CLE flow. You will descend into the instance *subcell* and do hierarchical editing on it.

1. In the layout canvas, select the instance *subcell* and choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to EIP.

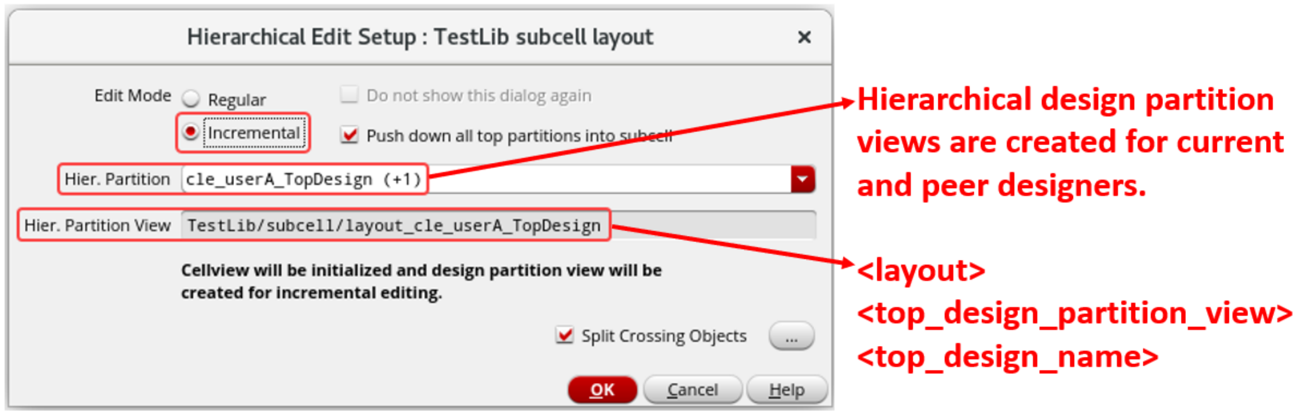



The Hierarchical Edit Setup form is displayed.

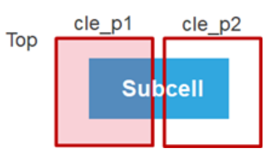
**Note:** If all the conditions for editing in *Incremental* mode are met, the Hierarchical Edit Setup form is displayed with **Edit Mode** as *Incremental*.

**Note:** In the Hierarchical Edit Setup form, CLE automatically detects the scenario and displays the option for *Incremental* editing, as shown here.




The Hierarchical Edit Setup form is displayed in designer mode when you *Descend Edit* or *Edit In Place* into a subcell, if the top-level design partitions are layer-based, or if the top-level design partitions are area-based and the following conditions are all met:

- The subcell straddles two or more design partitions.




- There is only one occurrence of this subcell in the top design (for performance reasons, checking is performed only to the display level or minimum 3).

In this case,

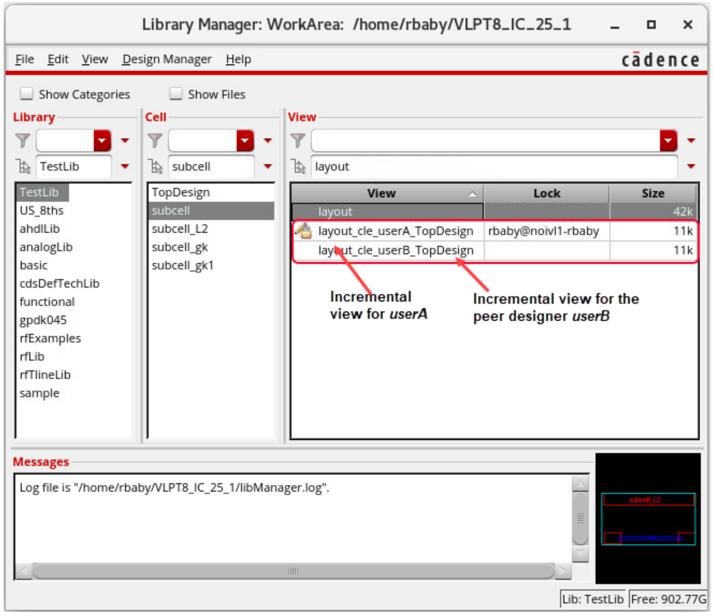
- In *Hier. Partition* name, the suffix (+1) with the name indicates the number of hierarchical design partition views that will be created. In this case, two hierarchical design partition views will be created, one for the current user *userA* and another one for the peer user *userB*.
- The *Hier. Partition View* name contains the name of the top design partition view, *userA*, and the name of the top design, *TopDesign*.

2. In the Hierarchical Edit Setup form, with the default options, click **OK** to create the hierarchical design partition views for the subcell.

   Two incremental hierarchical design partition views are added for the hierarchical instance *subcell*, as seen in the Library Manager.

   - layout_cle_userA_TopDesign is the incremental view for userA.

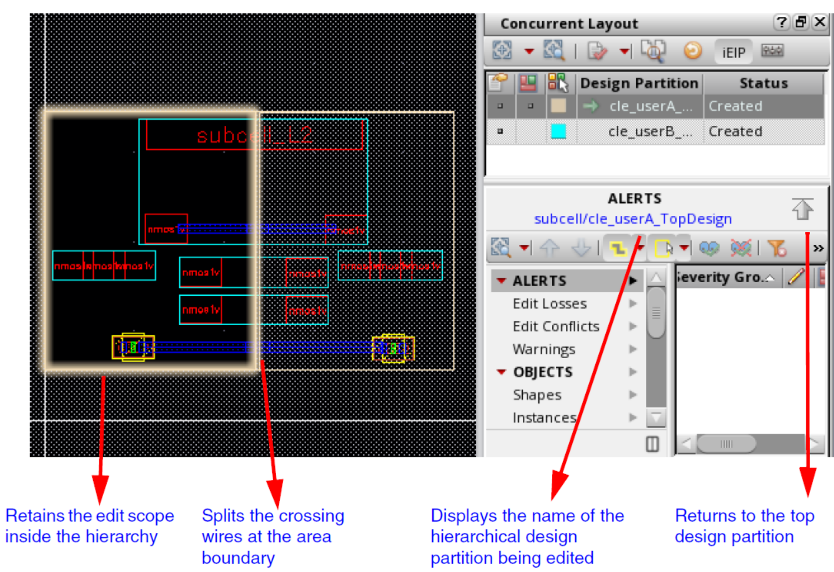
- layout_cle_userB_TopDesign is the incremental view for userB.



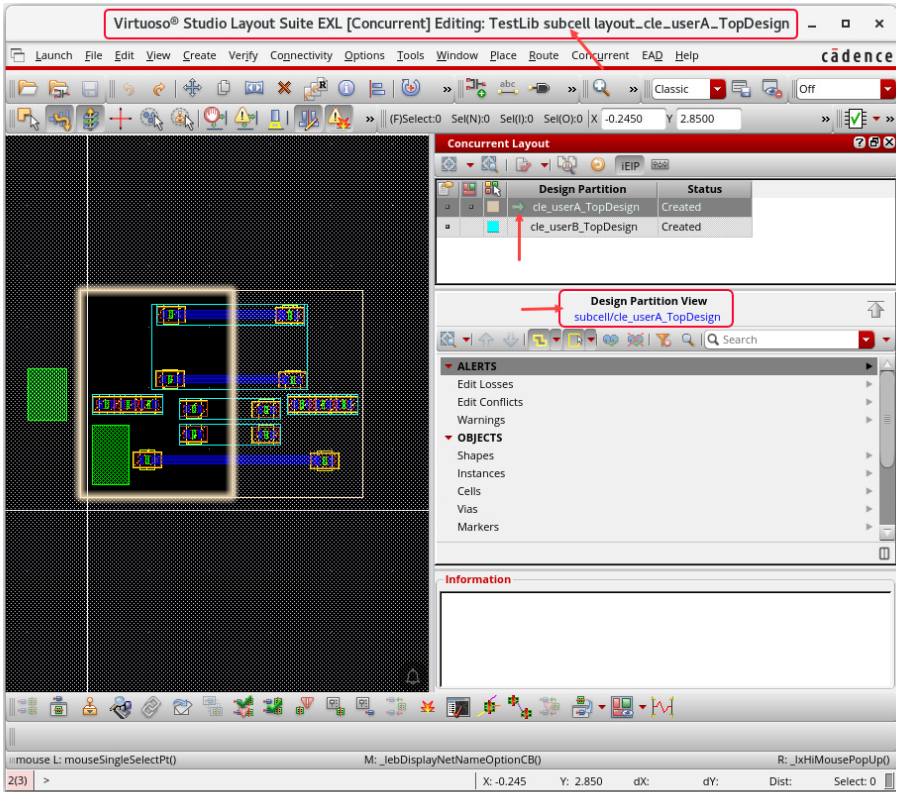

**Important:** When you EIP into the *layout_userB* hierarchical subcell from the second top design partition view, you are automatically redirected into the incremental hierarchical partition view *layout_cle_userB_TopDesign*, which was created by the first peer designer *userA*. Therefore, the Hierarchical Edit Setup form will NOT be displayed again.

**Note:** The *Hier. Partition View* (child) stores a reference back to the top-level design partition (parent) and all the incremental descend edits. The parent design partition area is now pushed down to the child, so the designer can continue to edit in the same design partition area across the hierarchy.

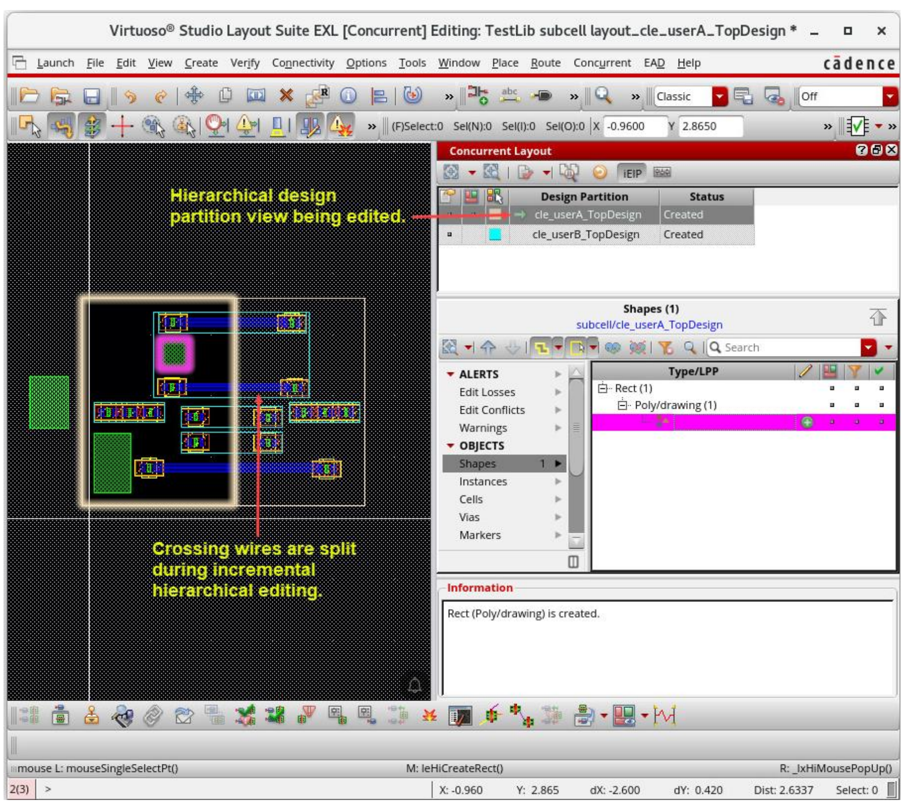
**Note:** When a hierarchical design partition is loaded into memory, it is also applied to the subcell in the virtual memory. This can affect other non-CLE cellviews if they refer to the same subcell.



You are now descended into *TestLib/subcell/layout_cle_userA_TopDesign* cellview.




3. Create a shape in the hierarchical design partition view *layout_cle_userA_TopDesign*.



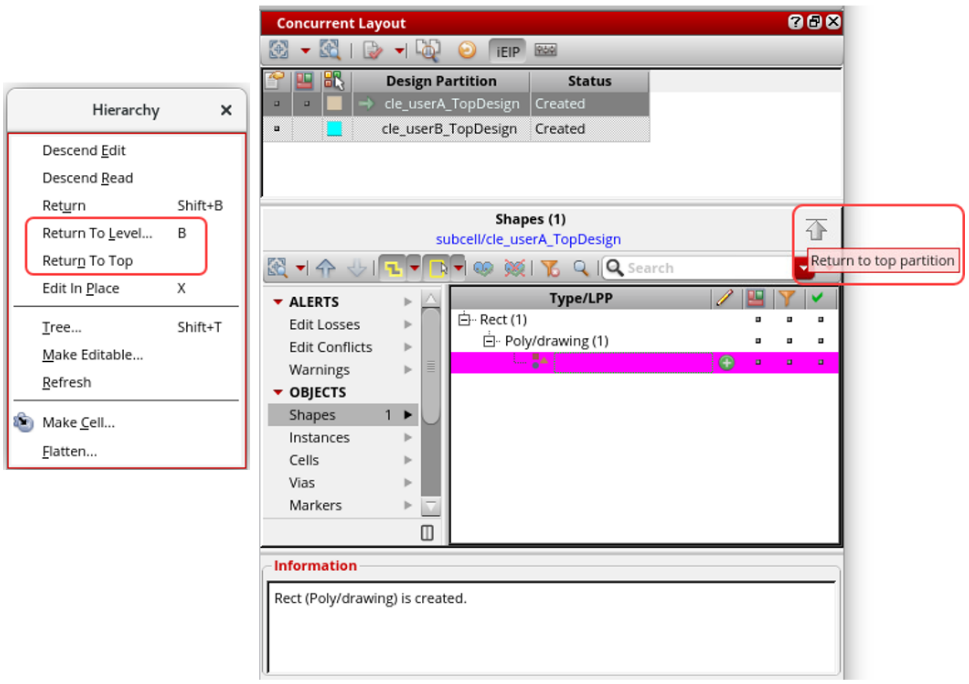

 

**Note:** Hierarchical design partition view *layout_cle_userA_TopDesign* is being edited.

**Note:** Crossing wires are split during incremental hierarchical editing.

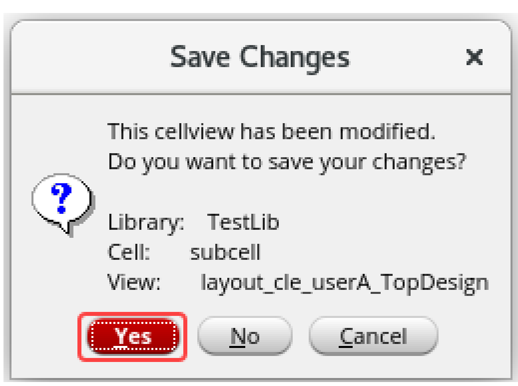
### Returning to the Top Partition Cellview *TestLib/TopDesign/layout_cle_userA*

1. In the layout canvas, choose the **Edit – Hierarchy – Return To Level (**bindkey **B**) / **Return To Top** menu command or click the **Return to top partition** button in the *Concurrent Layout* assistant.




You get the *Save Changes* form to inform you to save the changes you have made in the hierarchical design partition view *layout_cle_userA_TopDesign*.

2. In the *Save Changes* form, click the **Yes** button to save the changes in the hierarchical design partition view *layout_cle_userA_TopDesign*.




You are now returned to the top design partition view *TopDesign/userA*.

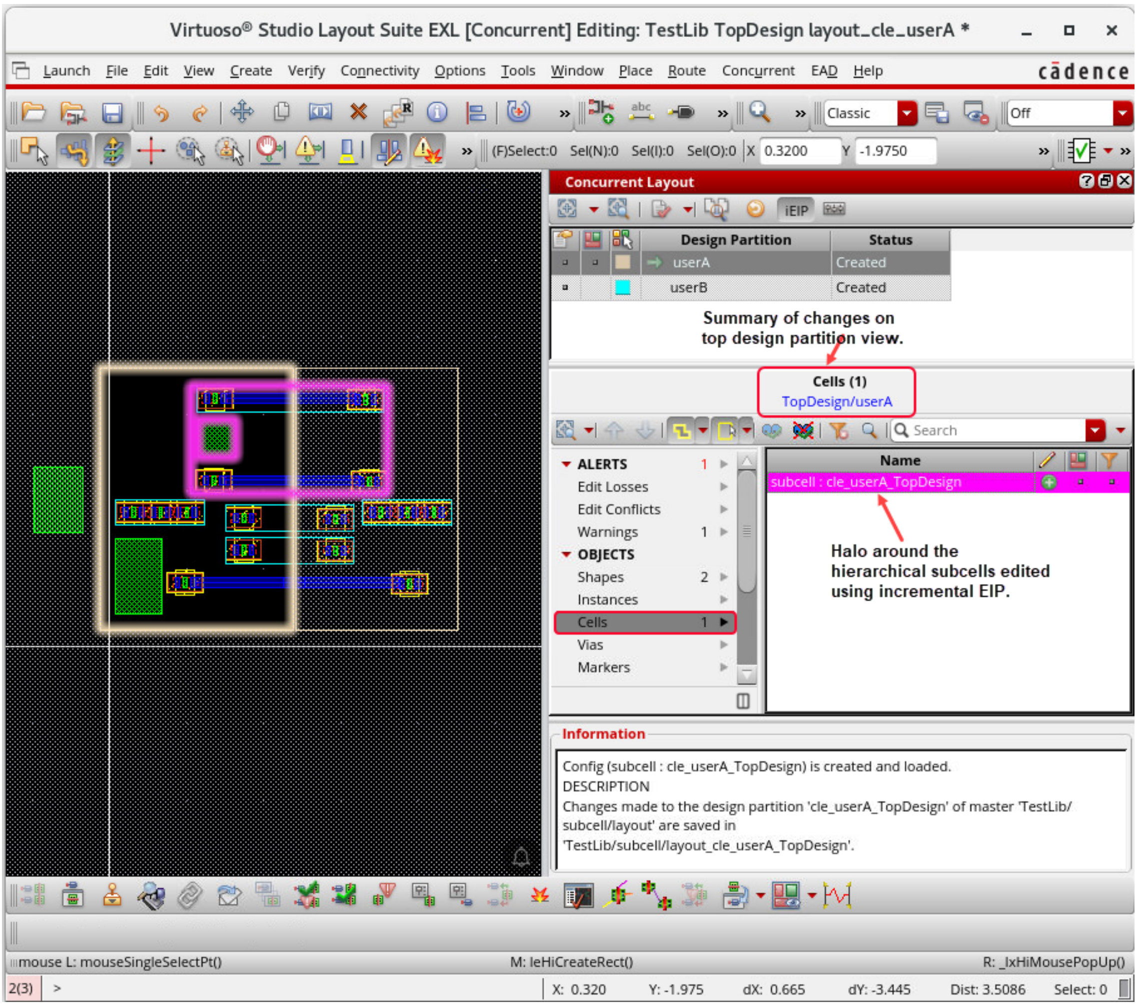
### Reviewing the Edit Changes in the Top Design Partition View *TopDesign/userA*

Now, you will review the edit changes in the top design partition view *TopDesign/userA*.

1. Review all the changes performed in the top design partition view *TopDesign/userA* and the first hierarchical design partition view *layout_cle_userA_TopDesign*.

   CLE assistant summarizes all the edits performed in the top design partition view (in this case *TopDesign/userA*) and incrementally edited hierarchical subcells (in this case *subcell/cle_userA_TopDesign*.

   **Note:** The hierarchical subcells edited in regular mode are excluded.



### Verifying the Incremental EIP Updates in *subcell/cle_userA_TopDesign*

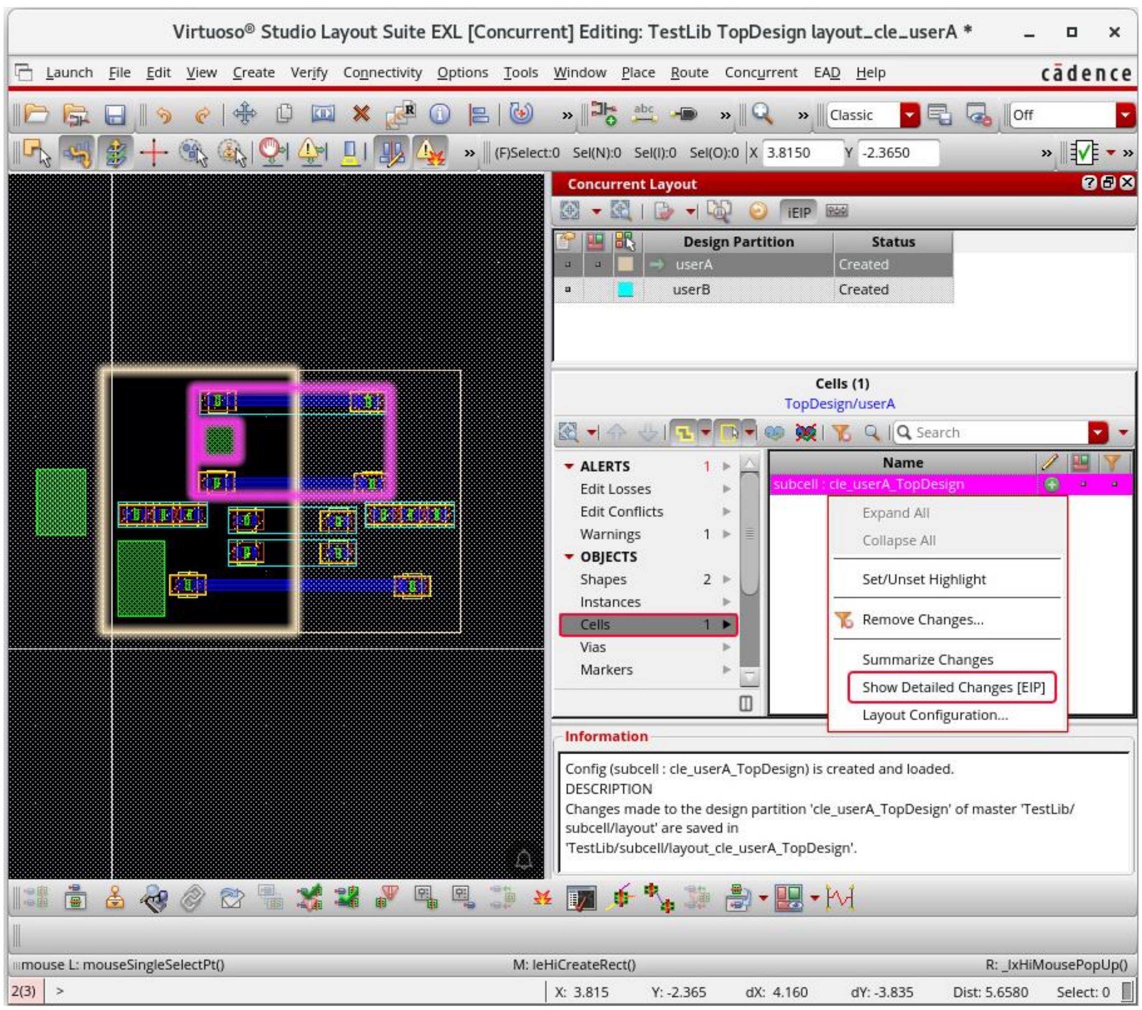
Changes made to the hierarchical design partition are saved in the hierarchical design partition view (child) pointed by a layout configuration. To verify incremental updates, you need to open or import a parent top-level design partition view with layout configuration. You can verify your updates, including all the changes in the child design partition view. These changes are applied to all occurrences of the subcell in the top-level design partition view.

Now, you will verify the updates made in the hierarchical design partition view.

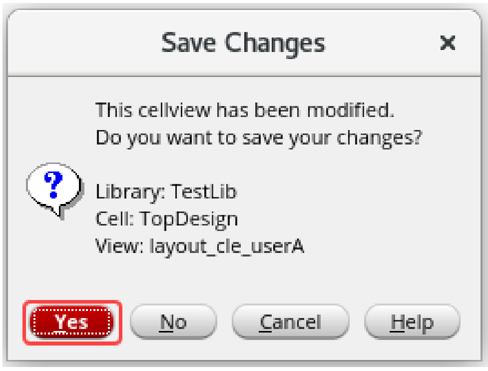
1. To view the details of the edit changes inside the incrementally edited hierarchical subcell, **right**-click the object representing the hierarchical design partition view *subcell:cle_userA_TopDesign* and choose the **Show Detailed Changes (EIP)** option.

   **Note:** This will perform EIP in read-only mode. The object is shown in the *Cells* category in the *OBJECTS* section of the *Summary* Pane.

   **Note:** Updates made in *Regular* mode are not displayed here.



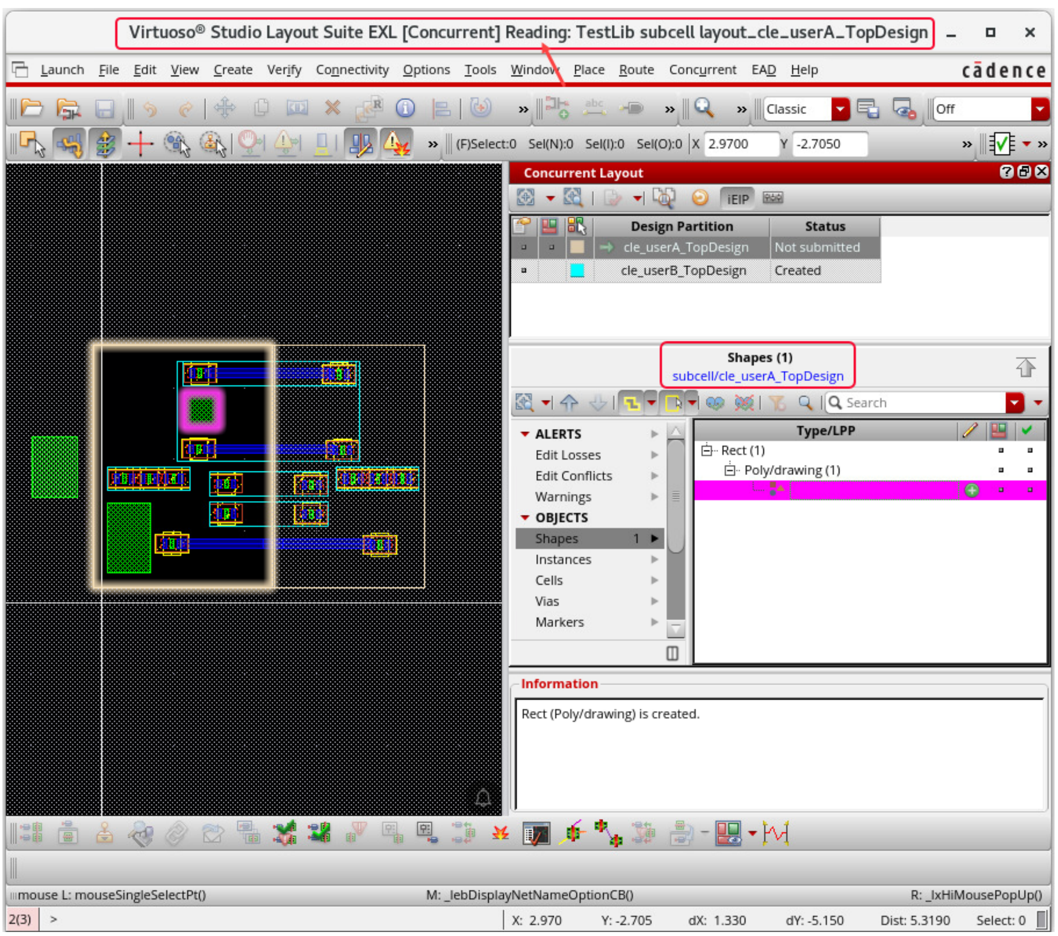
2. Save the changes in the top design partition view *TopDesign/userA* by clicking the **Yes** button.




**Note:** When you choose **Show Detailed Changes [EIP]**, EIP is done into the hierarchical design partition view *subcell/cle_userA_TopDesign* in read mode.

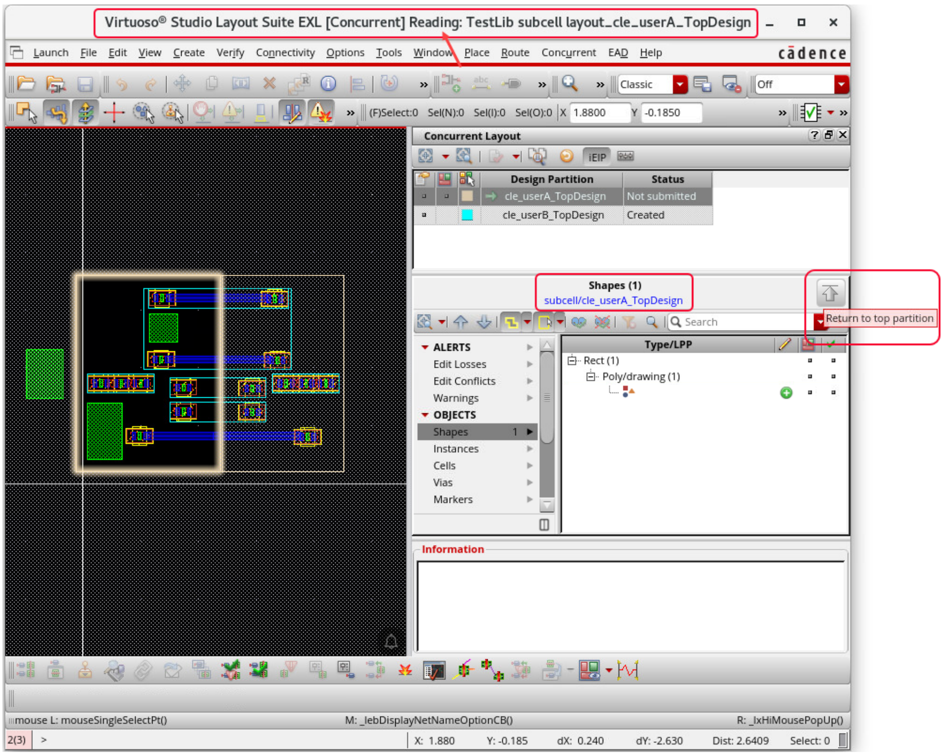
## Reviewing the Edit Changes in the Hierarchical Design Partition View *subcell/cle_userA_TopDesign*

1. On the *Concurrent Layout* assistant, expand the correct category to view the edit changes saved in the hierarchical design partition view *subcell/cle_userA_TopDesign*.

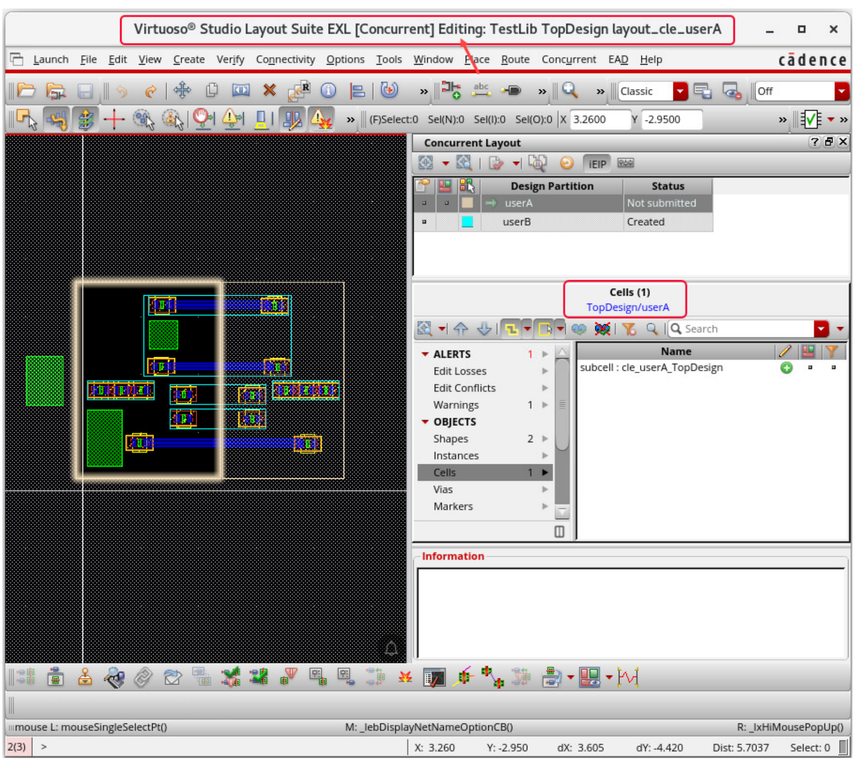


### Returning to the Top Design Partition View *TopDesign/userA*

1. After reviewing all the changes, click the **Return to top partition** button on the *Concurrent Layout* assistant to return to the top design partition view.



Now, you are returned to the top design partition view *TopDesign/userA*.



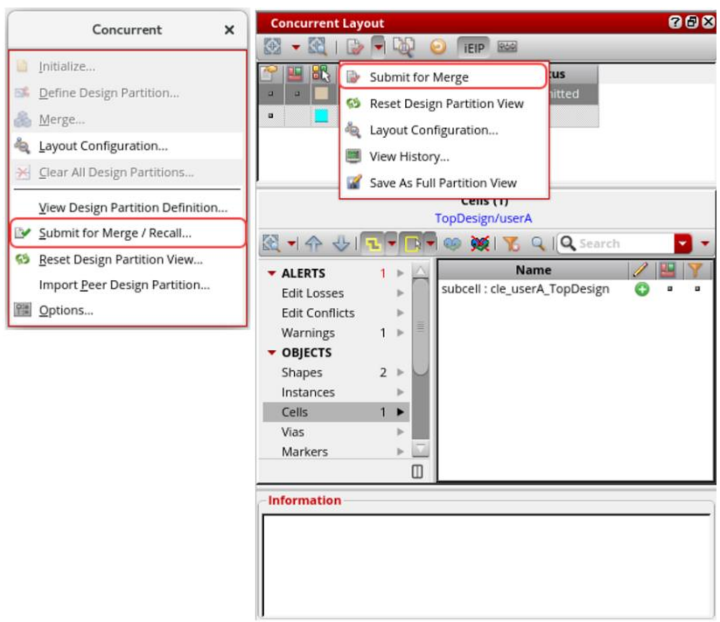
### Submitting the Incremental EIP Updates for Merge: *TopDesign/layout_cle_userA* and *subcell/layout_cle_userA_TopDesign*

Merging with the top design is easy because the parent and the child are in sync. For example, submitting the parent design partition view for merge automatically includes all child design partition views. Similarly, opening a submitted child design partition view for edit will automatically recall both the parent and child design partition views. Merging a parent will automatically load the children into the memory. To verify DRC, ensure to stream out the entire design hierarchy from VM.

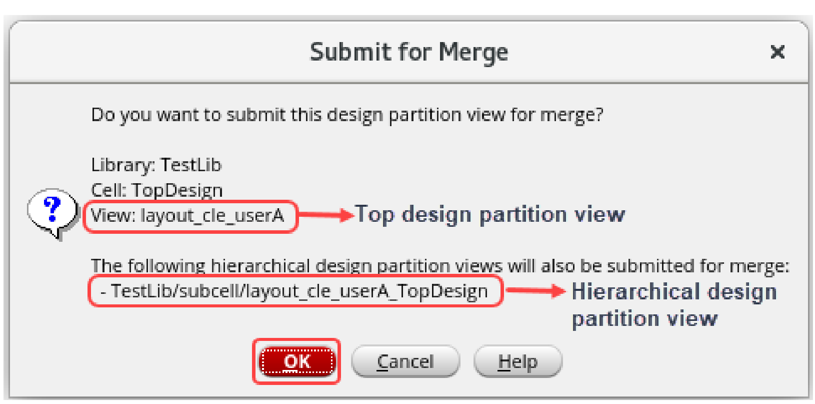
After you have completed and reviewed all the edits in the top design view, you can submit the hierarchical design partition for merge.

> **Note:** Make sure you have returned to the top-level design partition.

1. In the layout canvas, choose the **Concurrent – Submit for Merge / Recall** menu command or click the **Submit for Merge** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to submit the updates made in the subcell.




The top-level design partition *TopDesign/userA* is submitted for merge, and *Submit for Merge* dialog box is displayed, as shown here.

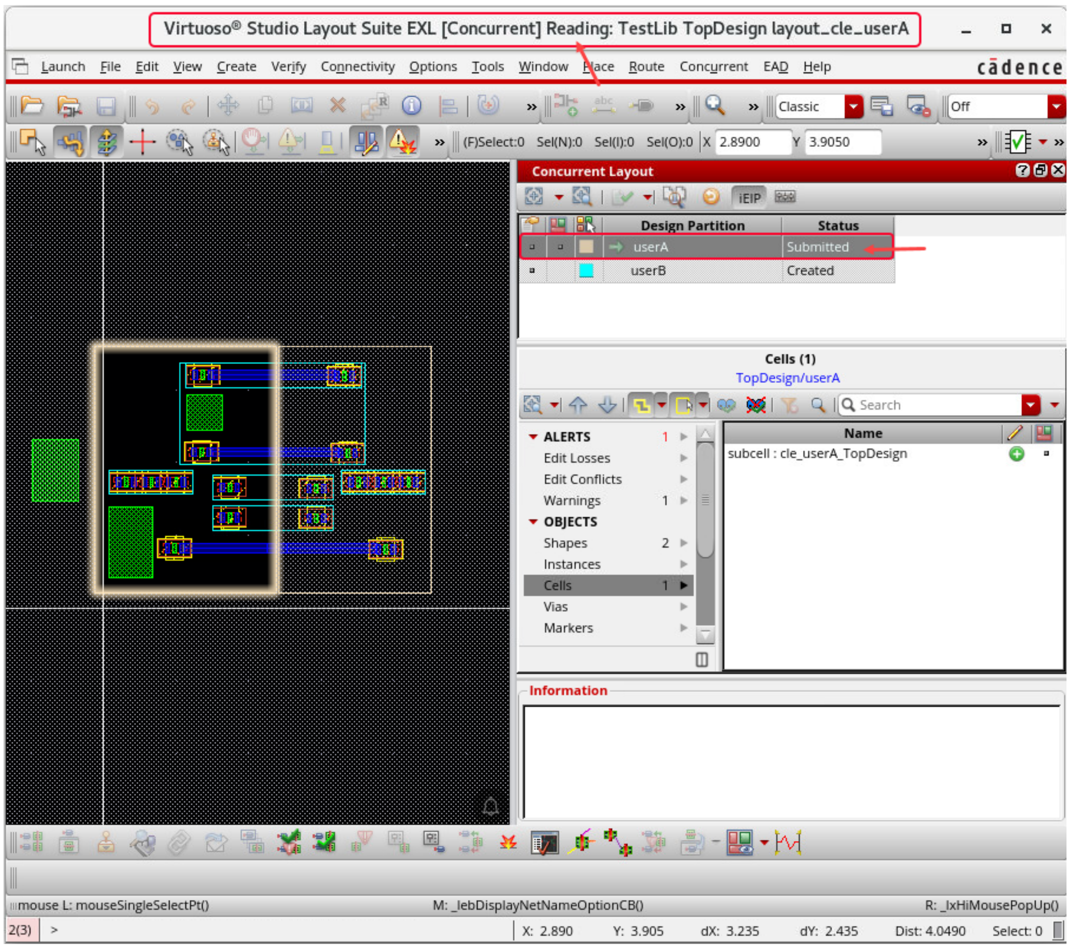


**Note:** The top design partition view *TestLib/TopDesign/layout_cle_userA* is submitted for merge.

**Note:** The hierarchical design partition view *TestLib/subcell/layout_cle_userA_TopDesign* is also submitted for merge.

2. Click **OK** in the *Submit for Merge* form.

   The status of the top design partition view *TestLib/TopDesign/layout_cle_userA* changes to *Submitted*, as shown here.




3. Observe the CIW window and CDS.log file for the top design partition view *TestLib/TopDesign/layout_cle_userA* merging details.

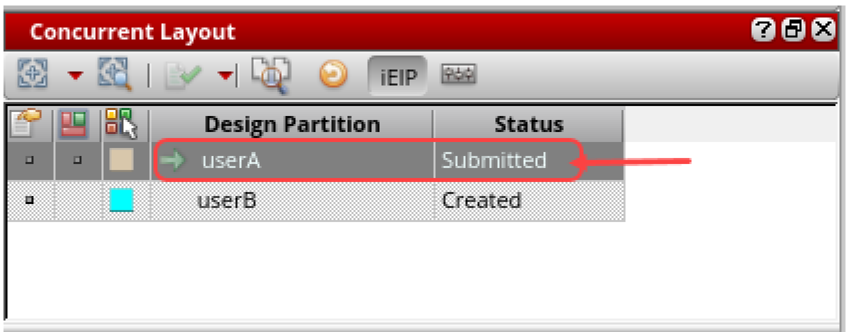

```
\o (CLE-101024): Design partition view 'TestLib/TopDesign/layout_cle_userA' submitted for merge.
```

**Note:** The top design partition view keeps track of all hierarchical design partition views that were edited from this design partition in EIP mode, and any change in its status is reflected across all hierarchical design partition views.

**Note:** For example, when the top design partition view is submitted for merge, its status changes to *Submitted*. In this case, the status of all hierarchical design partition views also changes to *Submitted*.

**Note:** Similarly, if any hierarchical design partition view or top design partition view is re-edited after submission, the top design partition is automatically recalled, and the status changes to *Not Submitted*.

The status changes to *Submitted* for the top design partition view and all the hierarchical partition views that were edited and were in the *Not Submitted* state.




**Note:** The *Submit for Merge* button on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant changes to *Recall Submission*.

4. Close the *TestLib/TopDesign/layout_cle_userA* cellview and keep the Virtuoso session running and continue with the next lab.

## Summary

In this lab, you learned how to:

- Edit the design partition **userA** hierarchically in designer mode using the CLE flow.

End of Lab

# Module 5: Editing the Design Partition userB in Designer Mode

# Lab 5-1 Setting the Edit Scope for the Design Partition userB in Designer Mode

**Objective: To set the edit scope for the design partition userB in designer mode.**

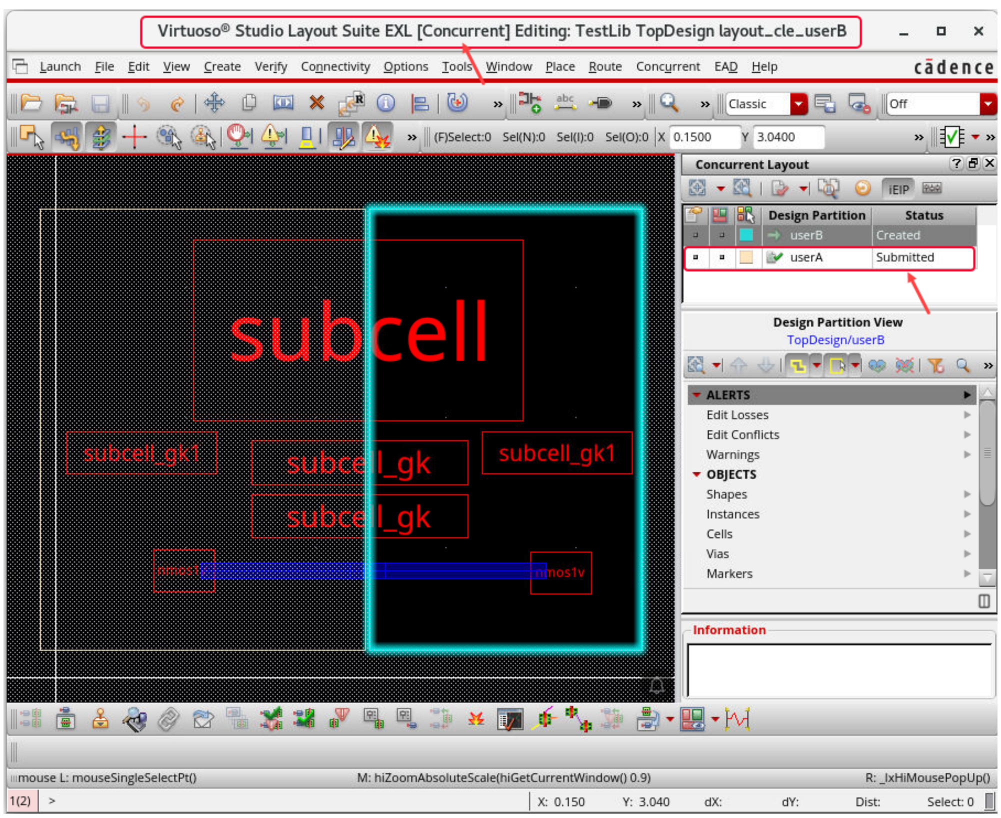
In this lab, you will see how to set the *Edit Scope* for the design partition **userB** in designer mode to use in the Concurrent Layout Editing flow.

1. In the following steps, you remain in designer mode and switch to using the second VNC session.

2. To start editing as a second designer **userB**, open the following second design partition view from the **Library Manager** or choose **File – Open** from the CIW, as shown here.

| | |
|---|---|
| **Library** | `TestLib` |
| **Cell** | `TopDesign` |
| **View** | `layout_cle_userB` |

The *TestLib/TopDesign/layout_cle_userB* cellview opens in the **EXL** mode, which is enabled for **Concurrent Editing**, as shown here.

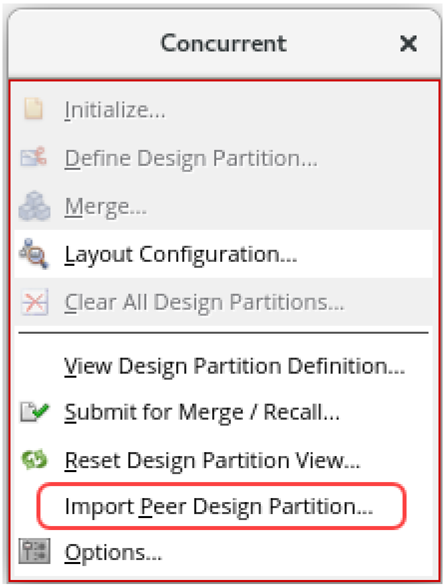



The status for **userA** is *Submitted*, as you see here.

### Importing the Peer Design Partition userA

To import the changes made in the peer design partitions and view the updated design, do the following:

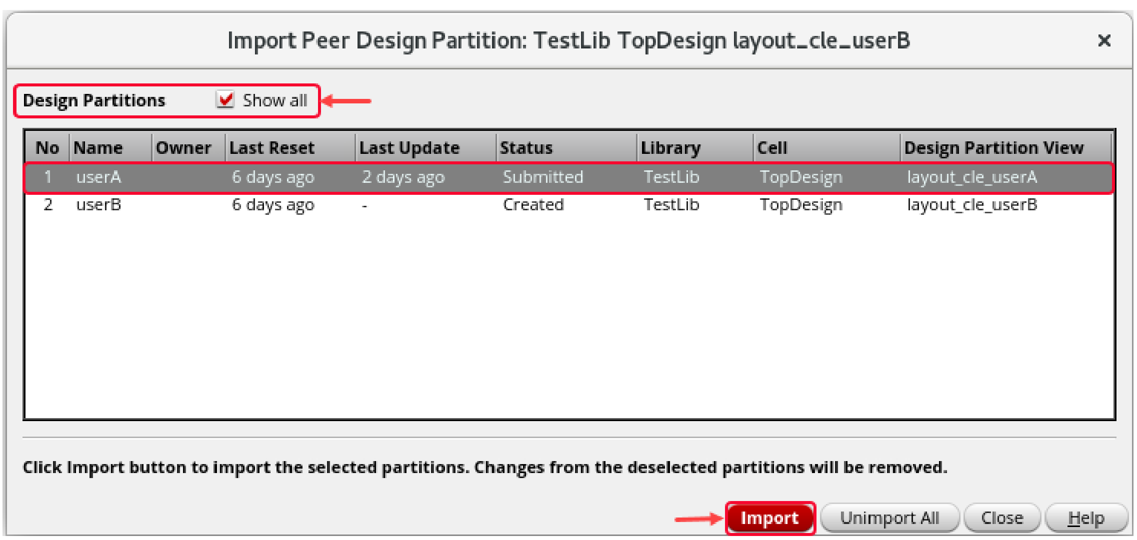
1. In the layout canvas, choose the **Concurrent – Import Peer Design Partition** menu command to display the *Import Peer Design Partition* form.




a. Select the **Show all** button to display all the edited peer design partitions.

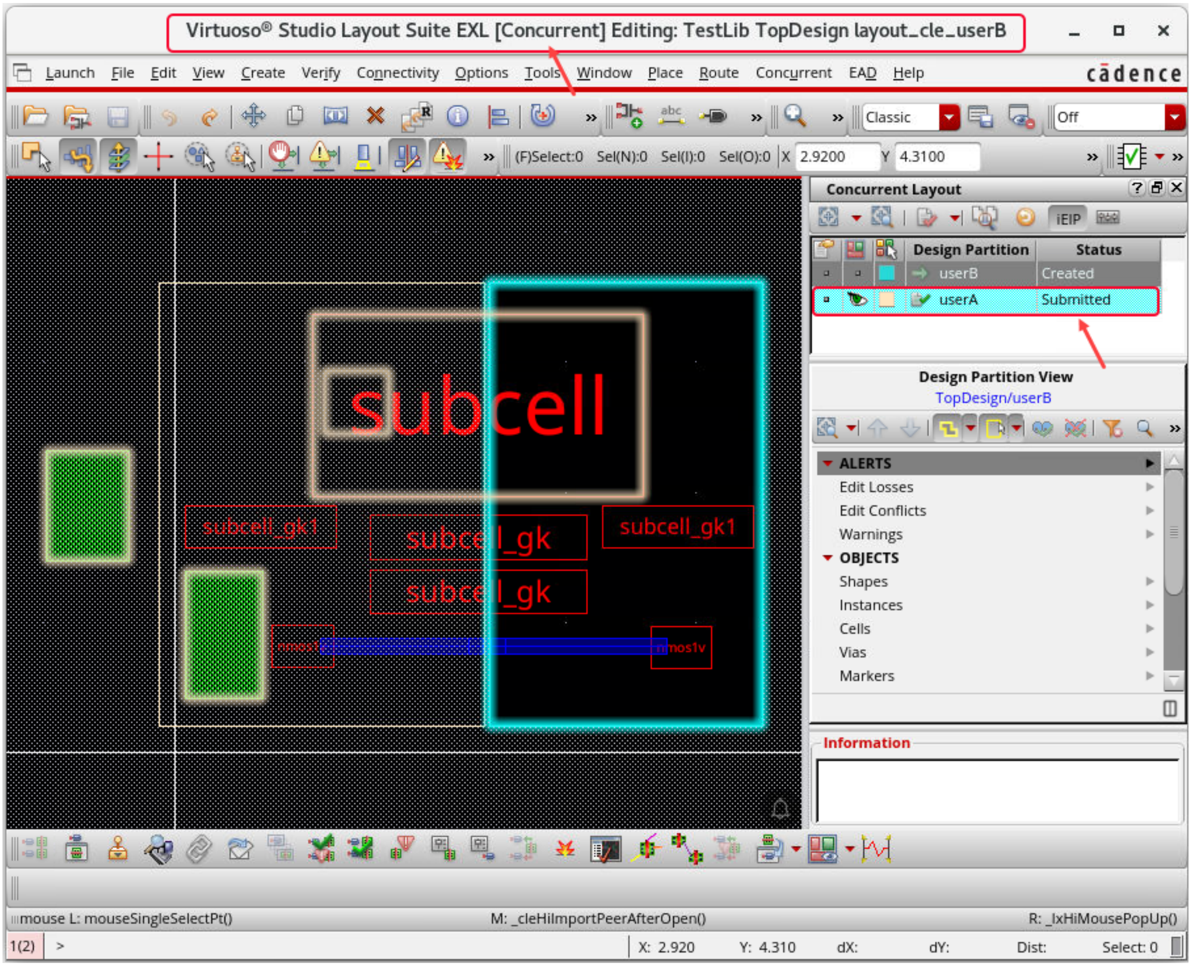
b. Select the design partition **userA** to import.

c. Click the **Import** button in the *Import Peer Design Partition* form.




The first designer **userA**'s edits are imported into the *TestLib/TopDesign/layout_cle_userB* design partition view, and it is opened in the **EXL** mode, which is enabled for **Concurrent Editing**, as shown here.

**Note:** When you open the second design partition, **userB**, CLE detects that **userA** has already submitted some changes. The *Import Peer Design Partition* form lets you import these changes from the peer design partition **userA** into the current design partition being opened for editing by **userB**.




The status for **userA** is *Submitted*, as you see here.

You will set the *Edit Scope* and start editing in the *TestLib/TopDesign/layout_cle_userB* design partition view.

## Setting the Edit Scope for the Design Partition userB

Now, you will set up the *Edit Scope* for the Design Partition **userB** using the options you had seen in the previous lab. The Edit Scope for each design partition is set to select only those objects that are inside the current partition by default.

You can change these settings using one of the following two methods in designer mode:

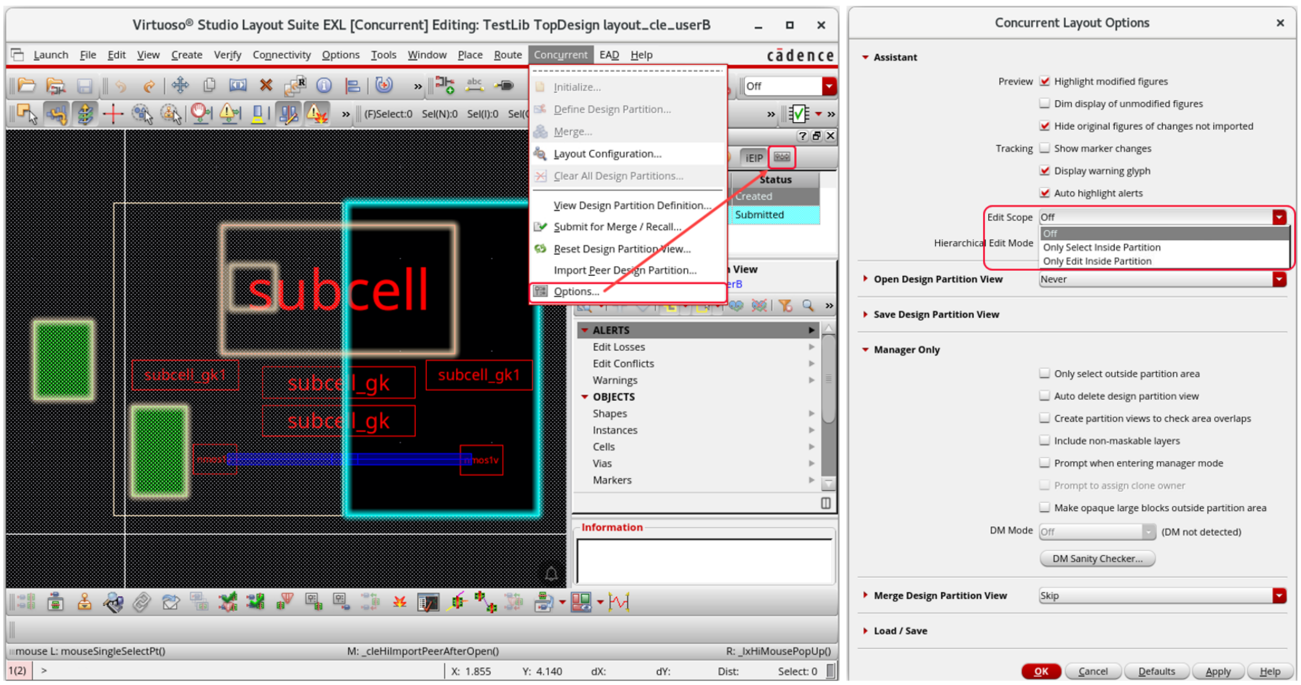
- Using the *Concurrent Layout Options* form.
- Using the *Concurrent Layout* assistant.

If the *Edit Scope* is set to *Off*, editing objects outside the partition generates alerts in the *Concurrent Layout* assistant. You will have to individually sign off each alert and warning.

### Method 1: Setting the Edit Scope for userB Using the Concurrent Layout Options Form

You will set the *Edit Scope* for **userB** using the *Concurrent Layout Options* form.

1. To set the Edit Scope for **userB**, do the following steps:




   a. From the layout window, choose the **Concurrent – Options** menu command or click the **Options** icon in the *Concurrent Layout* assistant to display the *Concurrent Layout Options* form.

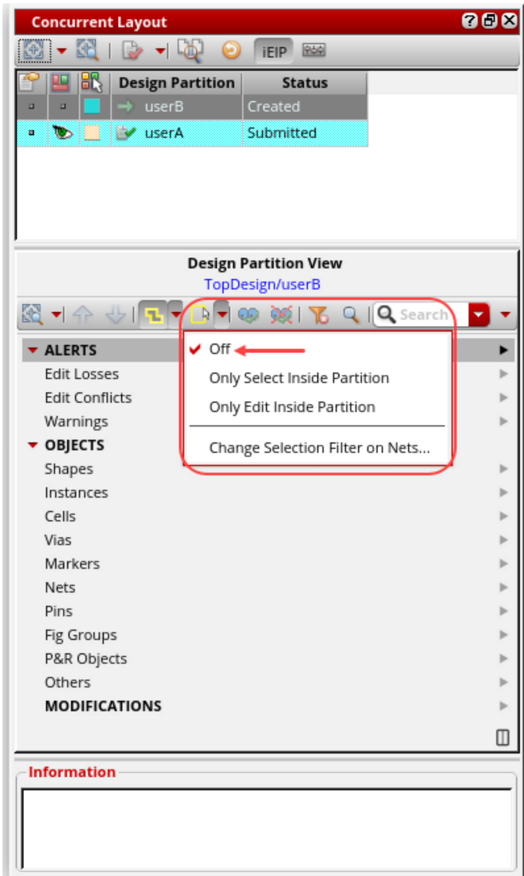
   b. In the *Concurrent Layout Options* form, in the *Edit Scope* cyclic field, choose the required option; in this case, say, select the option, *Off*.

   c. Click **OK** in the form.

### Method 2: Setting the Edit Scope for userB Using the Concurrent Layout Assistant

You will set the *Edit Scope* for **userB** using the *Concurrent Layout* assistant.

1. To set the Edit Scope for **userB**, perform the following steps:




   a. In the *Concurrent Layout* assistant, from the *Edit Scope* cyclic field, choose the required option; in this case, say, select the option, *Off*, as shown here.

2. Keep the *TestLib/TopDesign/layout_cle_userB* cellview open and the Virtuoso session running and continue with the next lab.

### Summary

In this lab, you learned how to:

- Set the *Edit Scope* for the design partition **userB** in designer mode to use in the Concurrent Layout Editing flow.

End of Lab

## Lab 5-2 Editing the Design Partition userB Using the CLE Flow

**Objective: To edit the design partition userB in designer mode using the CLE flow.**

In this lab, you will see how to edit the design partition **userB** in designer mode using the Concurrent Layout Editing flow.

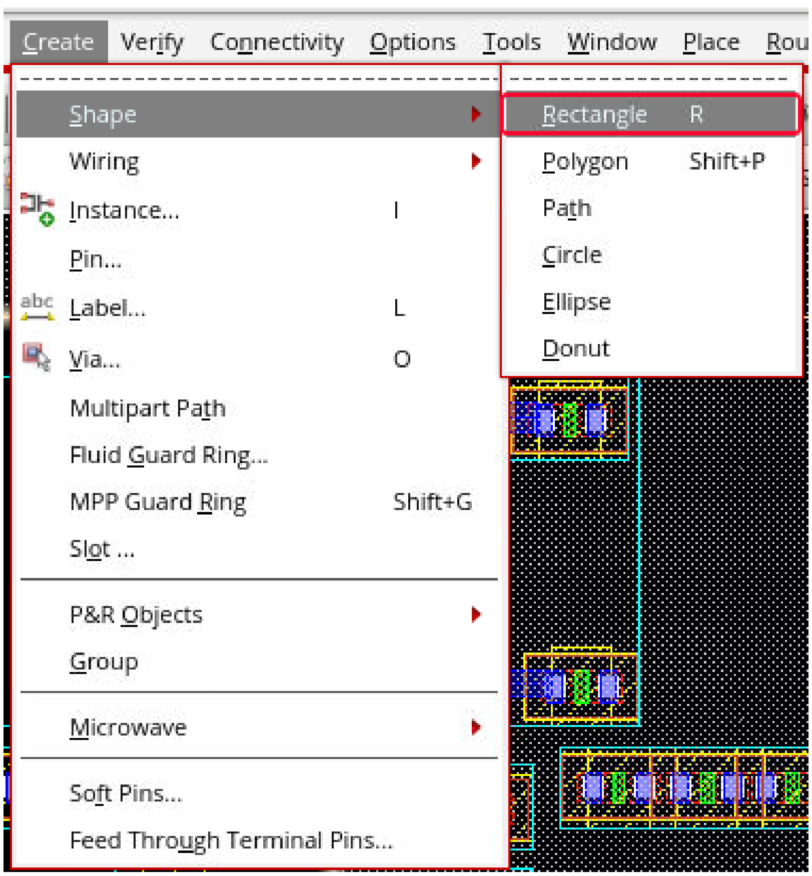
From the previous lab, the *TestLib/TopDesign/layout_cle_userB* cellview should be opened. If not, open it now.

1. In the following steps, you remain in designer mode and continue using the same second VNC session.

### Editing Inside the Design Partition userB Using the CLE Flow

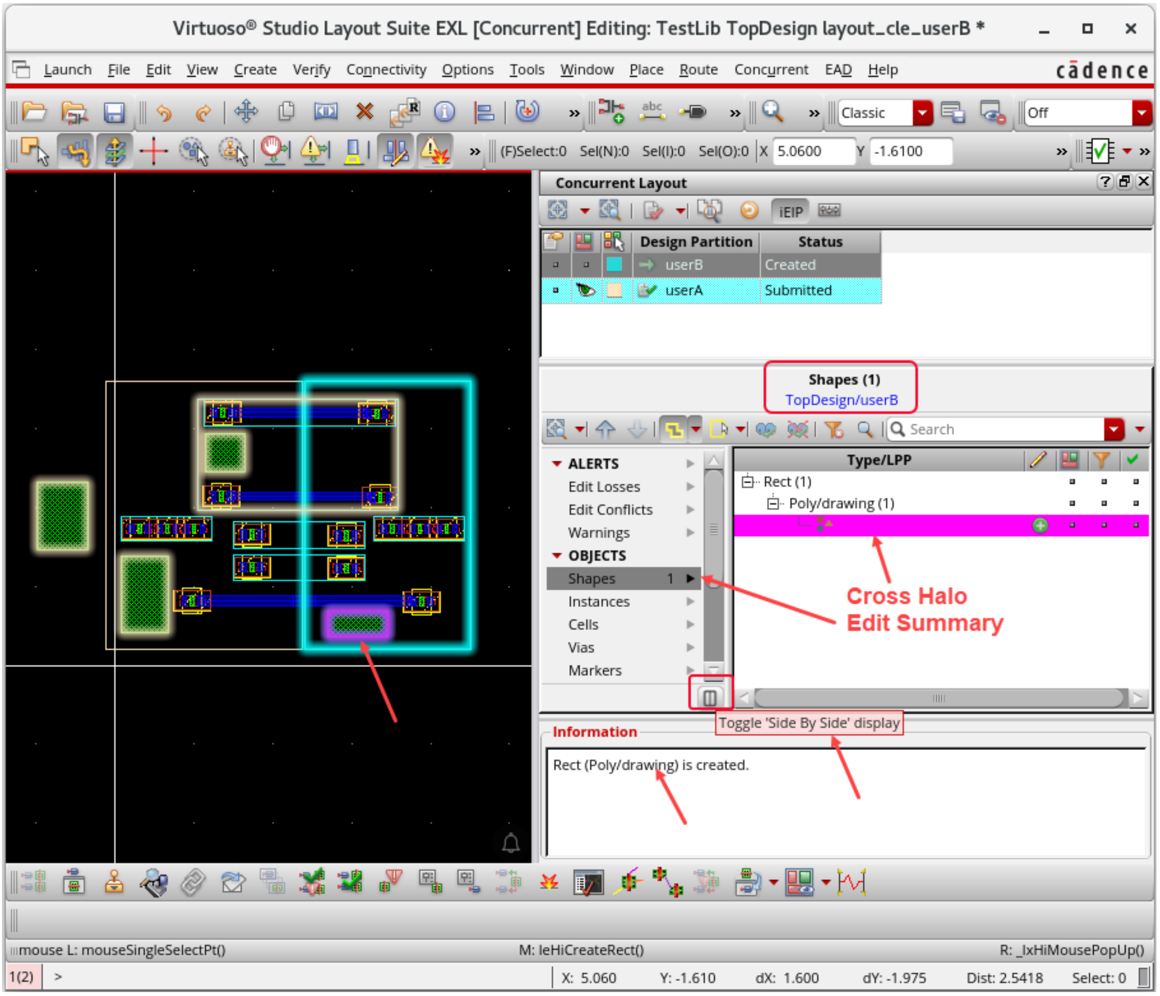
You will make some edits inside the design partition **userB**.

1. In the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.



2. Create a rectangle inside the design partition, as shown here.




Observe the *Edit Summary* and the *Cross Halo* results for the rectangle shape created inside the design partition **userB** in the *Concurrent Layout* assistant.

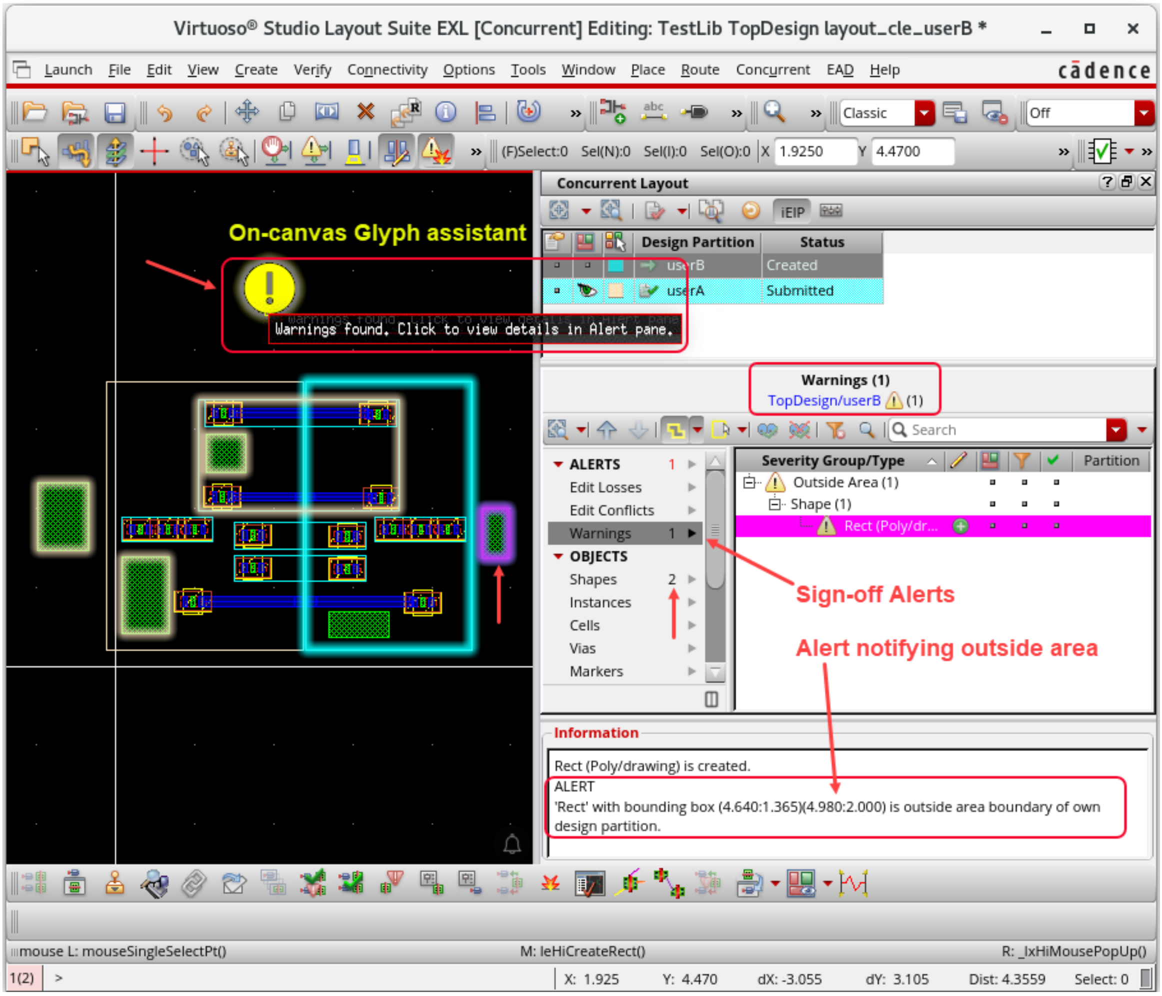
## Editing Outside the Design Partition userB Using the CLE Flow

You will make some edits outside the design partition **userB**.

1. In the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.

2. Create a rectangle outside the design partition, as shown here.

3. Click the *On-canvas Glyph assistant* to view the details in the *Alert* pane.



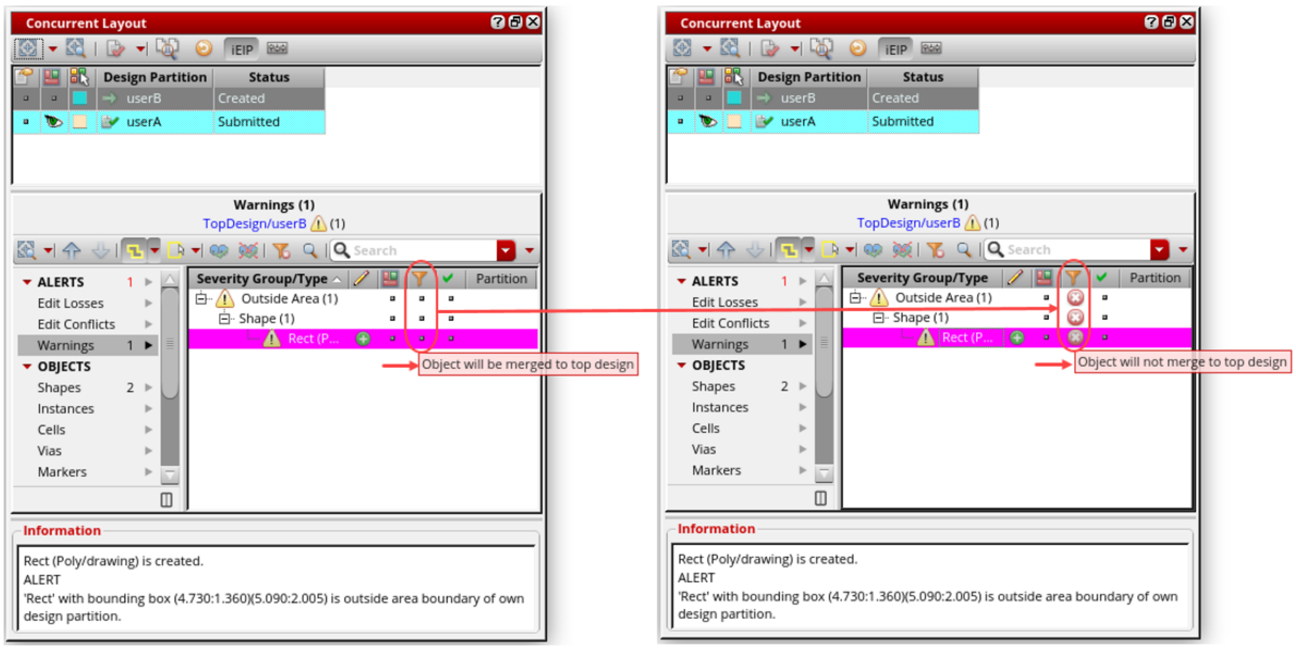

Observe the *Sign-off Alerts* and the *Alert notifying outside area* result for the rectangle shape created outside the design partition **userB** in the *Concurrent Layout* assistant.

**Tip:** Outside area edits need to be signed off to indicate if these edits are to be retained and were done intentionally. However, if any edits are not to be retained, click in the *Filter State* column to first mark these changes in the assistant and then use the *Remove Changes from Design Partition View* button to remove them.

### Removing the Outside Area Edits

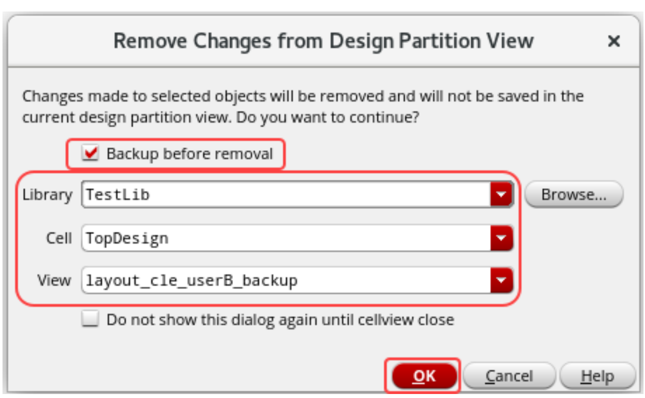
You will discard the last edit to remove the rectangle added outside the area boundary of your own design partition.

1. Click the *Filter State* column to mark from *Object will be merged to top design* to *Object will not merge to top design*, as shown here.




2. Click the *Remove Changes from Design Partition View* button to remove the rectangle added outside the area.

   The Remove Changes from Design Partition View form is displayed, as shown here.

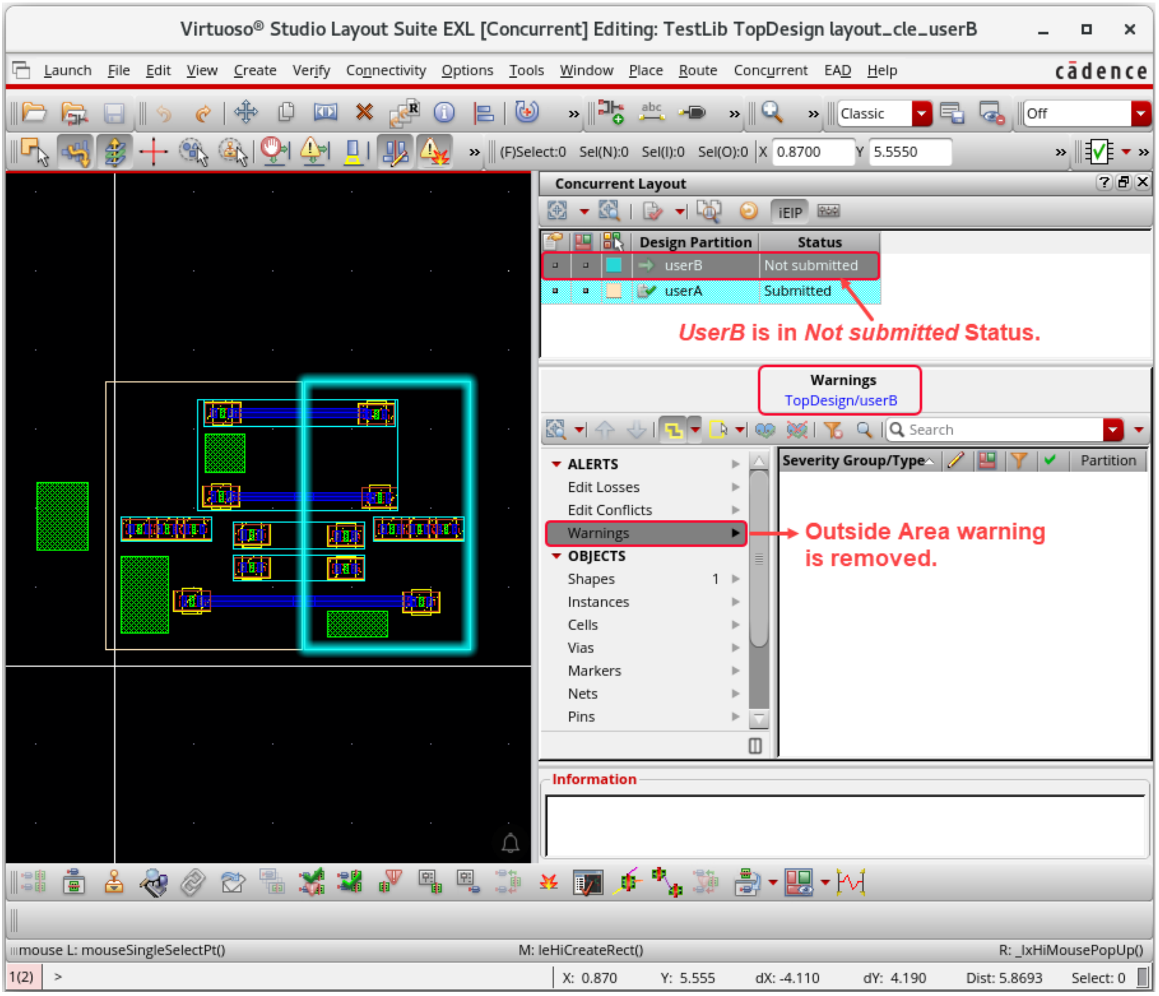



   a. Select the *Backup before removal* button to create a backup view.

   b. Keep the options as shown here.

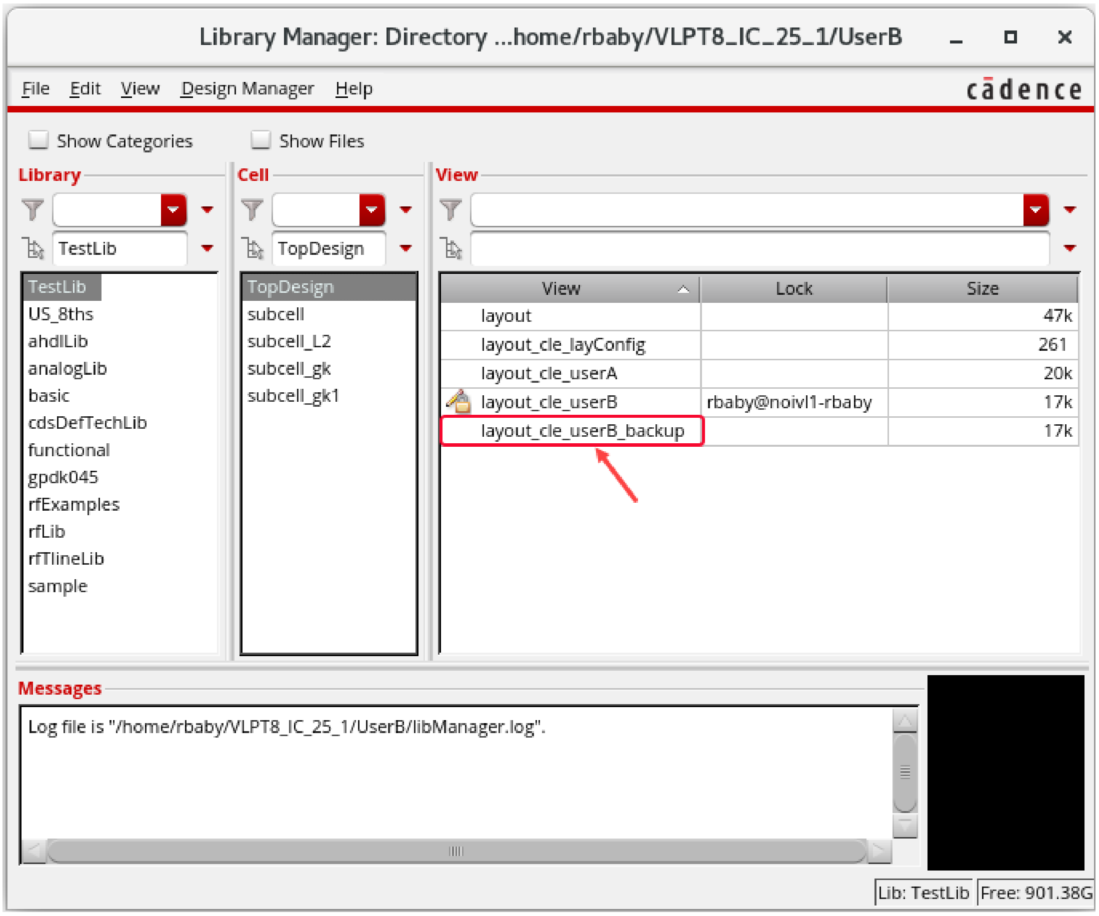
   c. Click **OK**.

The rectangle added outside the area boundary is removed in the layout canvas, and the *Alert notifying outside area* result for the rectangle shape created outside the design partition **userB** in the *Concurrent Layout* assistant is removed.



### Reviewing the Backup View *TestLib/TopDesign/layout_cle_userB_backup* in LM

1. Open the Library Manager and review the backup view *TestLib/TopDesign/layout_cle_userB_backup* created in the top design.



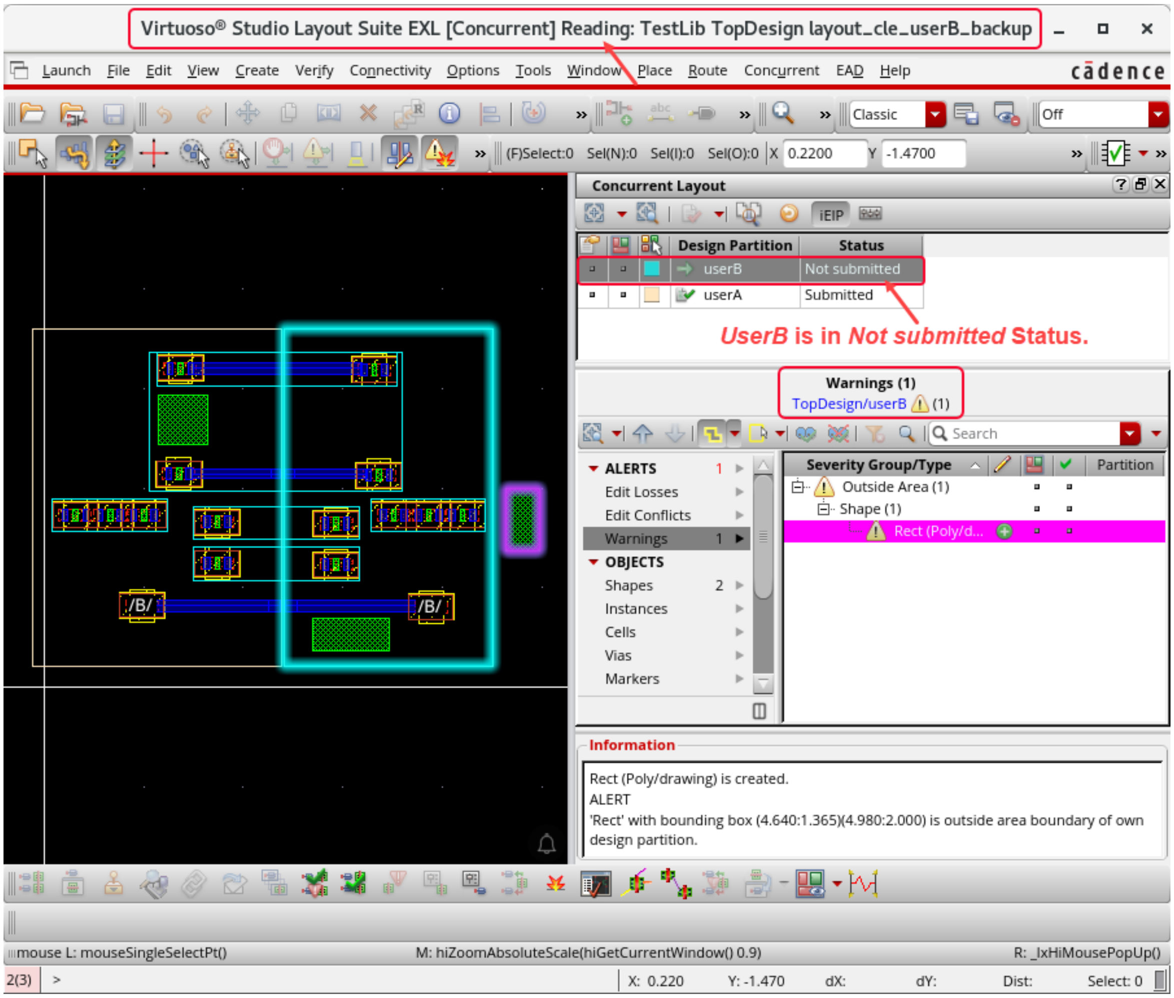

### Reviewing the Backup View *TestLib/TopDesign/layout_cle_userB_backup*

You will open the backup view *TestLib/TopDesign/layout_cle_userB_backup* and review it.

1. Open the backup view *TestLib/TopDesign/layout_cle_userB_backup* in **EXL** mode from the Library Manager.

   You get the **Info** window, as shown here.

2. Click the **Close** button to close the **Info** window and open the cellview *TestLib/TopDesign/layout_cle_userB_backup* in read-mode.




3. Close the TestLib/TopDesign/layout_cle_userB_backup cellview after reviewing it.

4. Keep the *TestLib/TopDesign/layout_cle_userB* cellview open and the Virtuoso session running, and continue with the next lab.

## Summary

In this lab, you learned how to:

- Edit the design partition **userB** in designer mode using the Concurrent Layout Editing flow.

End of Lab

# Lab 5-3 Editing the Design Partition userB Hierarchically Using the CLE Flow

**Objective: To edit the design partition userB hierarchically in designer mode using the CLE flow.**

In this lab, you will see how to edit the design partition **userB** hierarchically in designer mode using the CLE flow.

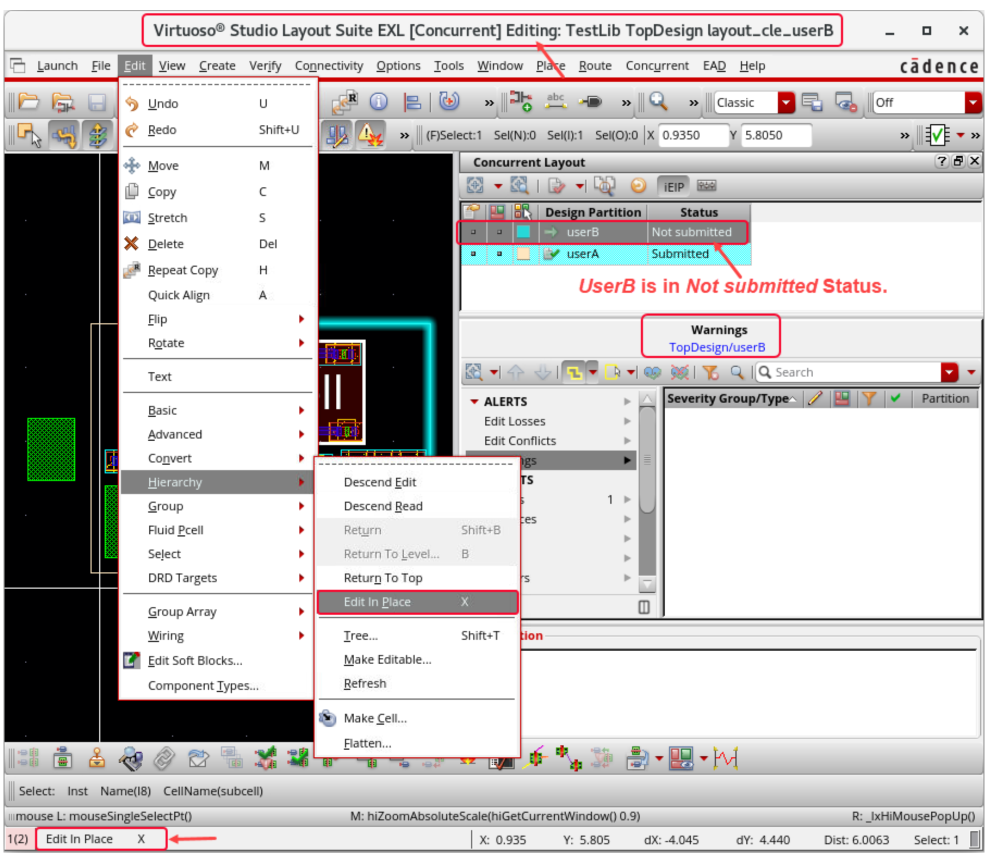
From the previous lab, the *TestLib/TopDesign/layout_cle_userB* cellview should be opened. If not, open it now.

1. In the following steps, you remain in designer mode and continue using the same second VNC session.

## Editing the Hierarchical Instance *subcell* Using the CLE Flow (EIP)

You will make hierarchical editing of the design partition **userB** using the CLE flow. You will descend into the instance *subcell* and do hierarchical editing on it.

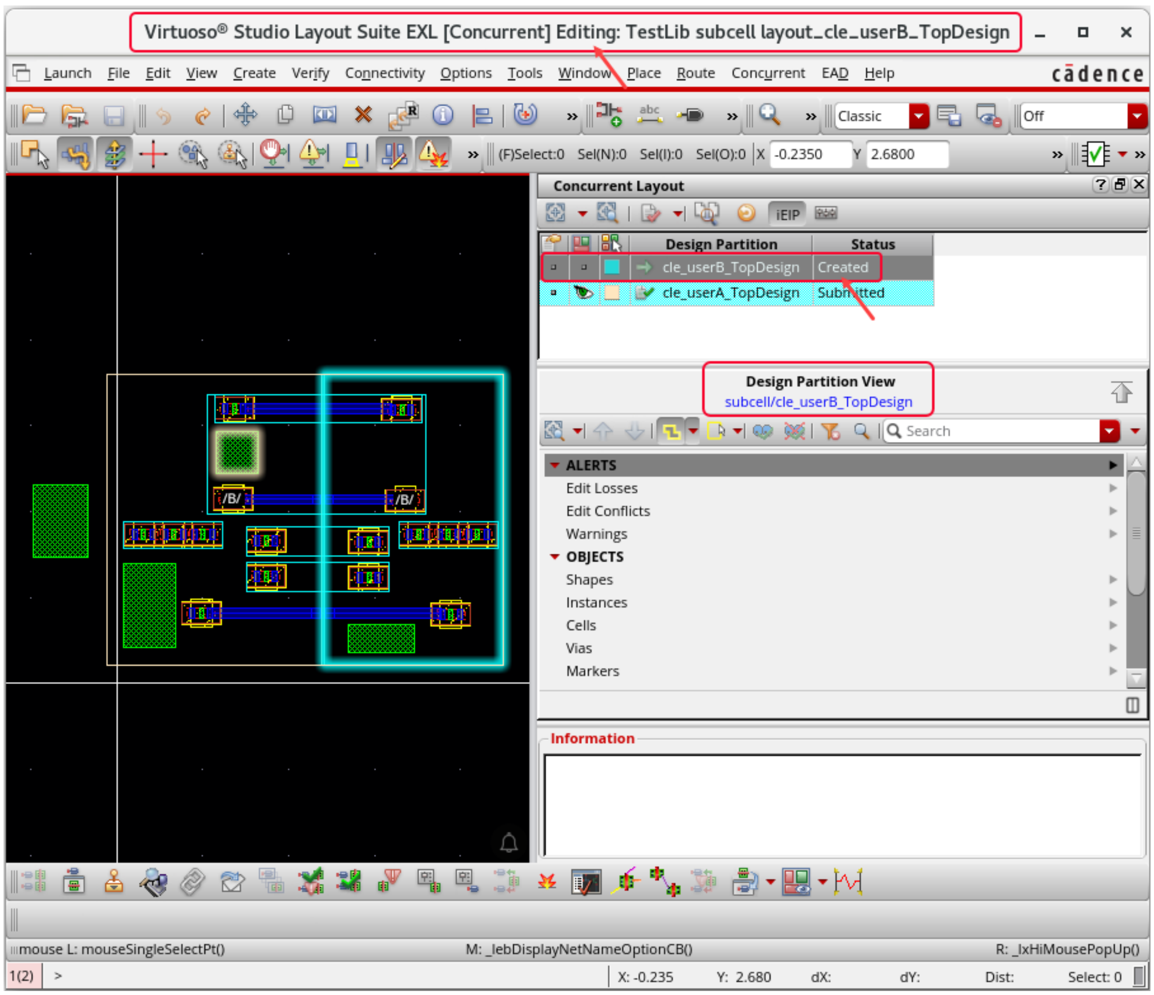
1. In the layout canvas, select the instance *subcell* and choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to EIP.



**Important:** When you EIP into the *layout_userB* hierarchical subcell from the second top design partition view, you are automatically redirected into the incremental hierarchical partition view *layout_cle_userB_TopDesign*, which was created by the first peer designer *userA*. Therefore, the Hierarchical Edit Setup form is NOT displayed again.

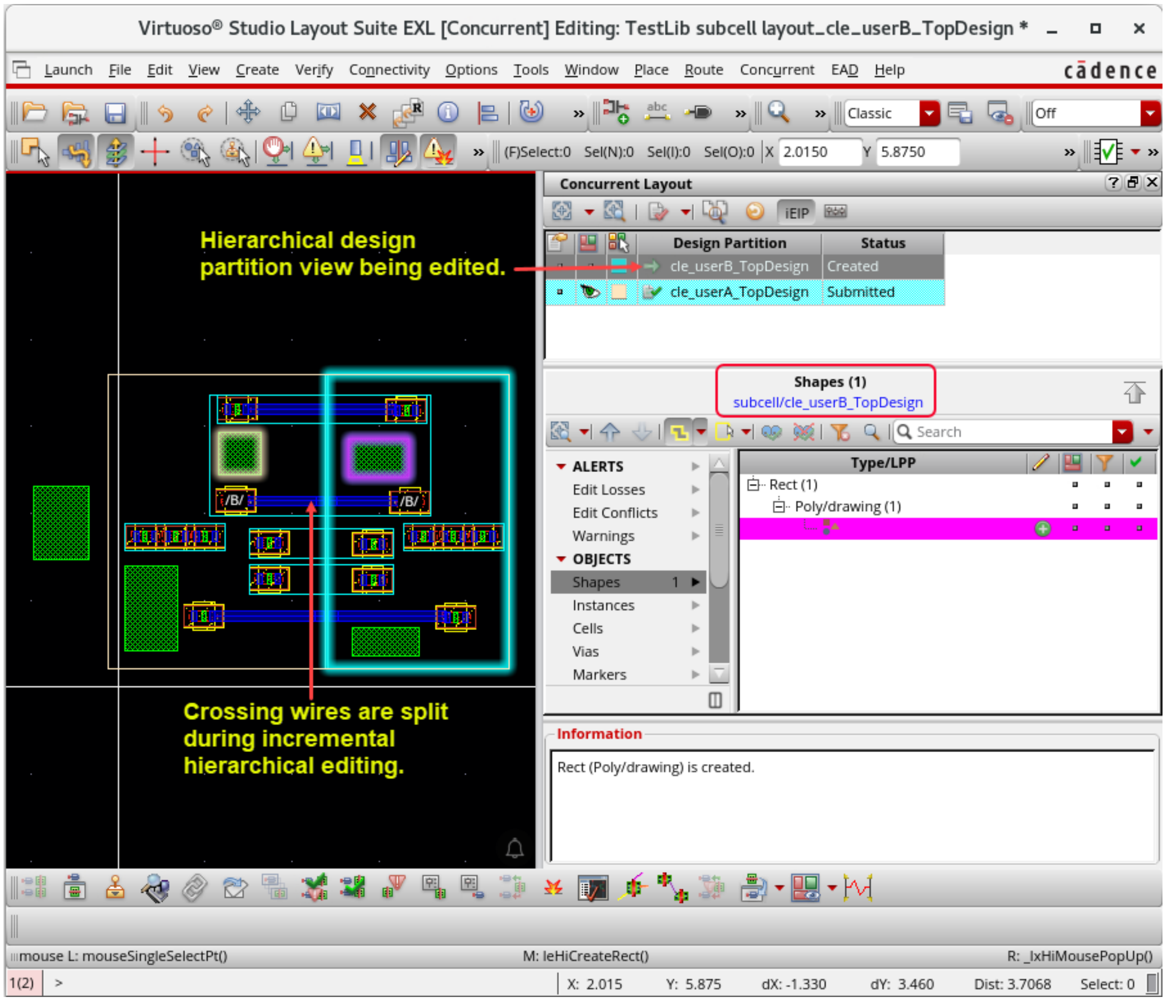
**Important:** The user who first EIPs into the hierarchical subcell in incremental mode creates hierarchical design partition views for themselves as well as for the peer designers. Therefore, *userB* directly opens the hierarchical design partition view that was created by *userA* in incremental mode.

You are now descended into *TestLib/subcell/layout_cle_userB_TopDesign* cellview



2. Create a shape in the hierarchical design partition view *layout_cle_userB_TopDesign*.



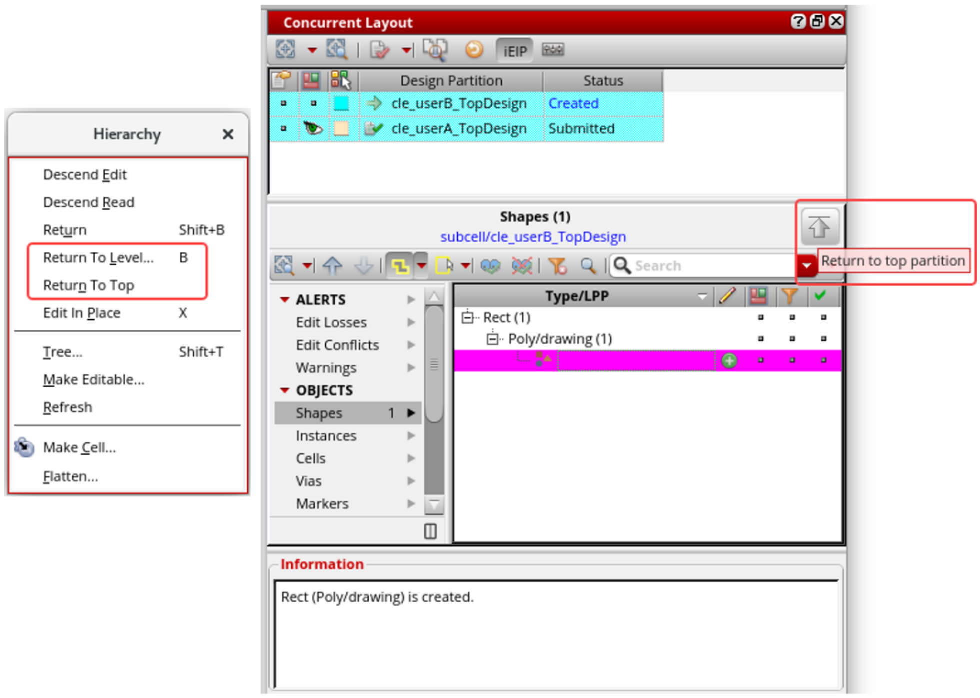

**Note:** Hierarchical design partition view *layout_cle_userB_TopDesign* is being edited.

**Note:** Crossing wires are split during incremental hierarchical editing.

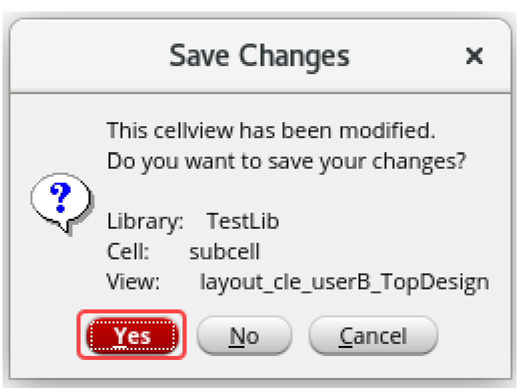
### Returning to the Top Partition Cellview *TestLib/TopDesign/layout_cle_userB*

1. In the layout canvas, choose the **Edit – Hierarchy – Return To Level (**bindkey **B**) / **Return To Top** menu command or click the **Return to top partition** button in the *Concurrent Layout* assistant.




You get the *Save Changes* form to inform you to save the changes you have made in the hierarchical design partition view *layout_cle_userB_TopDesign*.

2. In the *Save Changes* form, click the **Yes** button to save the changes in the hierarchical design partition view *layout_cle_userB_TopDesign*.




You are now returned to the top design partition view *TopDesign/userB*.

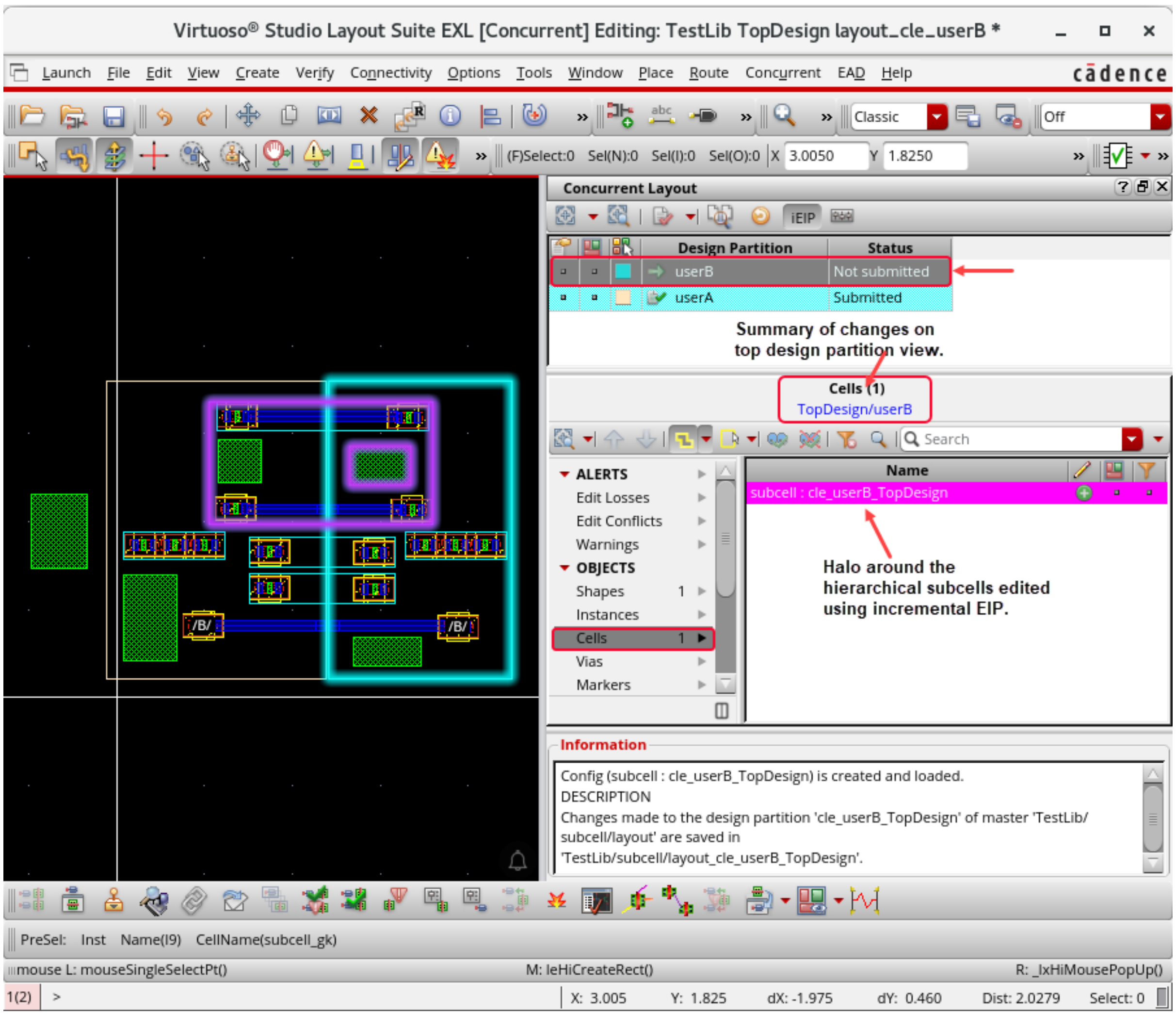
### Reviewing the Edit Changes in the Top Design Partition View *TopDesign/userB*

Now, you will review the edit changes in the top design partition view *TopDesign/userB*.

1. Review all the changes performed in the top design partition view *TopDesign/userB* and the second hierarchical design partition view *layout_cle_userB_TopDesign*.

   CLE assistant summarizes all the edits performed in the top design partition view (in this case *TopDesign/userB*) and incrementally edited hierarchical subcells (in this case *subcell/cle_userB_TopDesign*.

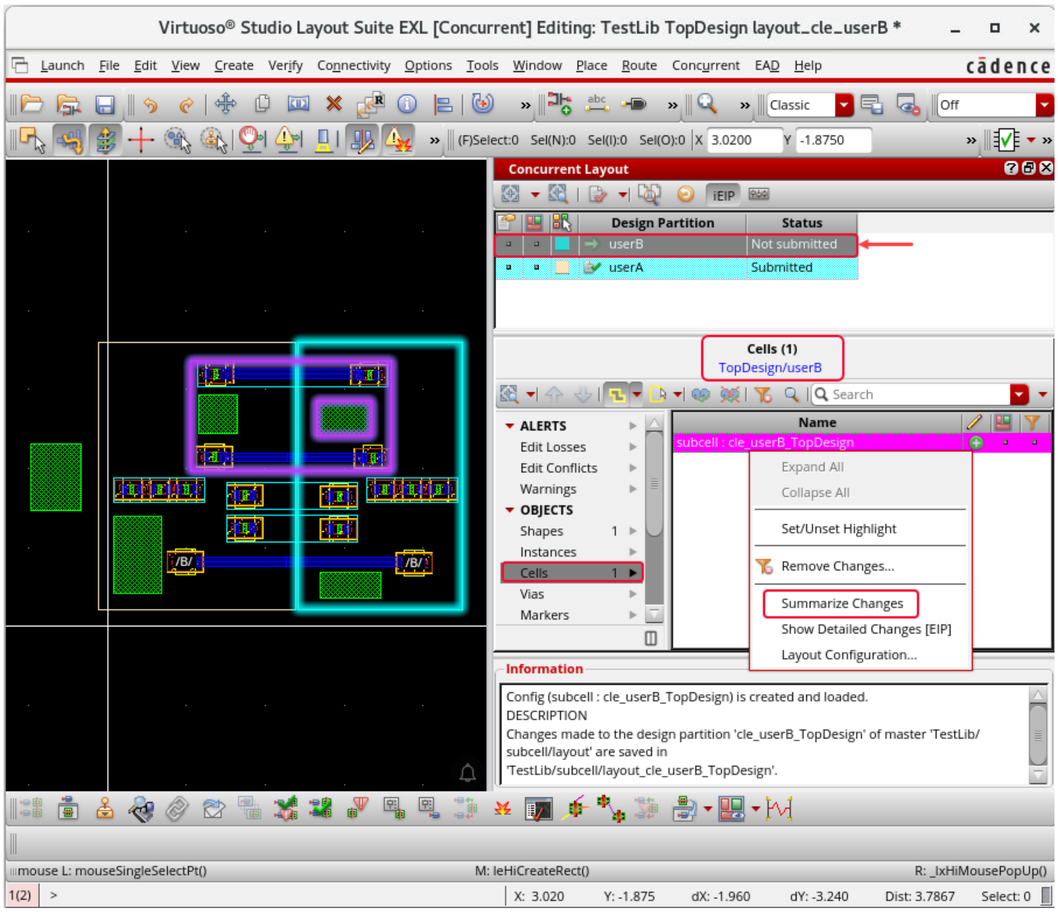
   **Note:** The hierarchical subcells edited in regular mode are excluded.



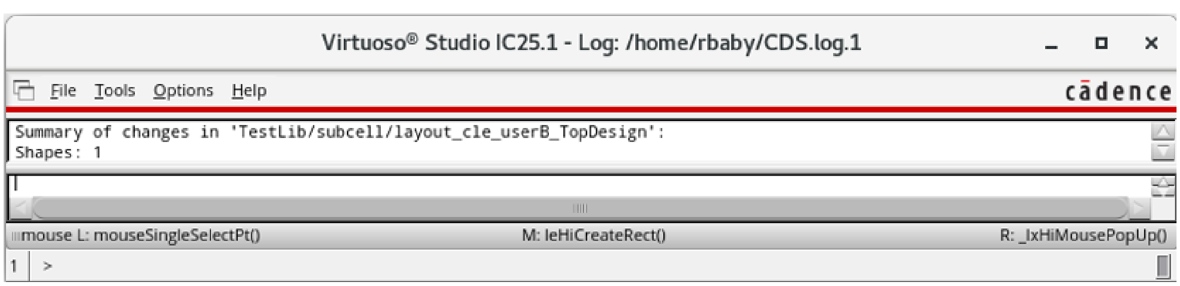
### Summarizing the Changes in the Hierarchical Design Partition View *subcell/cle_userB_TopDesign*

Now, you will observe the changes done in the hierarchical design partition view.

1. To summarize the changes inside the incrementally edited hierarchical subcell, **right**-click the object representing the hierarchical design partition view *subcell:cle_userB_TopDesign* and choose the **Summarize Changes** option.




2. Observe the CIW window and CDS.log file for the *Summary of changes* details in the hierarchical design partition view *TestLib/subcell/layout_cle_userB_TopDesign*.




```
\a _cleHiAssistantSummarizeChanges("TestLib" "subcell" "layout_cle_userB_TopDesign")
\o Summary of changes in 'TestLib/subcell/layout_cle_userB_TopDesign':
\o Shapes: 1
```

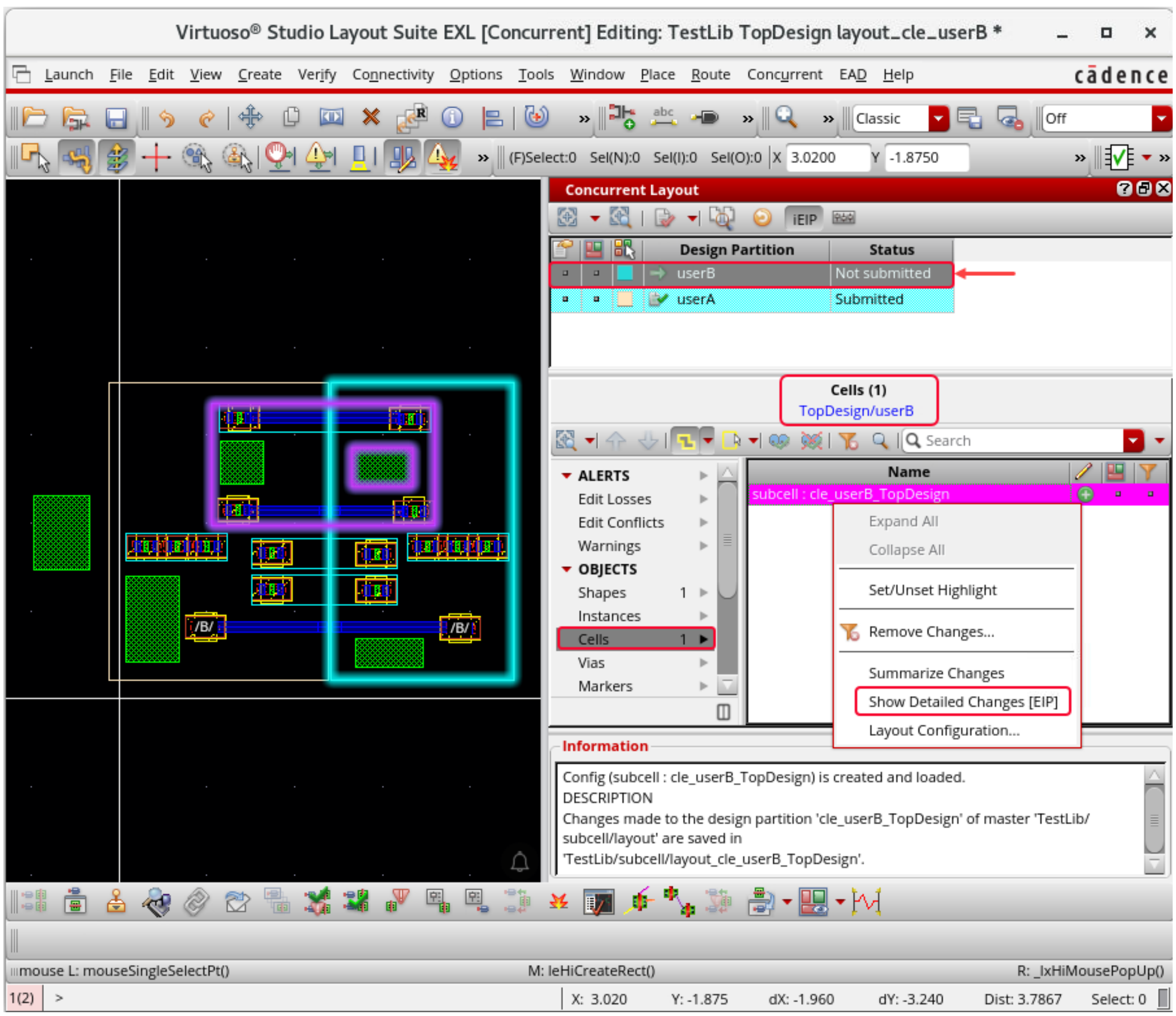
### Verifying the Incremental EIP Updates in *subcell/cle_userB_TopDesign*

Now, you will verify the updates made in the hierarchical design partition view.

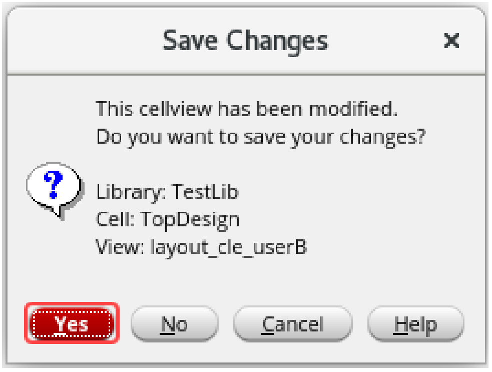
1. To view the details of the edit changes inside the incrementally edited hierarchical subcell, **right**-click the object representing the hierarchical design partition view *subcell:cle_userB_TopDesign* and choose the **Show Detailed Changes (EIP)** option.

   **Note:** This will perform EIP in read-only mode. The object is shown in the *Cells* category in the *OBJECTS* section of the *Summary* Pane.

   **Note:** Updates made in *Regular* mode are not displayed here.



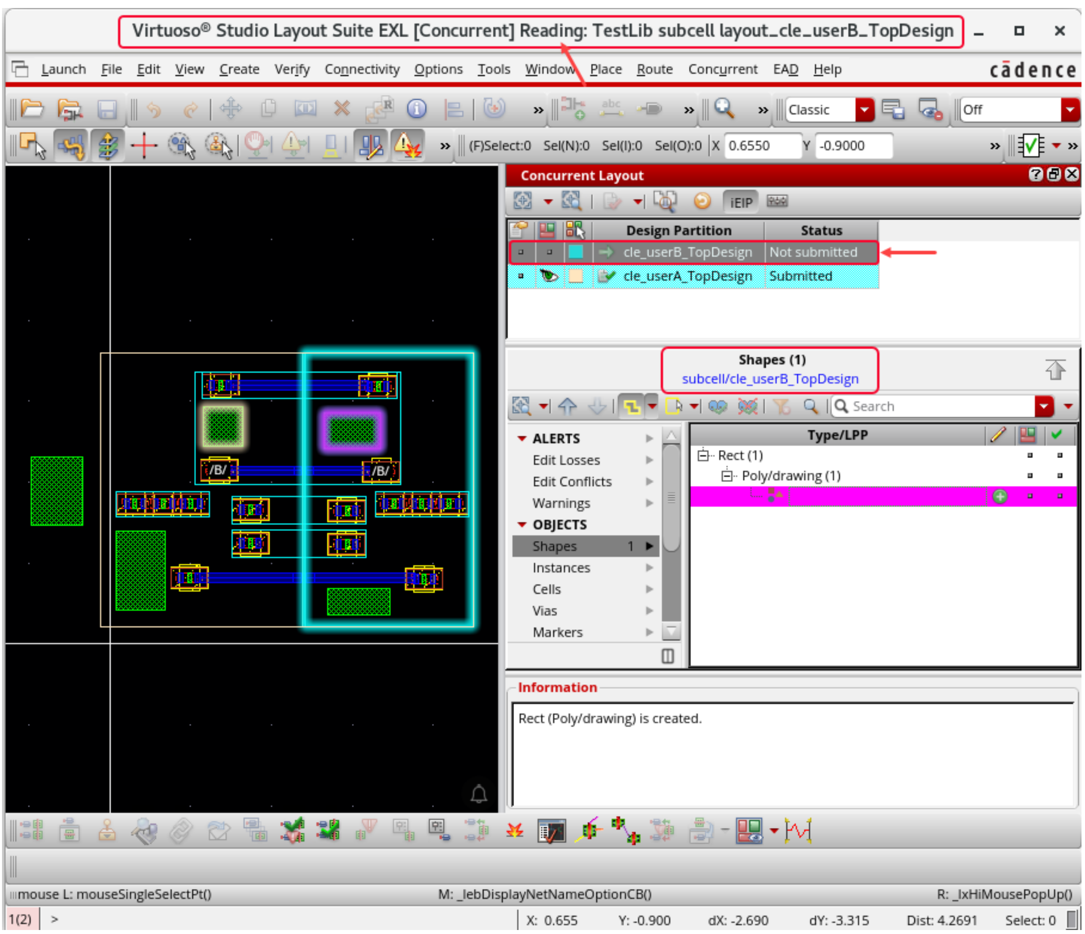
2. Save the changes in the top design partition view *TopDesign/userB* by clicking the **Yes** button.




**Note:** When you choose **Show Detailed Changes [EIP]**, EIP is done into the hierarchical design partition view *subcell/cle_userB_TopDesign* in read-mode.

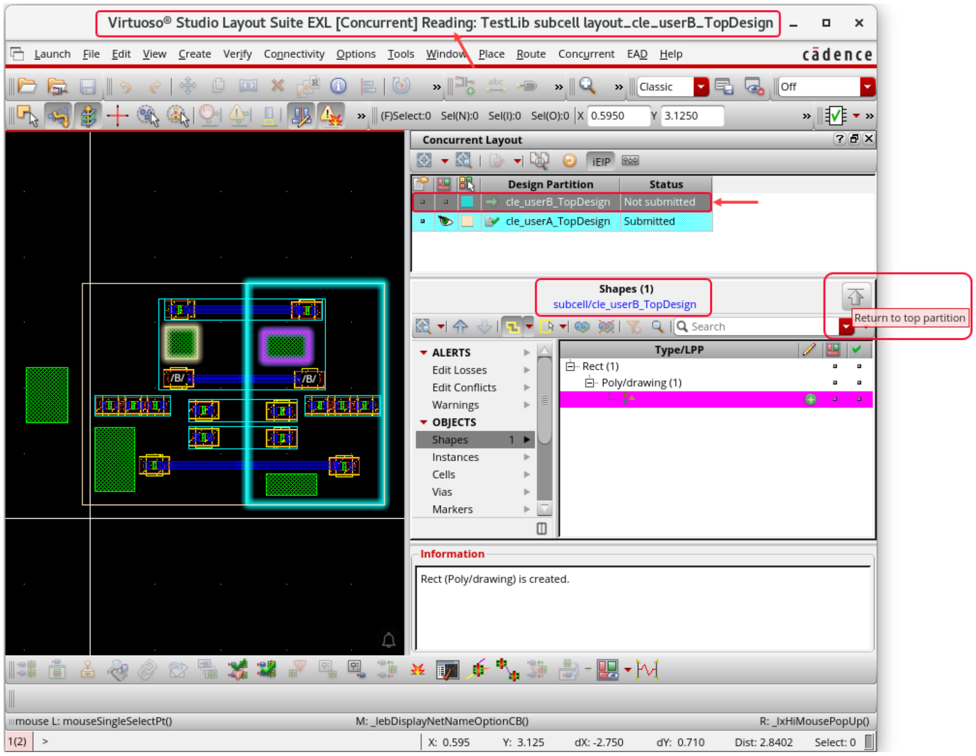
## Reviewing the Edit Changes in the Hierarchical Design Partition View *subcell/cle_userB_TopDesign*

1. On the *Concurrent Layout* assistant, expand the correct category to view the edit changes saved in the hierarchical design partition view *subcell/cle_userB_TopDesign*.

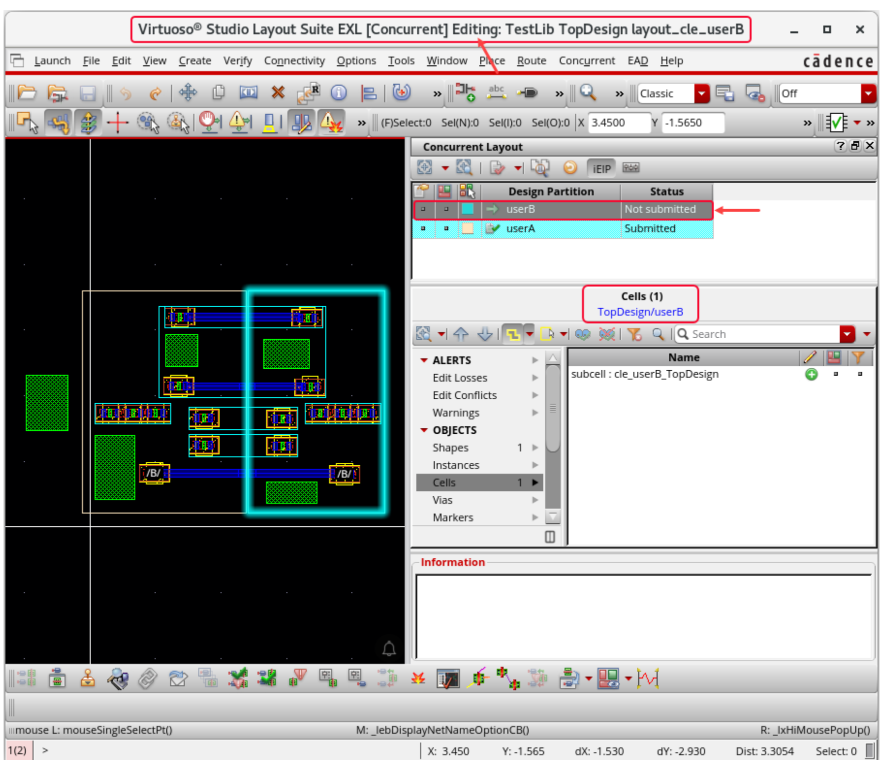


### Returning to the Top Design Partition View *TopDesign/userB*

1. After reviewing all the changes, click the **Return to top partition** button on the *Concurrent Layout* assistant to return to the top design partition view.




Now, you are returned to the top design partition view *TopDesign/userB*.



### Previewing the Changes Made by the Peer Users

You can preview or import the changes from your peer designers to check their progress.

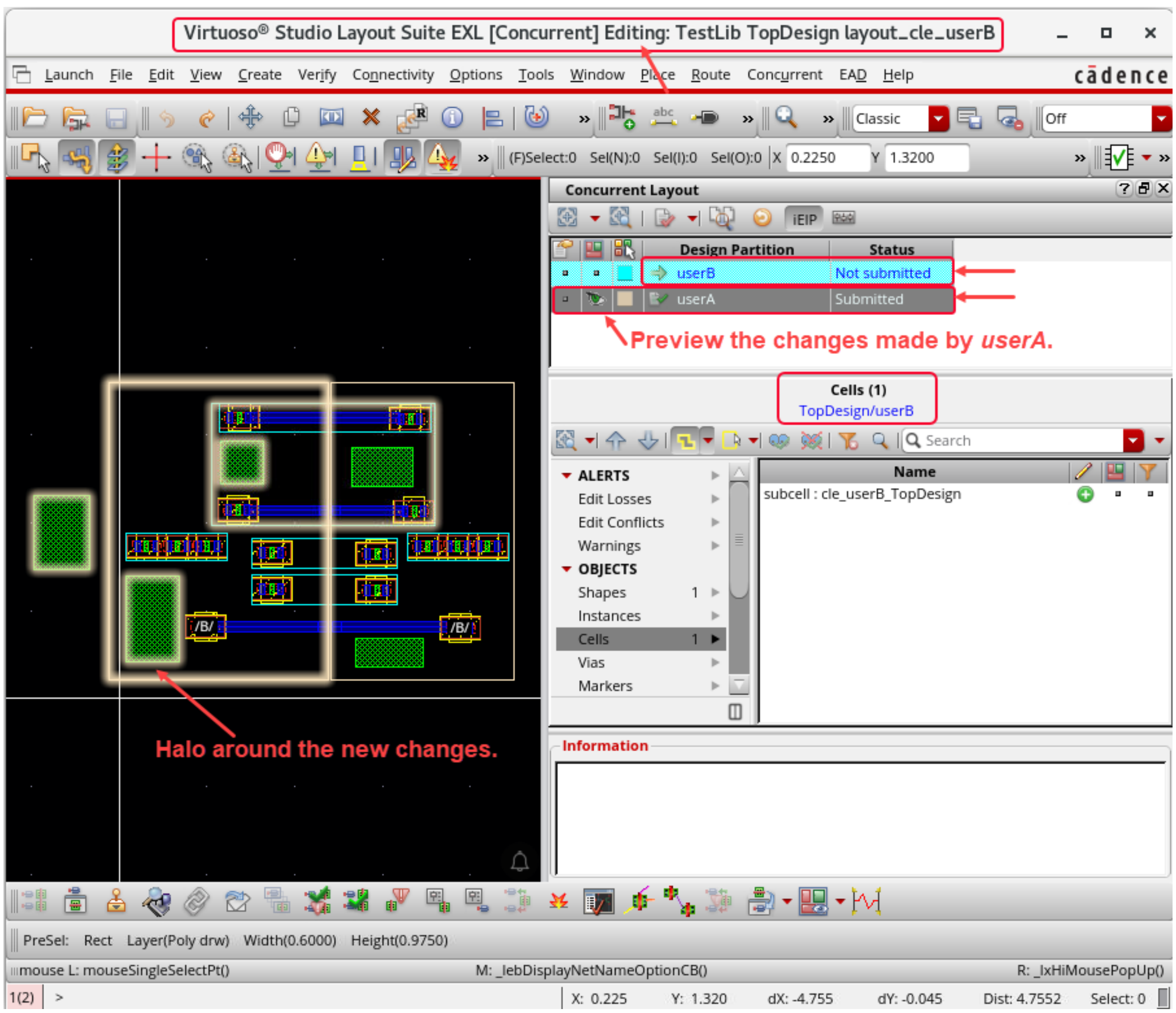
The design partition **userB** can preview the changes in the design partition **userA**. Similarly, the design partition **userA** can also preview the changes in the design partition **userB**.

**Important:** Before you import the changes, ensure that your peers have saved their latest changes.

1. To preview the changes in the design partition **userA**, do the following on the *Concurrent Layout* assistant.

   a. Select the design partition **userA**.

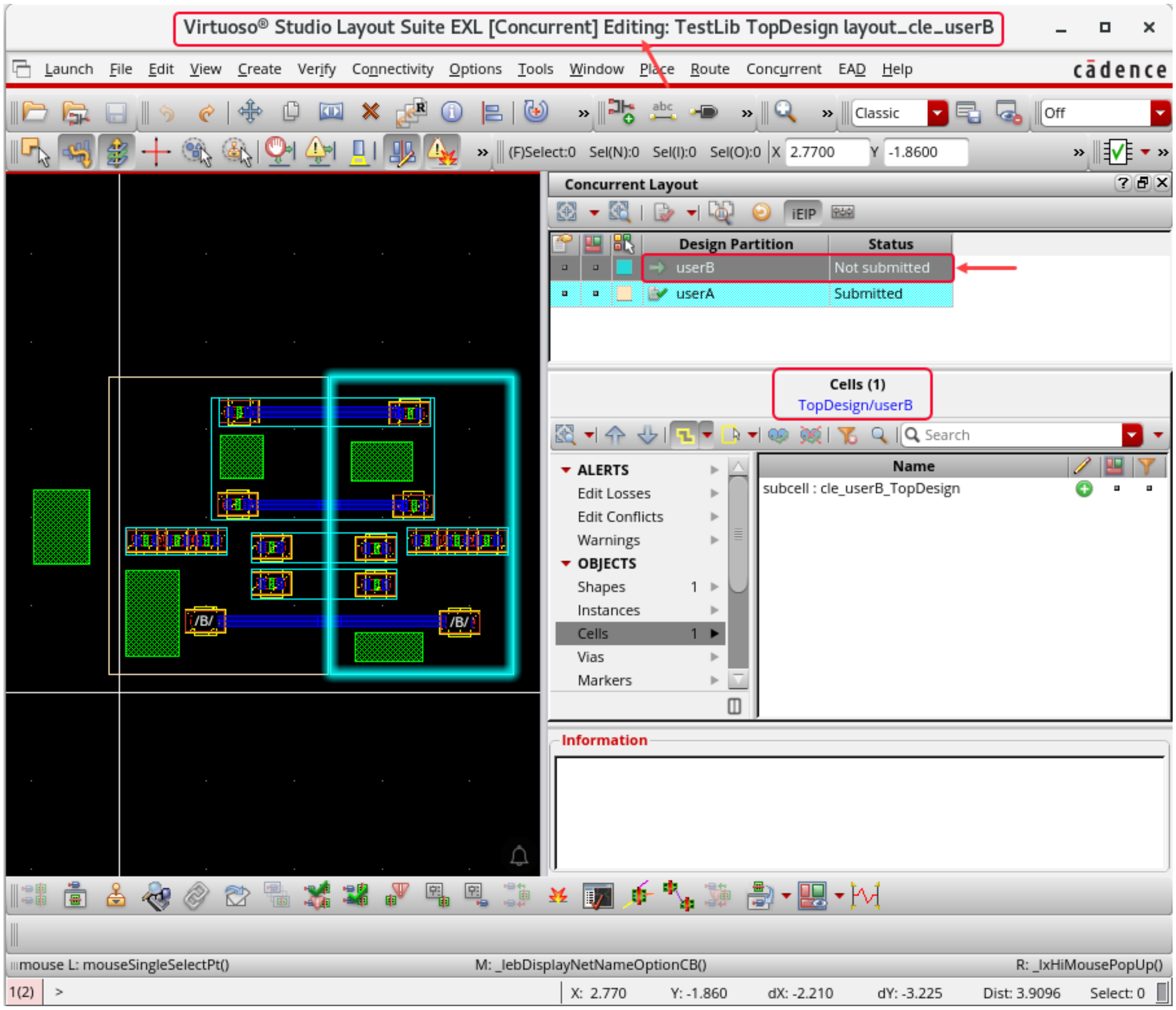
   b. Click the **Preview** button.




All the changes made by the design partition **userA** are marked by *Halo* in the canvas. These changes include the changes done in both the top design partition view and the hierarchical design partition view.

2. Select the design partition **userB** again.

3. Unselect the **Preview** button to deselect the selections made earlier.

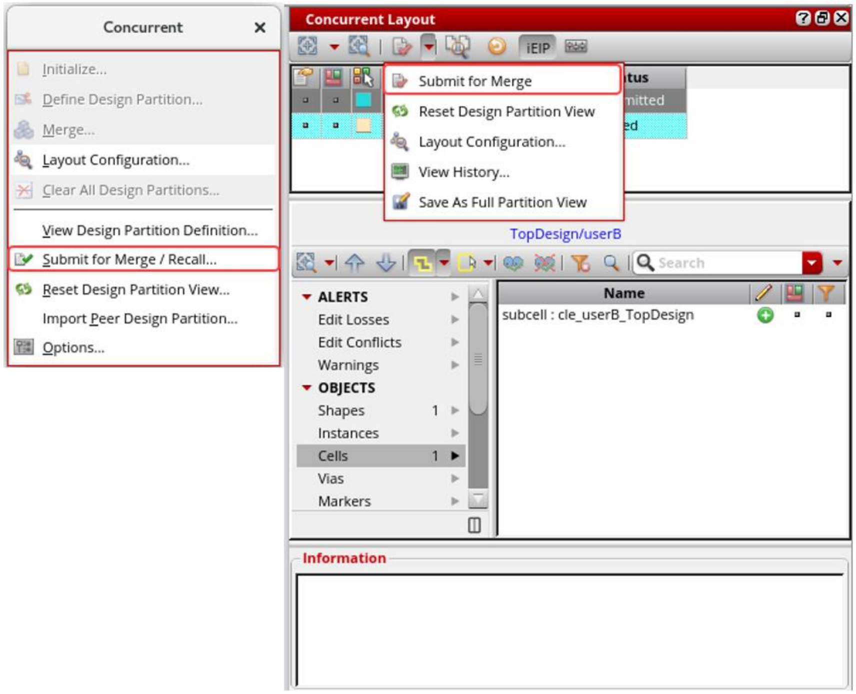



## Submitting the Incremental EIP Updates for Merge: *TopDesign/layout_cle_userB* and *subcell/layout_cle_userB_TopDesign*

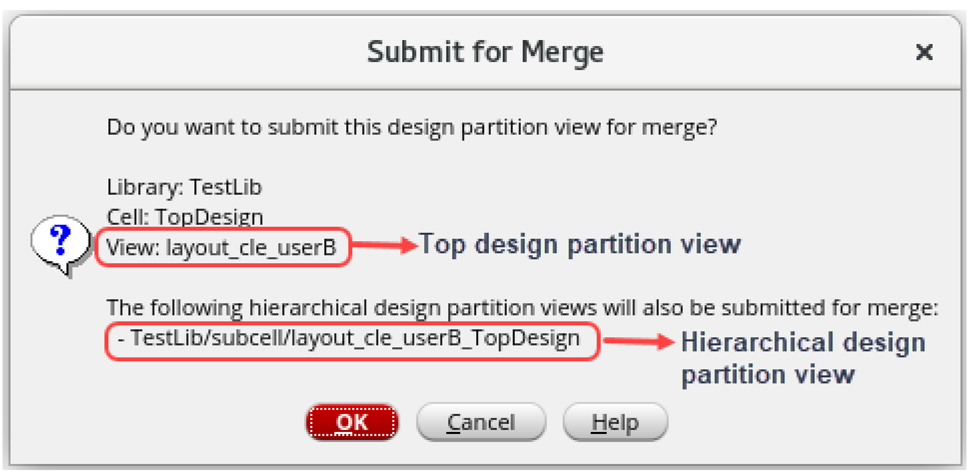
After you have completed and reviewed all the edits in the top design view, you can submit the hierarchical design partition for merge.

**Note:** Make sure you have returned to the top-level design partition.

1. In the layout canvas, choose the **Concurrent – Submit for Merge / Recall** menu command or click the **Submit for Merge** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to submit the updates made in the subcell.




The top-level design partition *TopDesign/userB* is submitted for merge, and *Submit for Merge* dialog box is displayed, as shown here.



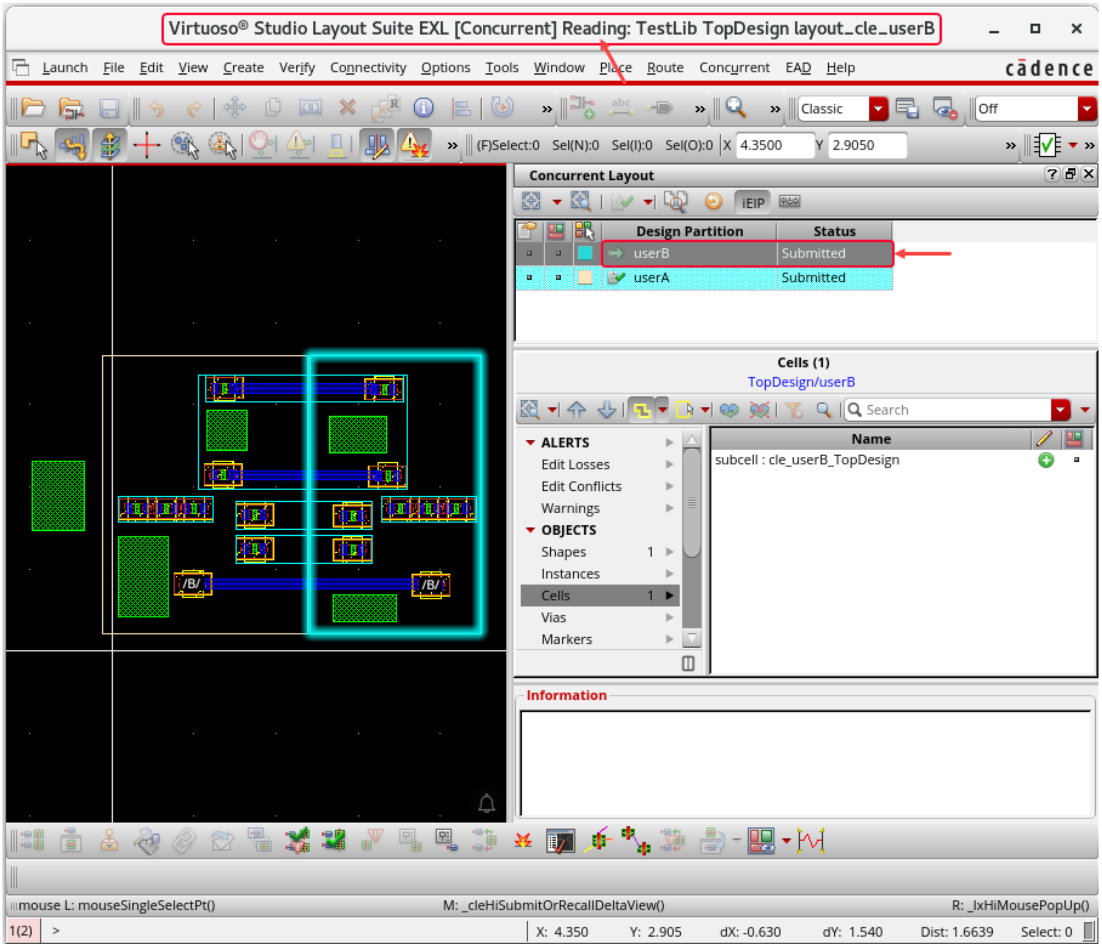

**Note:** The top design partition view *TestLib/TopDesign/layout_cle_userB* is submitted for merge.

**Note:** The hierarchical design partition view *TestLib/subcell/layout_cle_userB_TopDesign* is also submitted for merge.

2. Click **OK** in the *Submit for Merge* form.

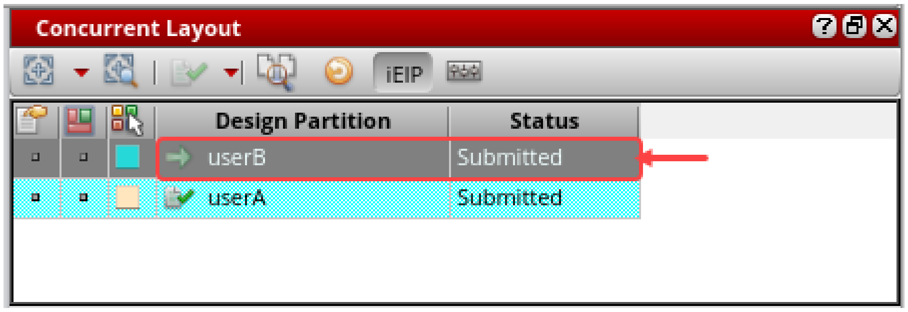
The status of the top design partition view *TestLib/TopDesign/layout_cle_userB* changes to *Submitted*, as shown here.



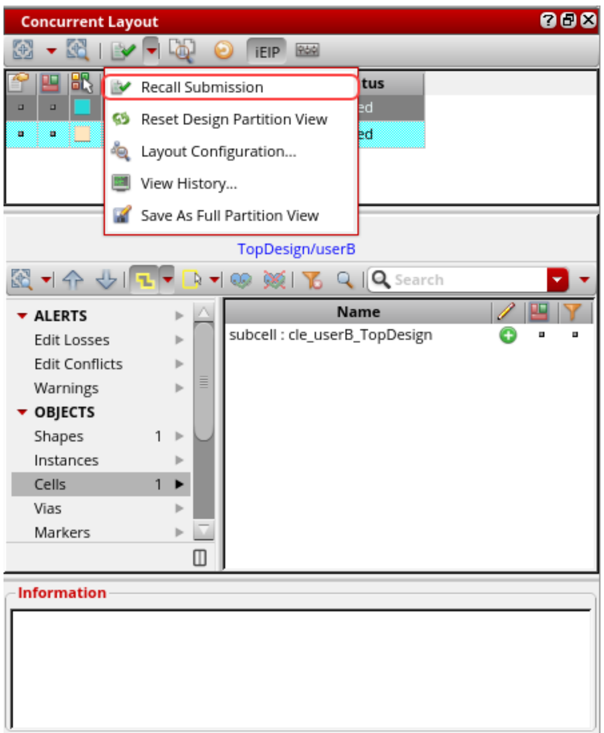

3. Observe the CIW window and CDS.log file for the top design partition view *TestLib/TopDesign/layout_cle_userB* merging details.


```
\o (CLE-101024): Design partition view 'TestLib/TopDesign/layout_cle_userB' submitted for merge.
```

The status changes to *Submitted* for the top design partition view and all the hierarchical partition views that were edited and were in the *Not Submitted* state.




**Note:** The *Submit for Merge* button on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant changes to *Recall Submission*, as shown here.




4. Close the *TestLib/TopDesign/layout_cle_userB* cellview and keep the Virtuoso session running and continue with the next lab.

## Summary

In this lab, you learned how to:

- Edit the design partition **userB** hierarchically in designer mode using the CLE flow.

End of Lab

# Module 6: Merging/Committing the Top Design in Manager Mode

## Lab 6-1 Merging/Committing the Top Design in Manager Mode

### Objective: To merge and commit the top design in manager mode.

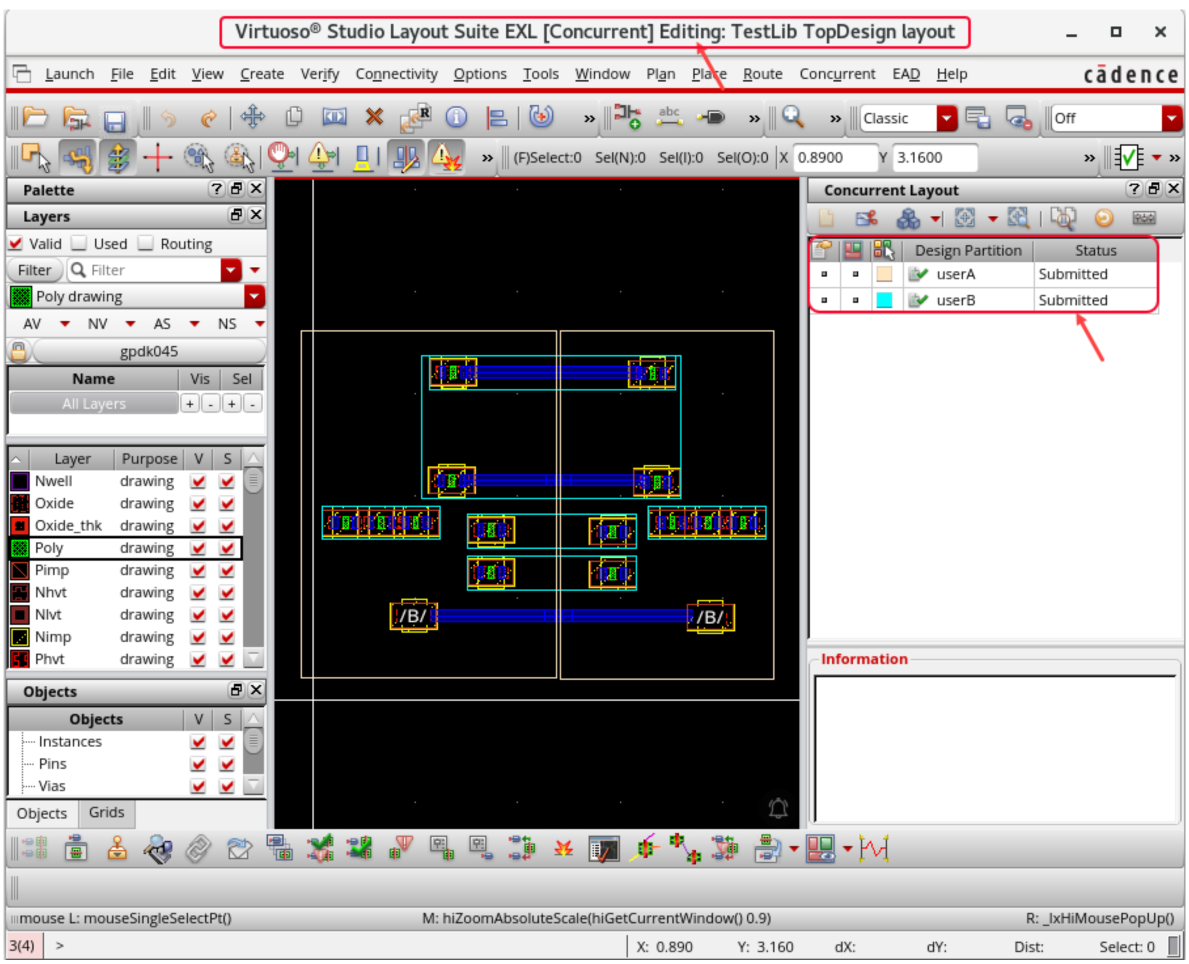
In this lab, you will see how to merge and commit the top design cellview by switching to manager mode using the Concurrent Layout Editing flow.

1. In the following steps, you switch to manager mode and switch to using the first VNC session.

2. In the first VNC session, switch to manager mode by opening the top design cellview from the **Library Manager** or choose **File – Open** from the CIW, as shown here.

| | |
|---|---|
| **Library** | `TestLib` |
| **Cell** | `TopDesign` |
| **View** | `layout` |

The *TestLib/TopDesign/layout* window opens in the **EXL** mode with CLE enabled.



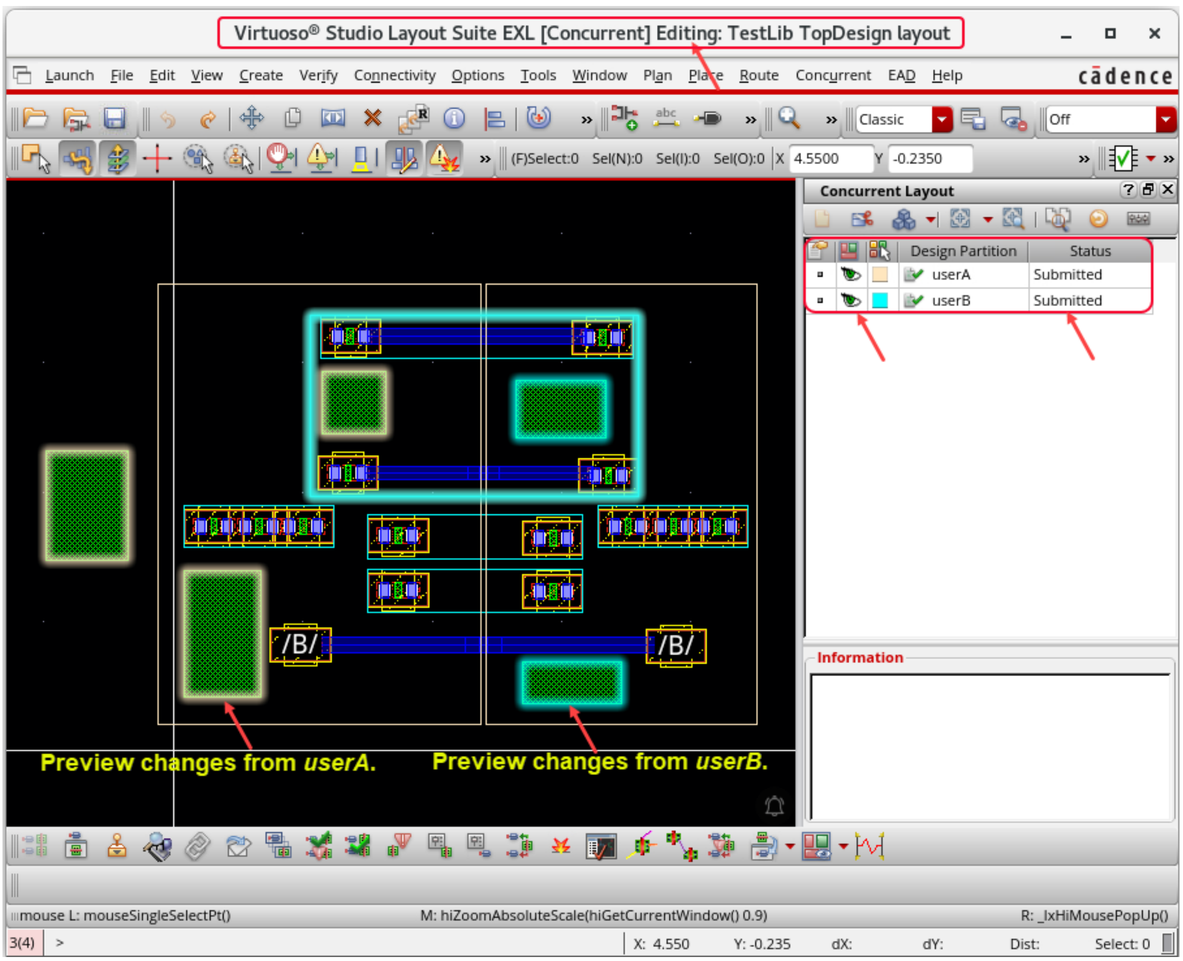
### Previewing the Design Partitions *userA* and *userB* in the Top Design

Now, you will preview the design partitions **userA** and **userB** in the top design.

1. In the layout canvas, select the first design partition **userA** and click the **Preview** button to review the changes.

2. Repeat this step for the second design partition, **userB**.

   The new changes appear transiently and are highlighted with a *Halo* in the layout canvas.



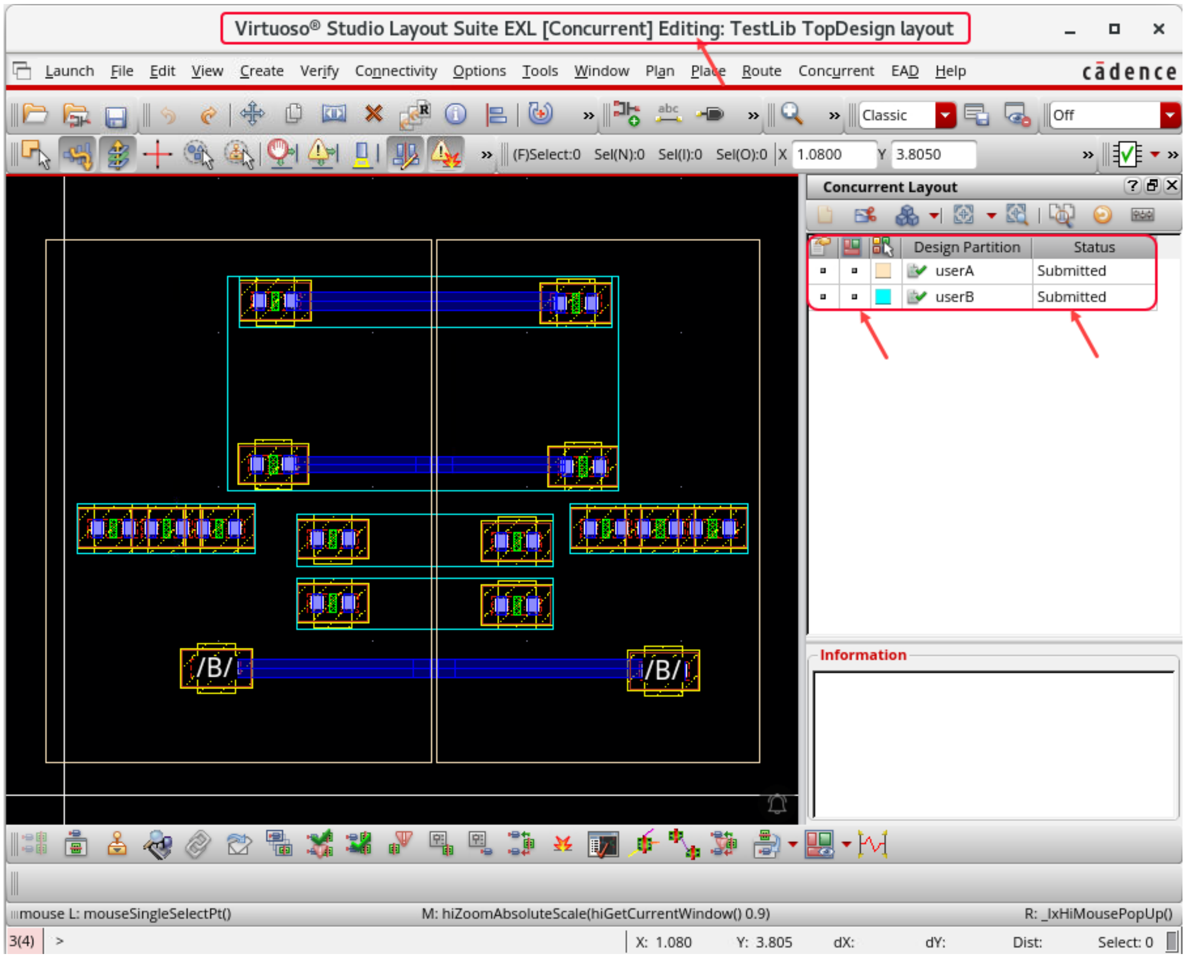

### Removing the Previews in the Design Partitions *userA* and userB in the Top Design

Now, you will remove the previews in the design partitions **userA** and **userB** in the top design.

1. Unselect the **Preview** button in the design partitions **userA** and **userB** to deselect the selections made earlier.

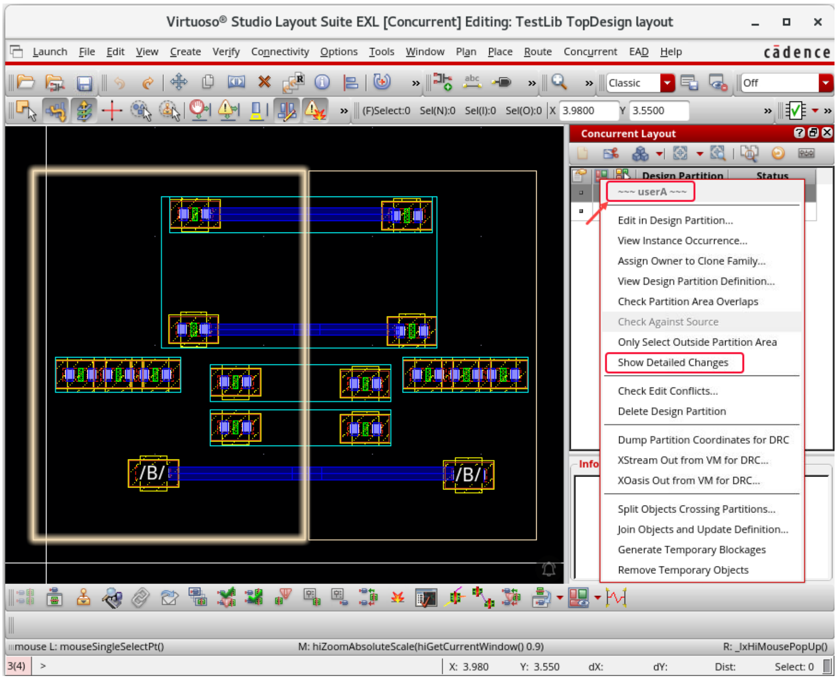
The previews in the design partitions **userA** and **userB** in the top design are removed.



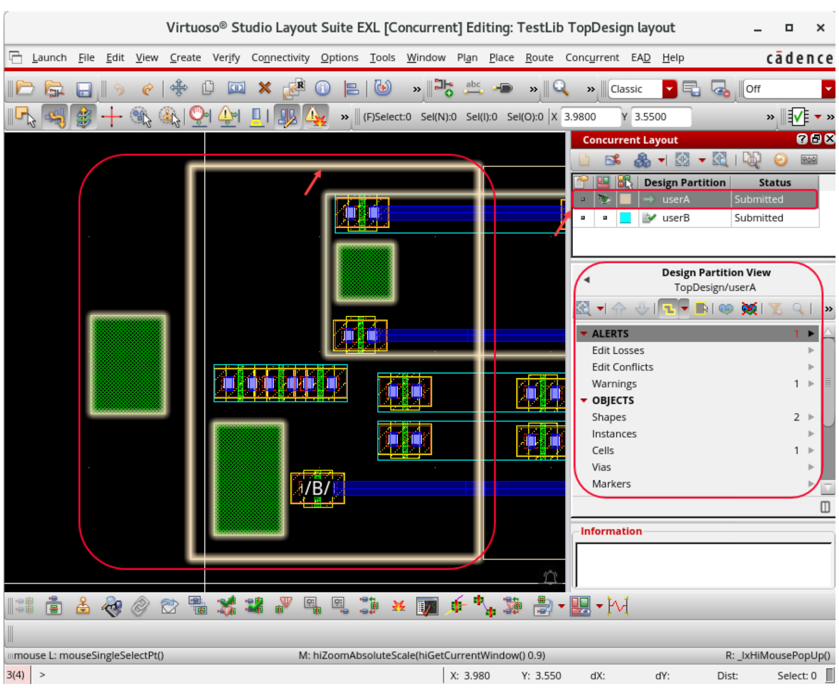

## Reviewing All the Changes in the Design Partition *userA* in the Top Design

Now, you will review all the changes in detail in the design partition **userA** in the Top Design.

1. In the layout canvas, on tte *Concurrent Layout* toolbar in the *Concurrent Layout* assistant, **right**-click the design partition **userA** and select the *Show Detailed Changes* option, as shown here.




The *Summary of the changes* is loaded in the *Concurrent Layout* assistant for each object type, as shown here.

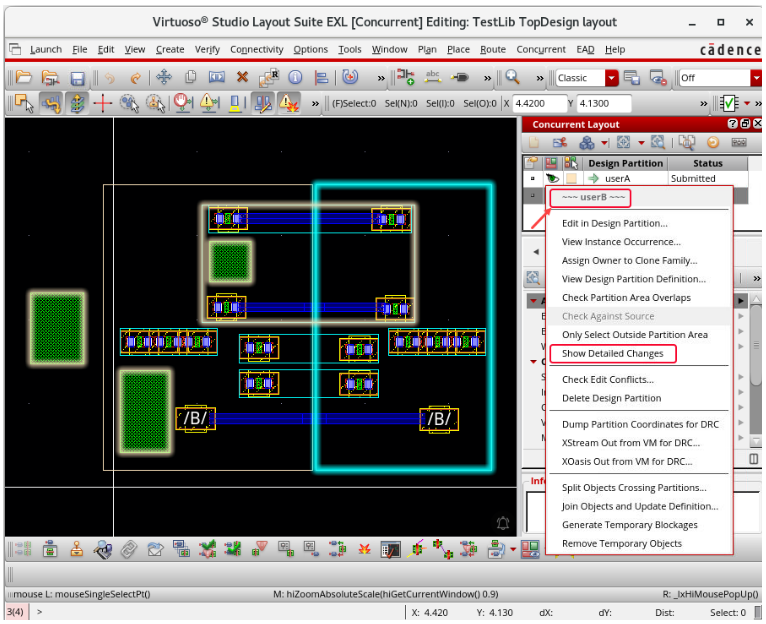


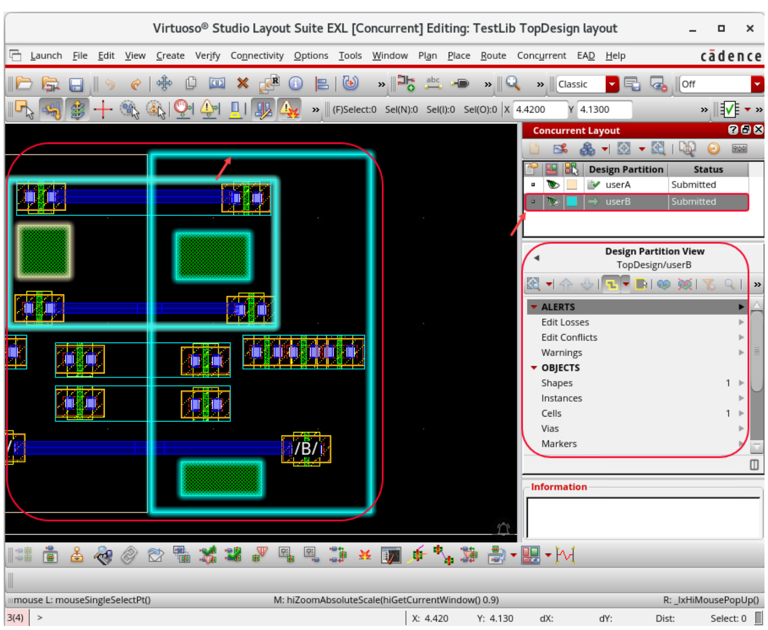
### Reviewing All the Changes in the Design Partition *userB* in the Top Design

Now, you will review all the changes in detail in the design partition **userB** in the Top Design.

1. In the layout canvas, on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant, **right**-click the design partition **userB** and select the *Show Detailed Changes* option, as shown here.

   The *Summary of the changes* is loaded in the *Concurrent Layout* assistant for each object type, as shown here.

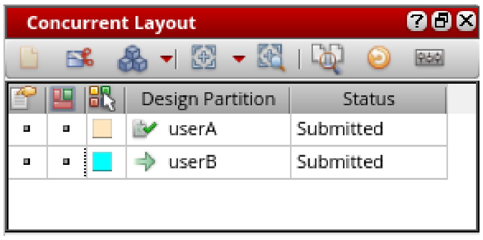
   
   

2. Unselect the **Preview** button in the design partitions **userA** and **userB** to deselect them.

### Merging a Submitted Design Partition

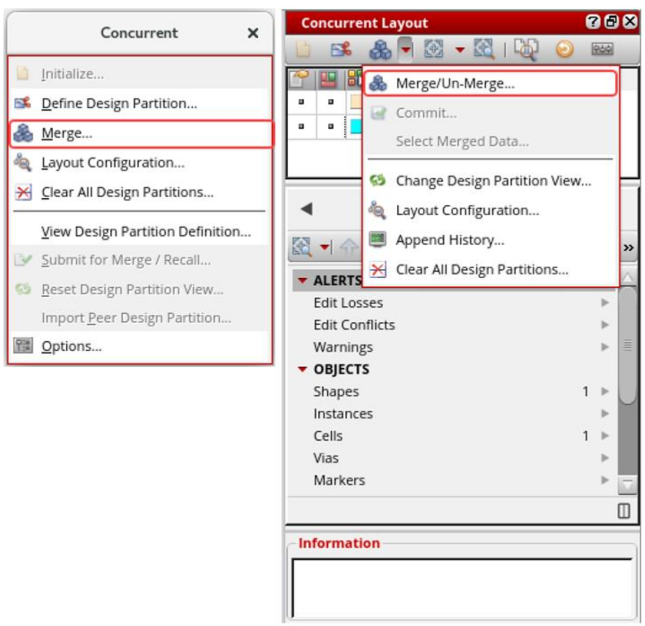
After the designers have completed editing the design and submitted their respective design partitions for merge, the manager needs to review these requests and then either approve or reject the merge requests.

When the merge process is in progress, the manager should make changes only in the design partition view. Any direct edits to the top design during merge are discarded when the top is reverted to the previously unmerged state.

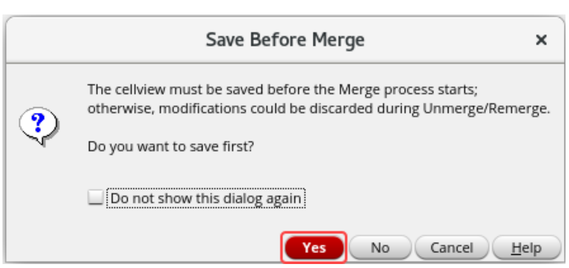



To approve the merge requests, perform the following steps.

1. In the layout canvas, choose the **Concurrent – Merge** menu command or click the **Merge/Un-Merge** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant.

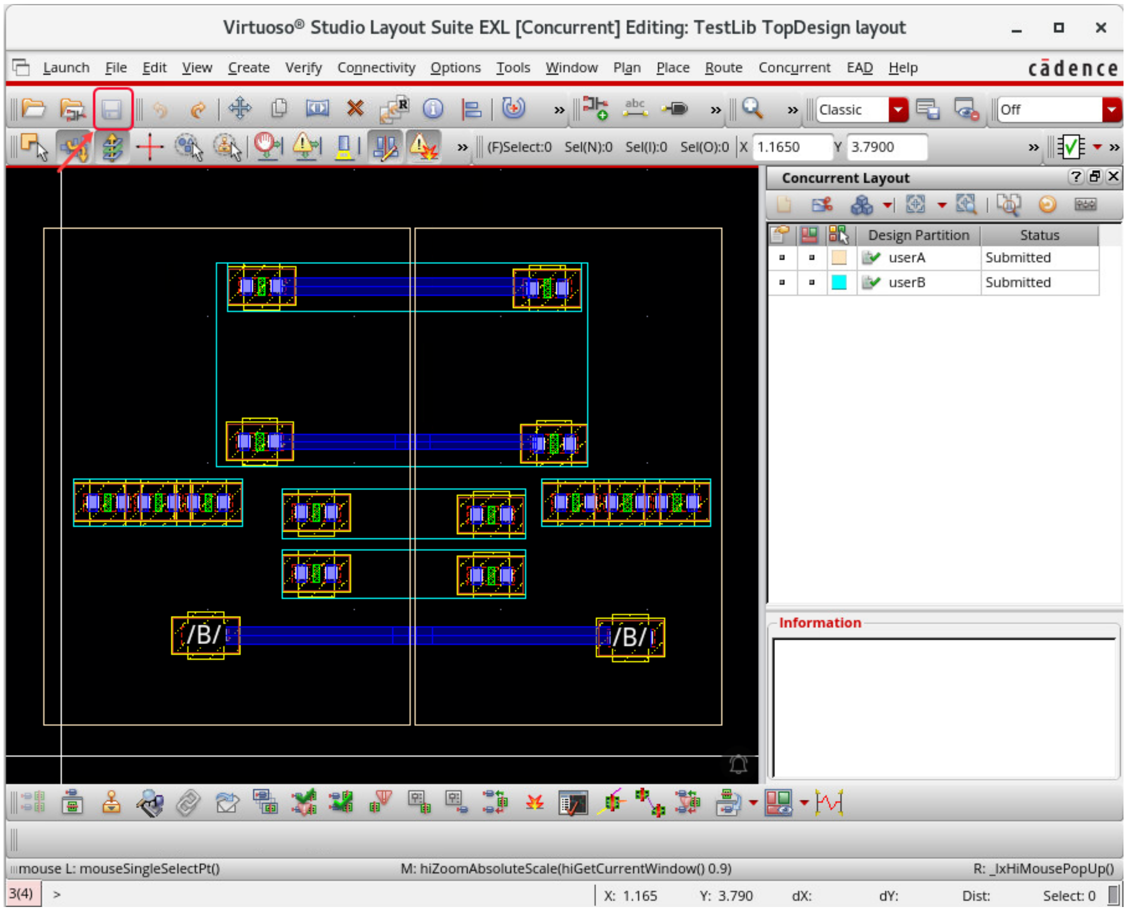


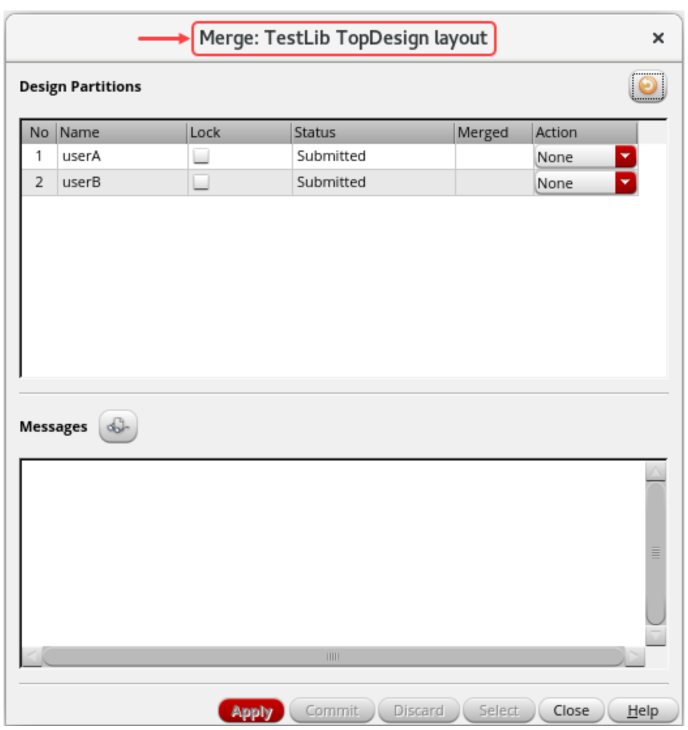

You get the *Save Before Merge* form to save the *TestLib/TopDesign/layout* cellview.



2. Click **Yes** in the *Save Before Merge* form.

   The *TestLib/TopDesign/layout* cellview is saved, and the *Merge* form is displayed.



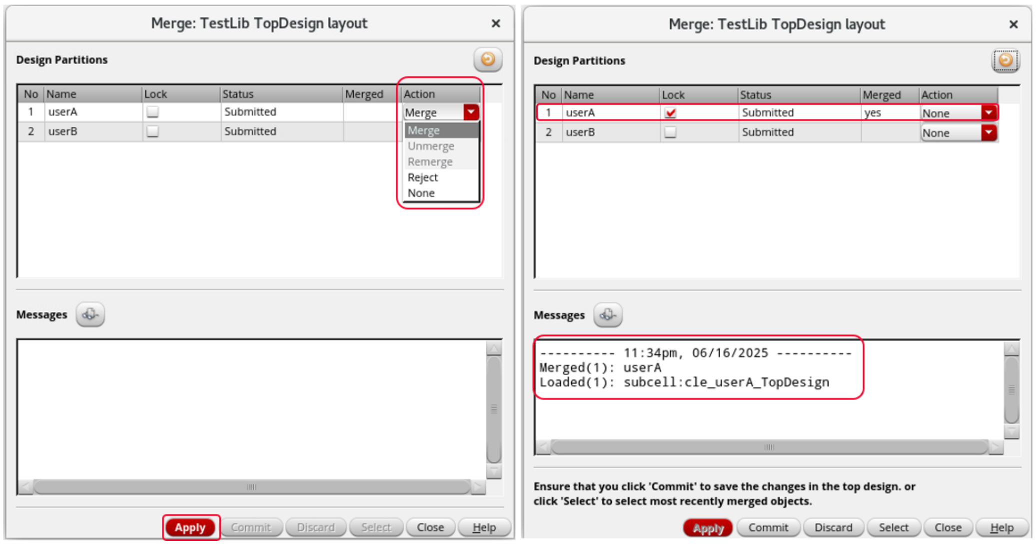



### Merging the Submitted Design Partitions *userA* and *userB*

You will merge the submitted design partitions **userA** and **userB** using the *Merge* form. In the *Merge* form, do the following steps to merge the design partitions.

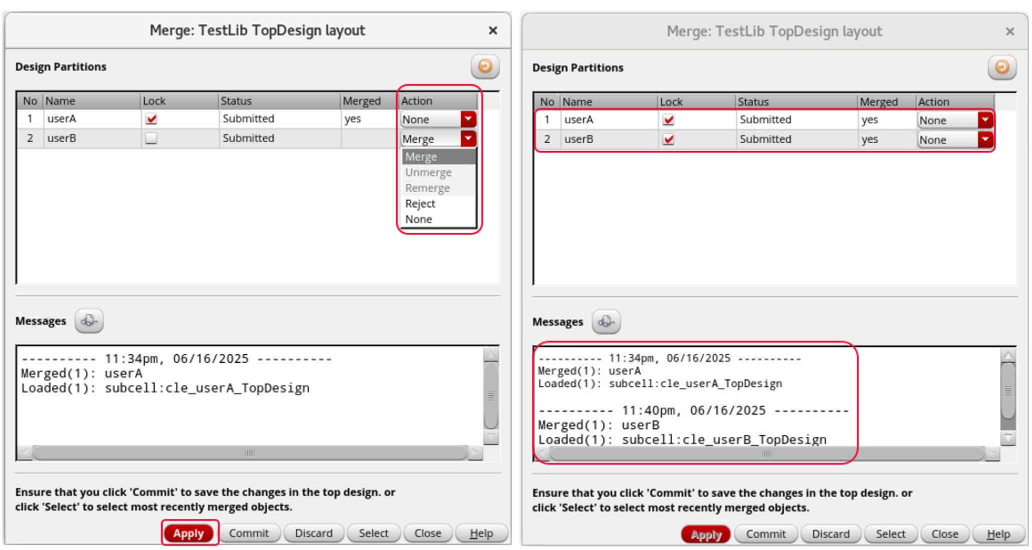
1. Click in the *Action* field of the design partition **userA**.




   a. Select the *Merge* option and click the *Apply* button.

   The design partition view *TestLib/TopDesign/layout_cle_userA* was successfully merged with design partition **userA**.

2. Now click in the *Action* field of the design partition **userB**.




   a. Select the *Merge* option and click the *Apply* button.

   The design partition view *TestLib/TopDesign/layout_cle_userB* was successfully merged with design partition **userB**.

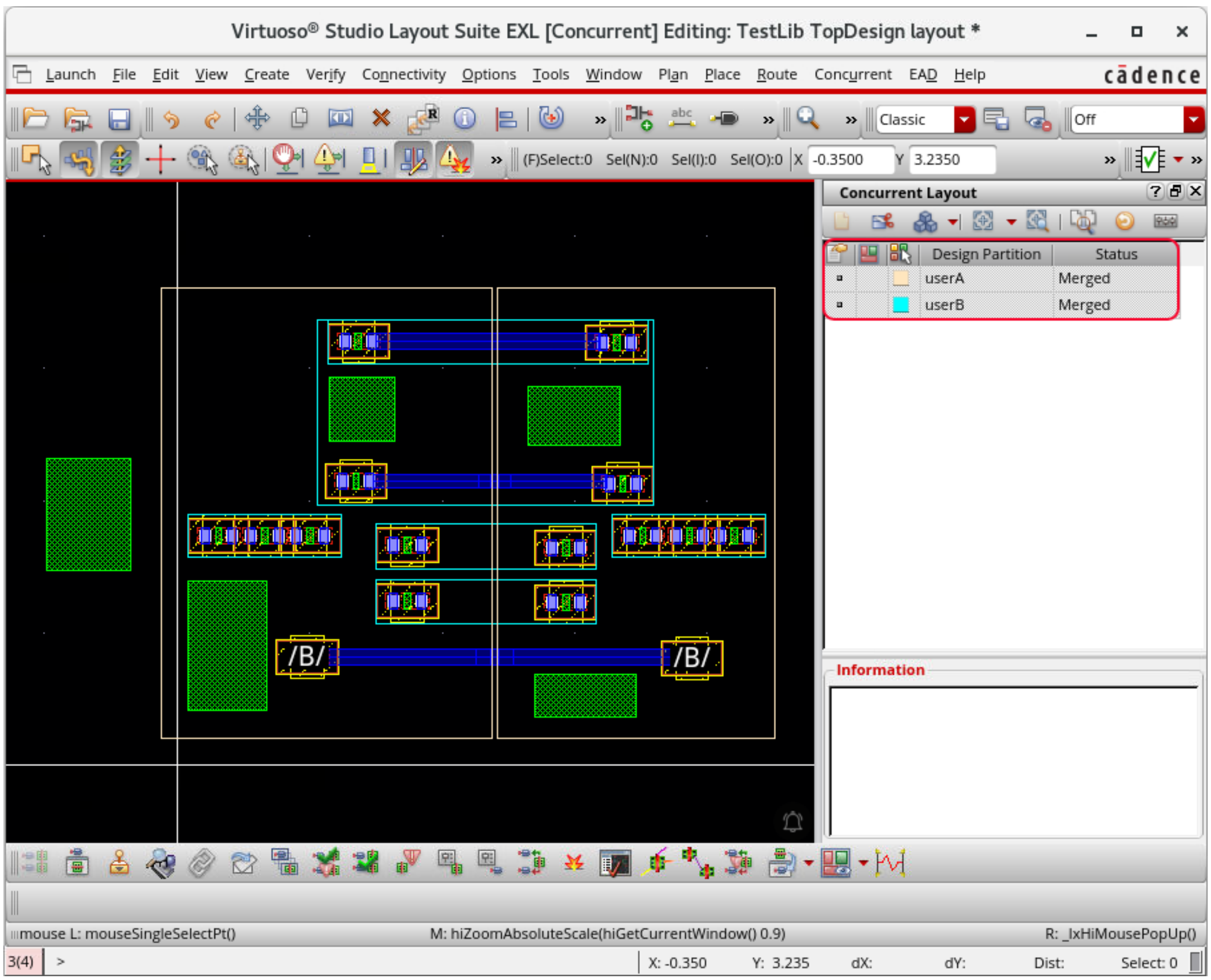
**Note:** The design partitions with action set as *Merge* are merged with the top design.

**Note:** To specify an action for each design partition, you can select each design partition separately or select all of them together.

**Important:** After the merge process is complete, the state of the top design partition view changes to *Merged*, and the hierarchical partition views are loaded in the virtual memory.

**Important:** The top design is now modified after the merge is complete. However, you can still unmerge the changes of one or more or all the design partition views. To do this, in the *Merge* form, select the design partitions to unmerge, set the *Action* as *Unmerge*, and click the *Apply* button.

The status of the design partitions **userA** and **userB** changes to *Merged*.



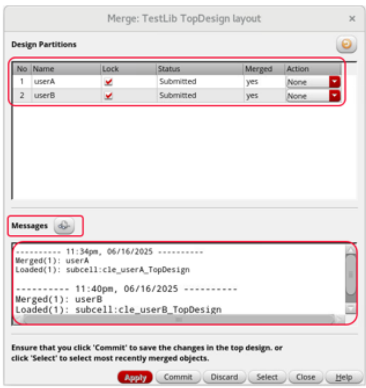

Edit lock is set on all design partition views and hierarchical design partition views to prevent designers from making any new changes on a submitted design partition view.

On merge, the edit lock is not set on the design partition views that are in the *Not Submitted* state to support the health-check flow.

### Viewing the Merge Log

Information regarding the actions taken is added to the **Messages** box in the *Merge* form. You will view the merge log now.

1. In the *Merge* form, click the **Messages** button to view the merge log, as shown here.



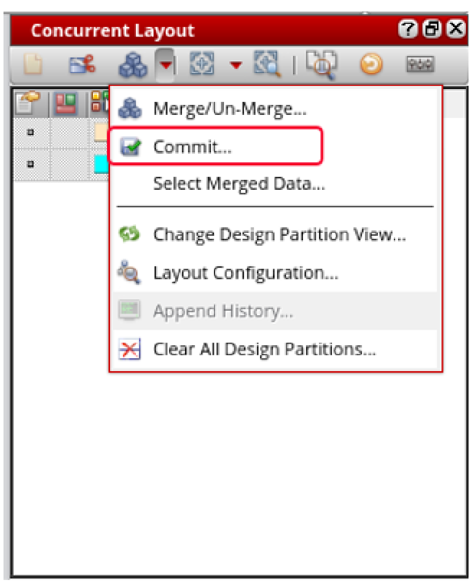


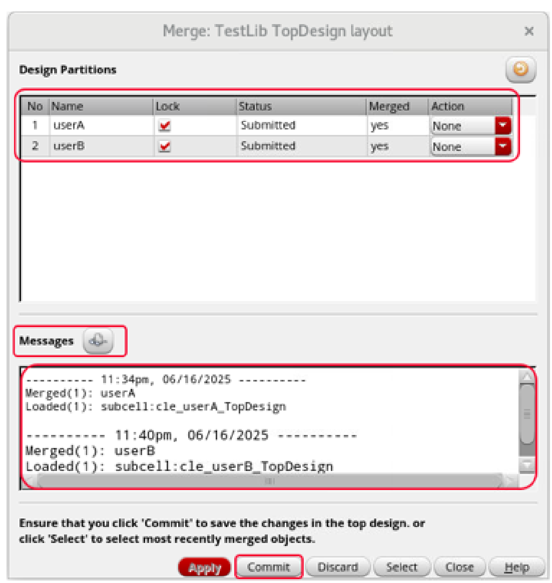
### Committing the Merged Design Partitions *userA* and *userB*

You need to run the **Commit** command to complete the merge process and save the merged changes into the top design.

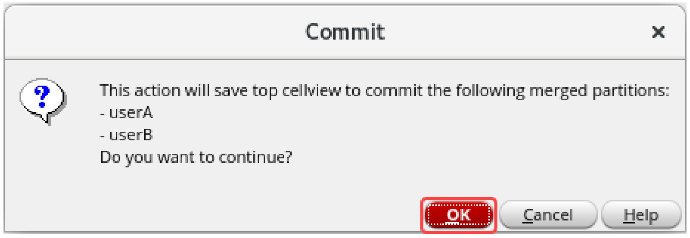
To commit the design partitions, perform the following steps.

1. In the layout canvas, click the **Commit** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant or click the **Commit** button in the *Merge* form.



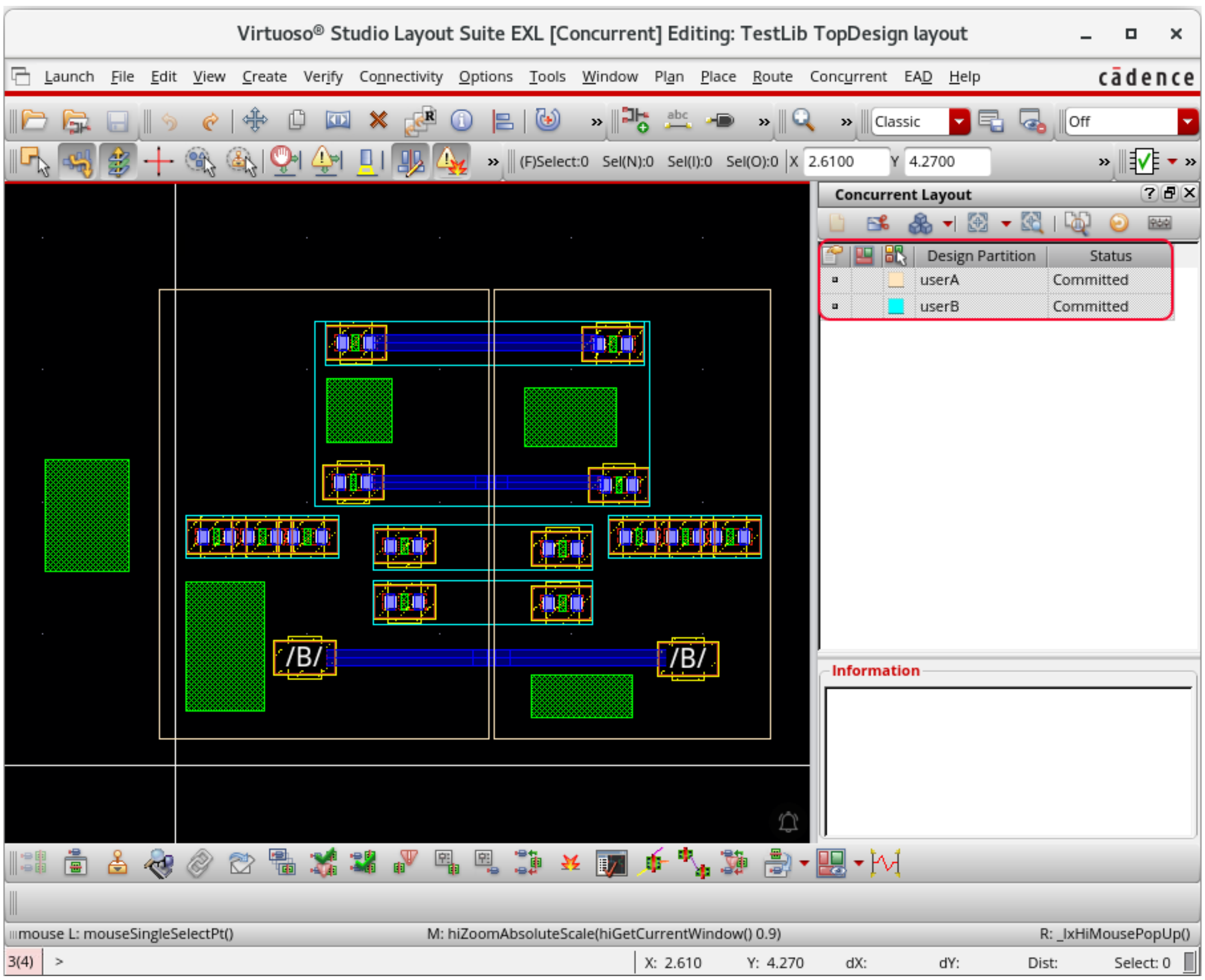
The *Commit* dialog box is displayed to confirm the merged design partitions **userA** and **userB** undergoing *Commit*.




2. Click **OK** in the *Commit* dialog box to continue.

   This step recursively merges the children's hierarchical design partition views into the subcell and commits the changes.

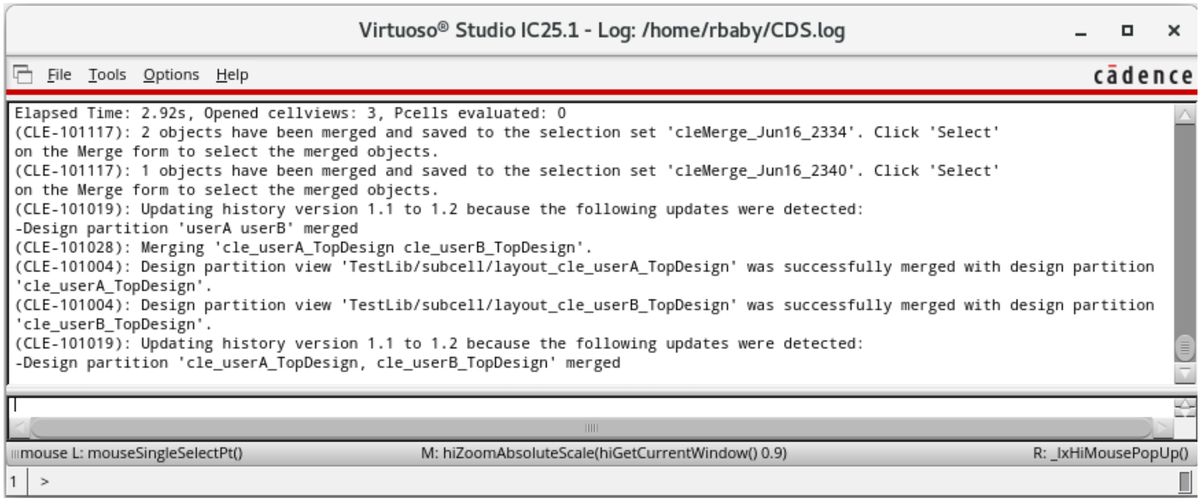
   The status of the design partitions **userA** and **userB** changes to *Committed*.




This means that the changes from respective design partition views have now been added to and saved in the top design.

After committing, the design partition views still exist. However, the edit locks on them are released. Designers can continue to work on these design partitions if needed.

3. Observe the CIW window for the merging details of the design, as shown here.



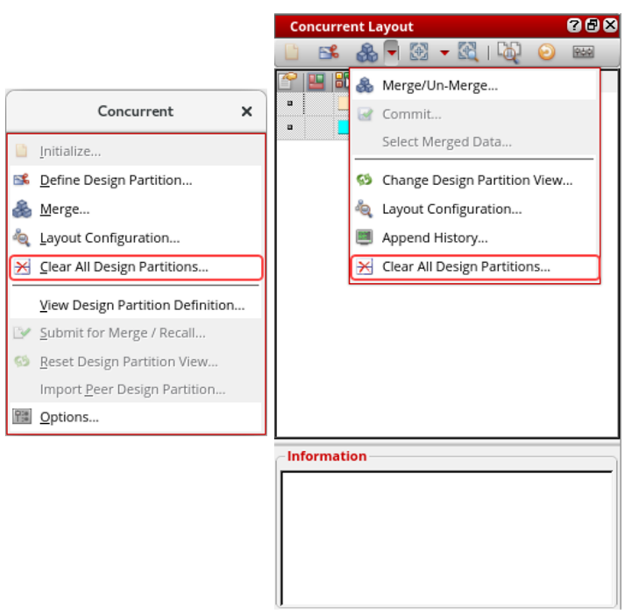

## Clearing the Design Partitions *userA* and *userB* in Manager Mode (Optional)

After all the design partitions are merged with the top design, you might want to delete all design partitions and exit the Concurrent Layout Editing environment.

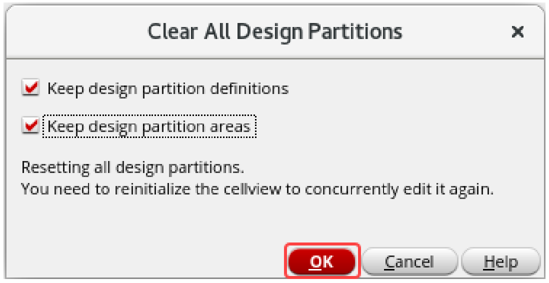
If your task is complete and there is no need for concurrent editing, you can delete the design partitions.

1. To clear all the design partitions, use the following steps.

   a. In the layout canvas, choose the **Concurrent – Clear All Design Partitions** menu command or click the **Clear All Design Partitions** option from the *Merge* drop-down list on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to clear all the design partitions.



The Clear All Design Partitions form is displayed.

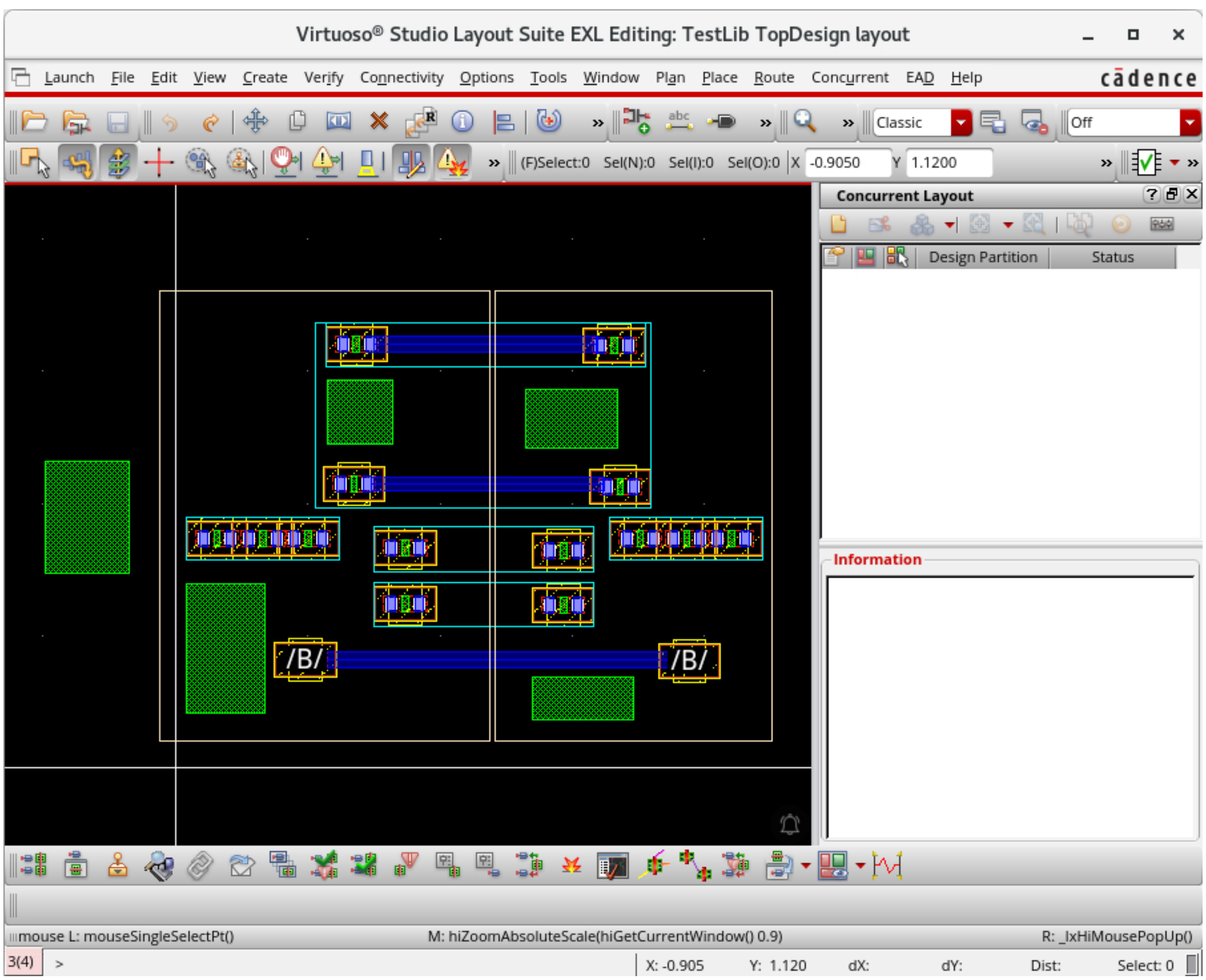



b. Select the **Keep design partition definitions** check box if you want to retain the design partition definitions.

c. Select the **Keep design partition areas** check box in the area-based design partitions if you want to retain the design partition areas created using the Define Design Partition form.

d. Click **OK**.

All the design partitions will be deleted, and you can edit the design without using Concurrent Layout.



### Summary

In this lab, you learned how to:

- Merge and commit the top design cellview by switching to manager mode using the Concurrent Layout Editing flow.

End of Lab